

Figure Comparisons

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Each page compares an original book illustration with its TikZ reconstruction. The figure number and full caption are taken from the accompanying main book; the heading links to that figure. When a book figure contains several panels, a panel number identifies the illustrated component in source order.

To refer to a comparison, use its **figure number and panel**, or the stable identifier printed in the footer. PDF bookmarks provide direct access to each comparison.

Figure 1.4

Main book, page 5

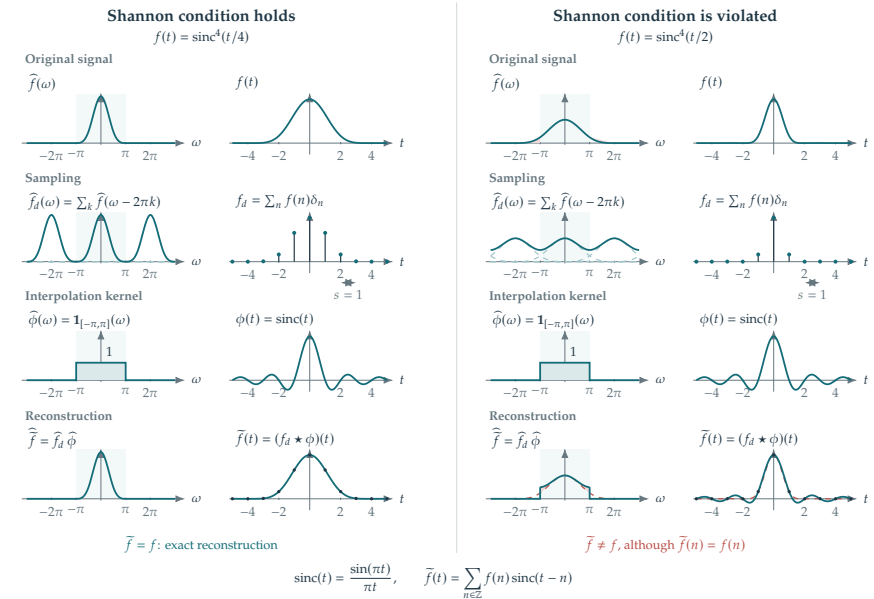
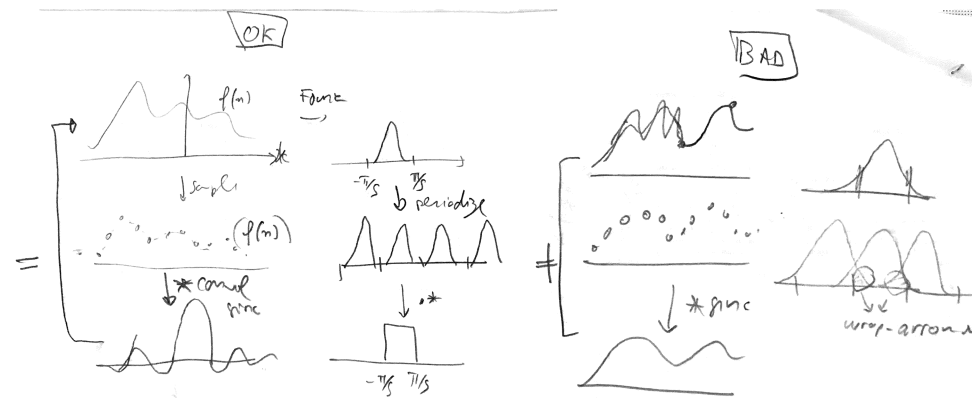
Sampling and Shannon interpolation in the frequency and time domains, with unit sample spacing. Left: the bandlimiting condition of Theorem 1.2 holds. Right: overlapping spectral replicas cause aliasing; the reconstructed signal still passes through every sample. The rows show the original signal, sampling, the interpolation kernel, and reconstruction.

Complete figure / Shannon interpolation with and without aliasing

The sampling theorem 1.2 and Poisson formula (1.7) connect the time and frequency domains at each of four stages.

Original

TikZ reconstruction



Pale band: $[-\pi, \pi]$. Dashed spectral curves: replicas. Red shading: aliased mass.
 Dashed red in the reconstruction row: the original signal and spectrum. Dots: the shared samples.

Reading the reconstruction. The paired layouts follow the supplied shannon-interp reference images. Unit sample spacing is used. Exact signal/spectrum pairs are $\text{sinc}(t/a)^4$ and a B3($a\omega/(2\pi)$), for $a=4$ and $a=2$. The rows show the original signal, its weighted Dirac comb and replicated spectrum, the sinc kernel and ideal low-pass filter, and the reconstruction. In the aliased case the bottom time curve is the exact inverse transform of the clipped replicated spectrum; it differs between samples while retaining every sample value. Red shading tracks folded spectral mass; the dashed red curves show the originals in the reconstruction row. The same vertical scales are used within each domain.

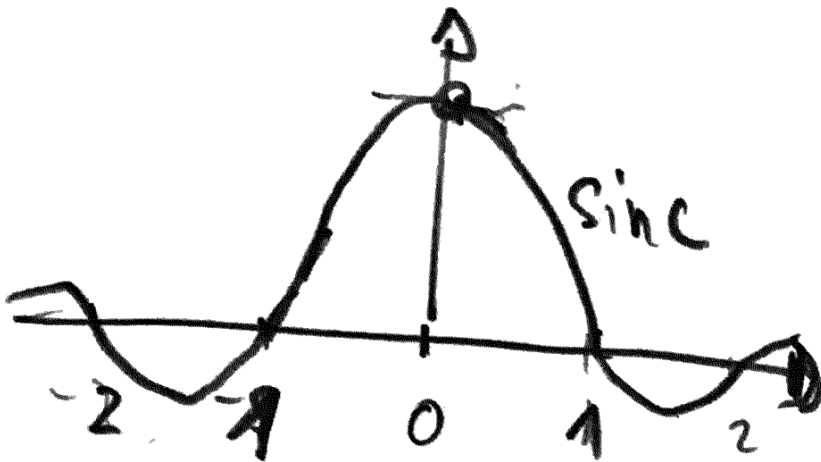
Figure 1.5

The sinc interpolation kernel.

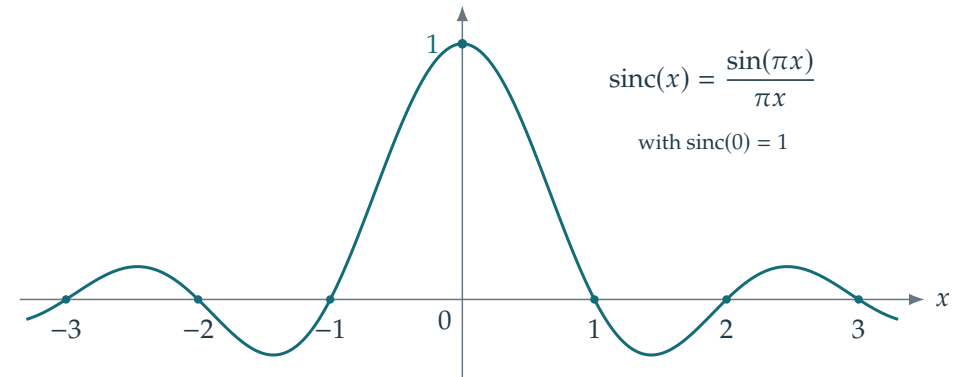
Complete figure / The cardinal sine

The reconstruction kernel in Shannon interpolation uses the normalized cardinal sine.

Original



TikZ reconstruction



Reading the reconstruction. The value at zero is the continuous extension, equal to one. Every nonzero integer is a zero. The plotted normalization is $\sin(\pi x)/(\pi x)$, as in the theorem.

Figure 1.6

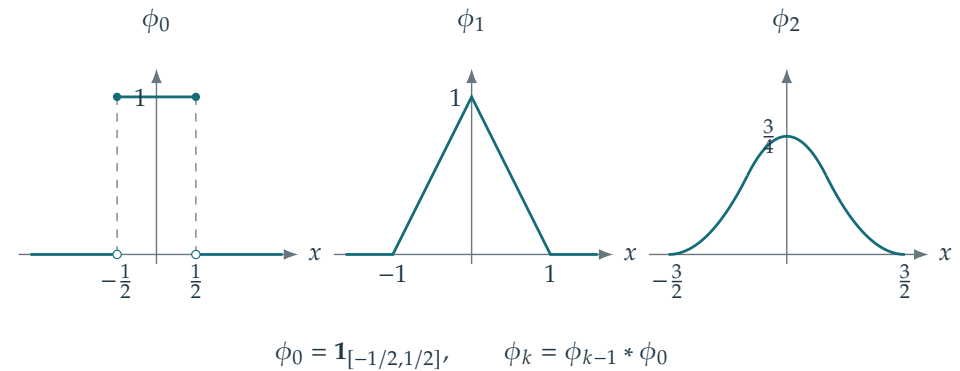
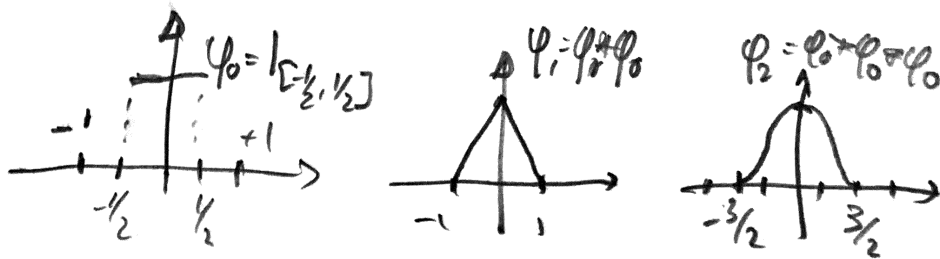
Main book, page 6

Cardinal splines as basis functions for interpolation.

Complete figure / Cardinal B-splines of degrees zero, one, and twoThe chapter defines ϕ_0 as the centered unit box and obtains each subsequent spline by convolution with that box.

Original

TikZ reconstruction



Reading the reconstruction. Supports and heights follow the convolution definition exactly. The quadratic spline peaks at $3/4$, so its integer translates are not an interpolating basis with the samples as coefficients. Endpoint values of the box follow the closed-interval convention in the text.

Figure 1.7

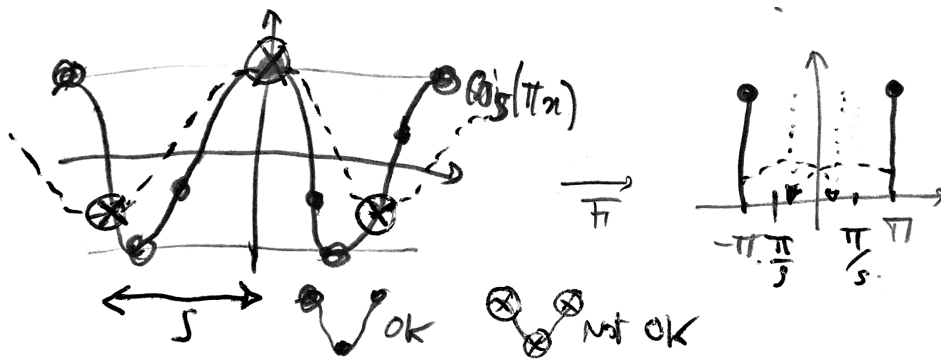
Main book, page 7

Aliasing of a sine wave. This periodic example is interpreted through its discrete spectral lines rather than as an $L^2(\mathbb{R})$ signal.

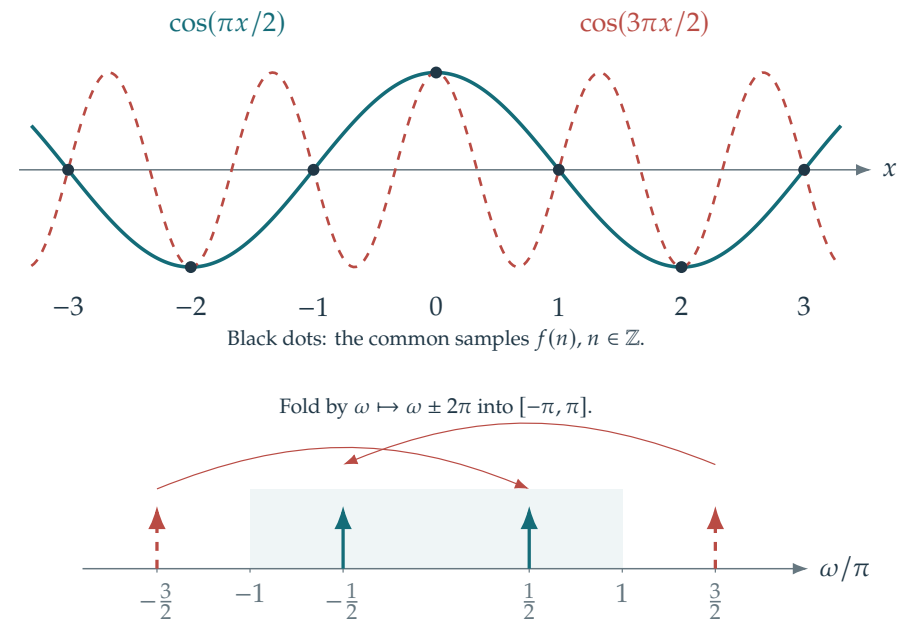
Complete figure / Two frequencies with identical samples

A periodic sinusoidal example explains aliasing through discrete spectral lines.

Original



TikZ reconstruction

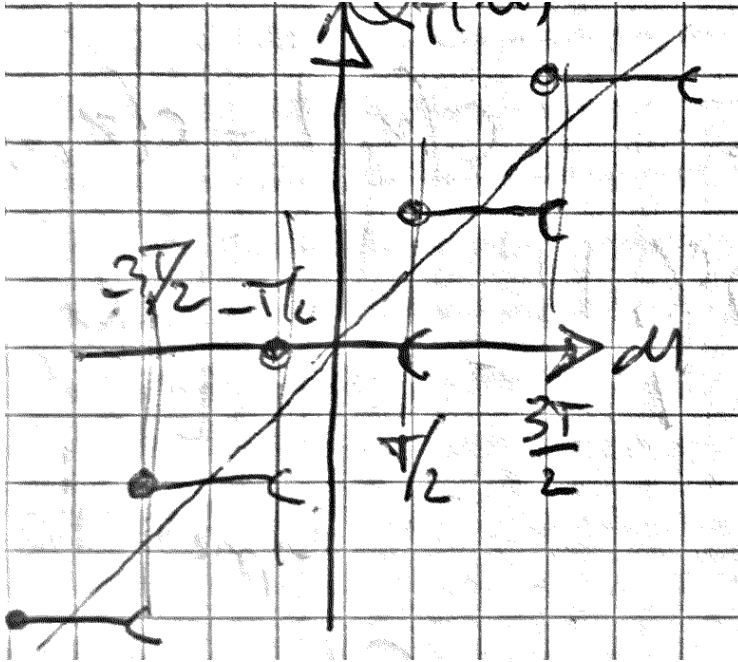
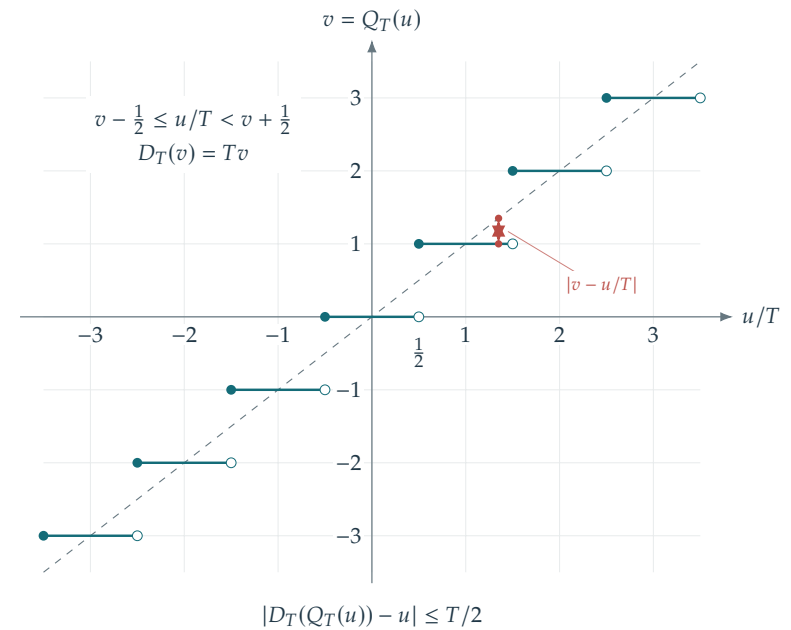


Reading the reconstruction. For unit sample spacing, $\cos(\pi x/2)$ and $\cos(3\pi x/2)$ agree at every integer. Frequencies are identified modulo 2π ; the high-frequency pair folds to the low-frequency pair. This example is periodic and is not an L^2 signal on the real line.

Figure 1.8

Main book, page 8

A uniform scalar quantizer.

Complete figure / Uniform scalar quantizationThe quantizer returns an integer index; dequantization multiplies that index by the step size T .**Original****TikZ reconstruction**

Reading the reconstruction. The axes are the normalized input u/T and integer index v . Filled left and open right endpoints implement the stated half-open cells exactly. The red vertical segment measures quantization error at a fixed input; rescaling gives the bound $T/2$ after dequantization. The diagonal represents the unquantized value.

Figure 2.3

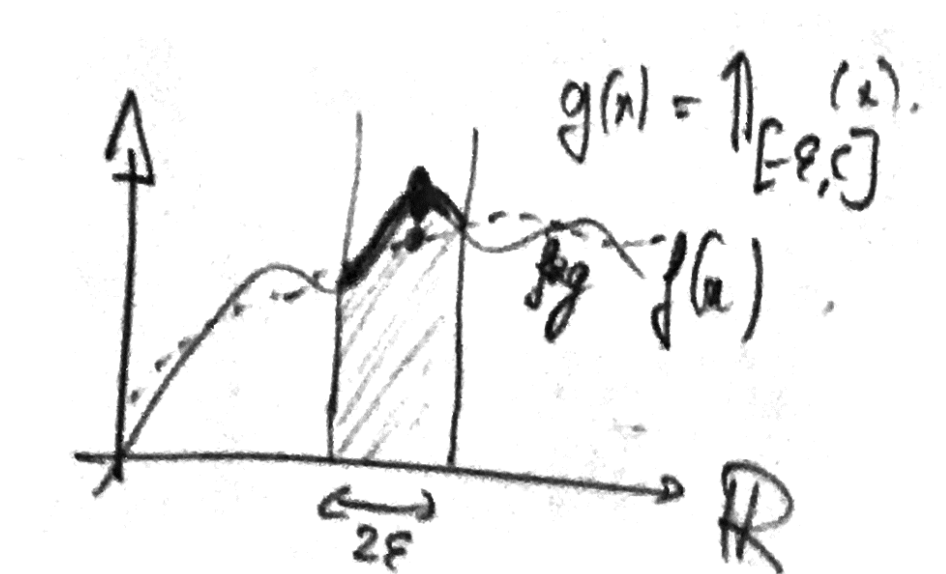
Main book, page 11

Convolution on $\mathbb{R}$. The dashed curve is $f \star g$; the marked value $(f \star g)(x)$ equals the shaded integral.

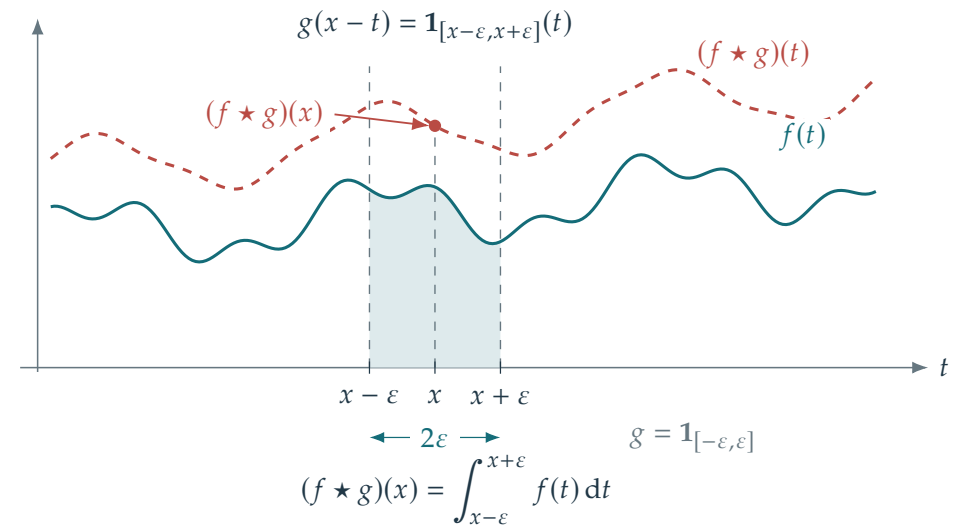
Complete figure / Convolution with a box window

Convolution; equation (2.4).

Original



TikZ reconstruction



Reading the reconstruction. The shaded interval is $[x-\epsilon, x+\epsilon]$, the support of $g(x-t)$, with g the unnormalized box on $[-\epsilon, \epsilon]$. The dashed red curve is the exact sliding integral of the same displayed signal, computed by integrating its affine and sinusoidal terms; the red dot is at $(x, (f \star g)(x))$. The illustrated width is $2\epsilon = 1.5$, so the convolution is an integral, not a unit-mass average. The curve and shaded area use the same normalization.

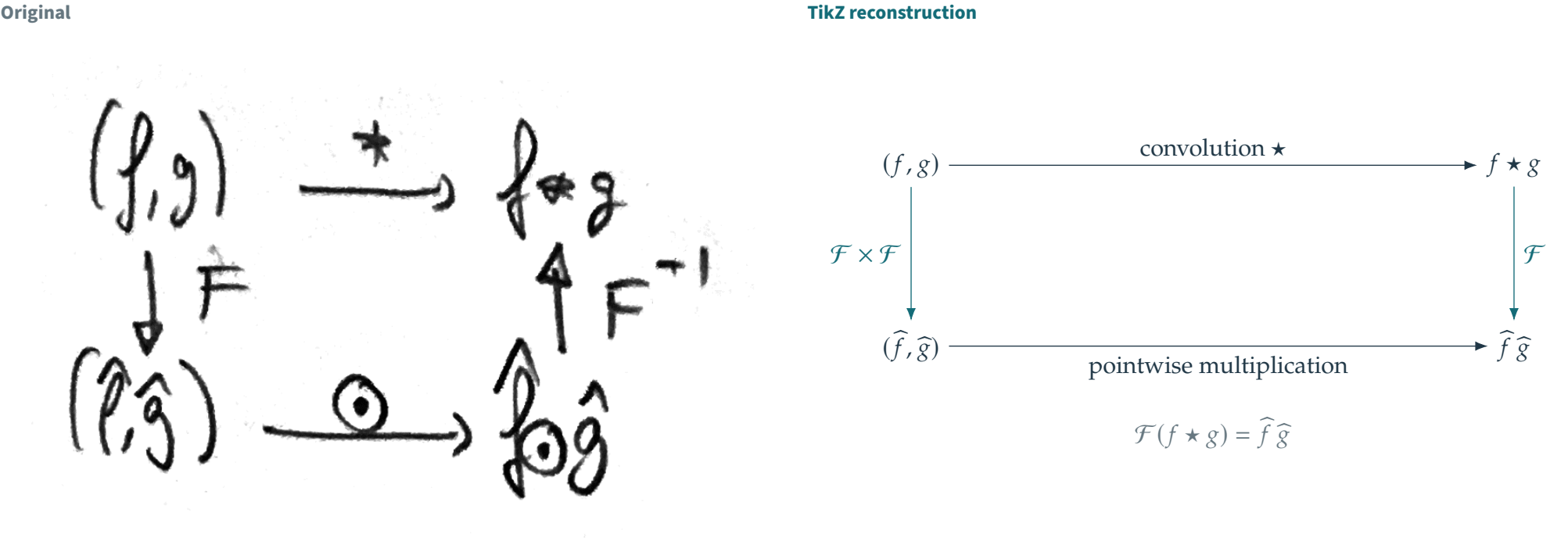
Figure 2.6

Main book, page 13

How the Fourier transform converts convolution into multiplication.

Complete figure / Convolution becomes multiplication

Convolution theorem; equation (2.6).



Reading the reconstruction. The diagram uses the chapter convention $\mathcal{F}(f \star g) = \widehat{f} \widehat{g}$. Both vertical arrows apply the Fourier transform, making the square commute; the original reverse Fourier arrow is equivalently written in the forward direction.

Figure 2.7

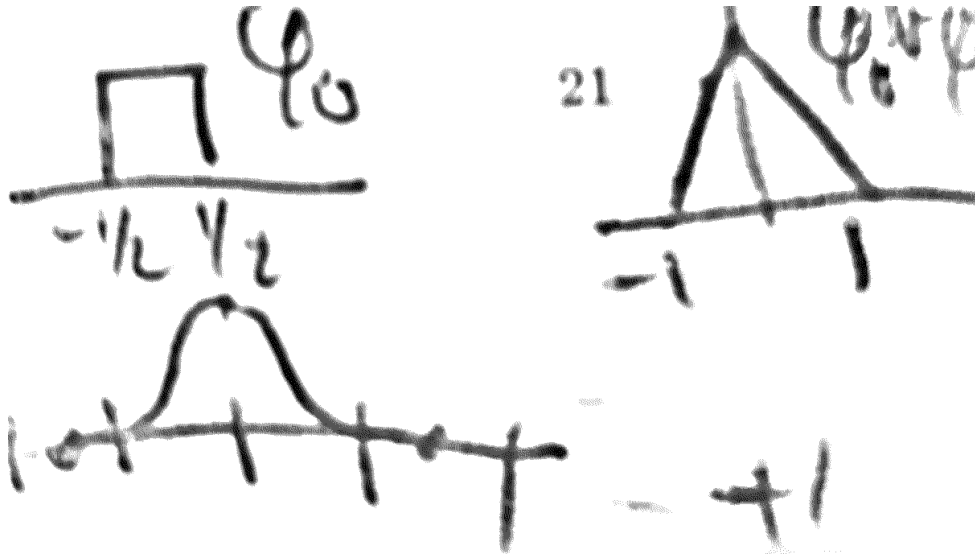
Main book, page 14

Cardinal splines are defined by successive convolutions.

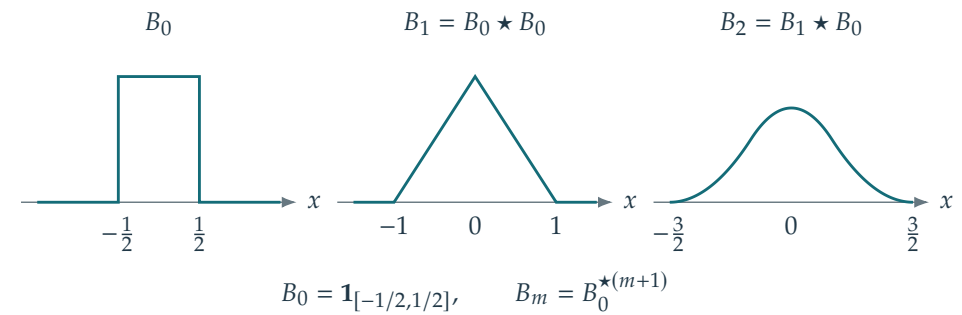
Complete figure / Cardinal splines by convolution

Convolution; Figure 2.7.

Original



TikZ reconstruction



Reading the reconstruction. Interpretation to check: The box, triangle, and quadratic spline are exact centered cardinal B-splines B_0 , $B_1=B_0 \star B_0$, and $B_2=B_1 \star B_0$. The original final handwritten symbol is unclear; explicit degree and convolution labels replace it.

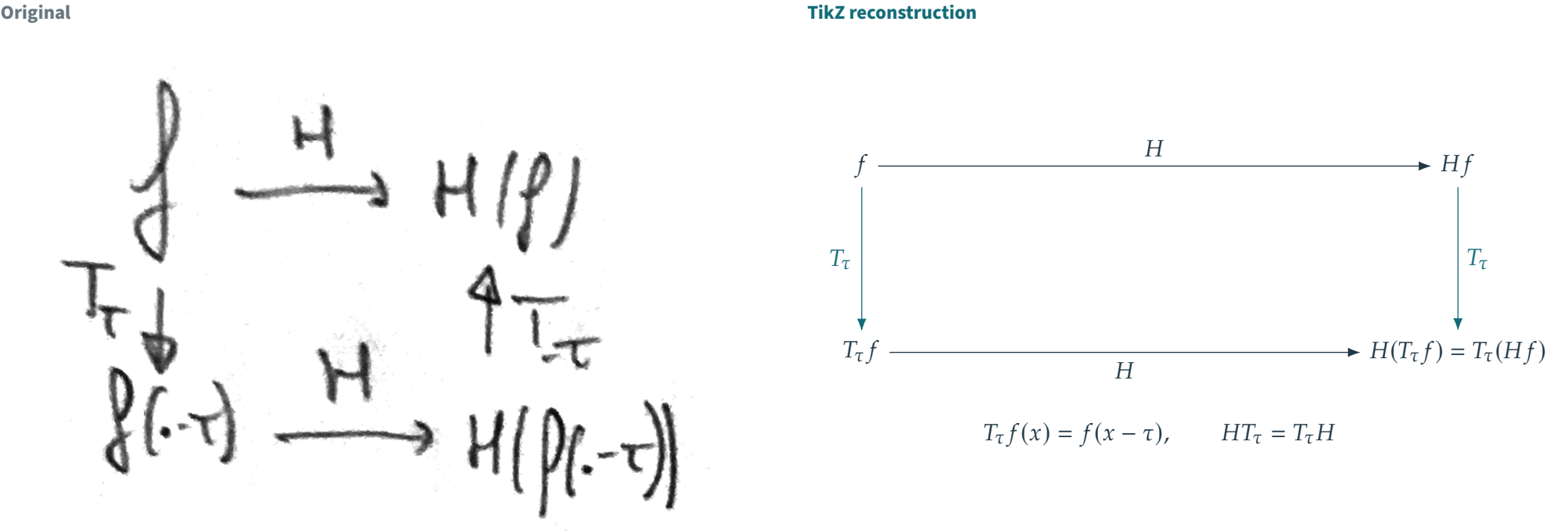
Figure 2.8

Main book, page 14

Commutative diagram for translation invariance.

Complete figure / Translation equivariance

Translation-invariant operators; $T_\tau f(x) = f(x - \tau)$.



Reading the reconstruction. The original square is made explicit with $T_\tau f = f(\cdot - \tau)$. The bottom-right value is $H(T_\tau f) = T_\tau(Hf)$; this is equivariance, called translation invariance in the chapter.

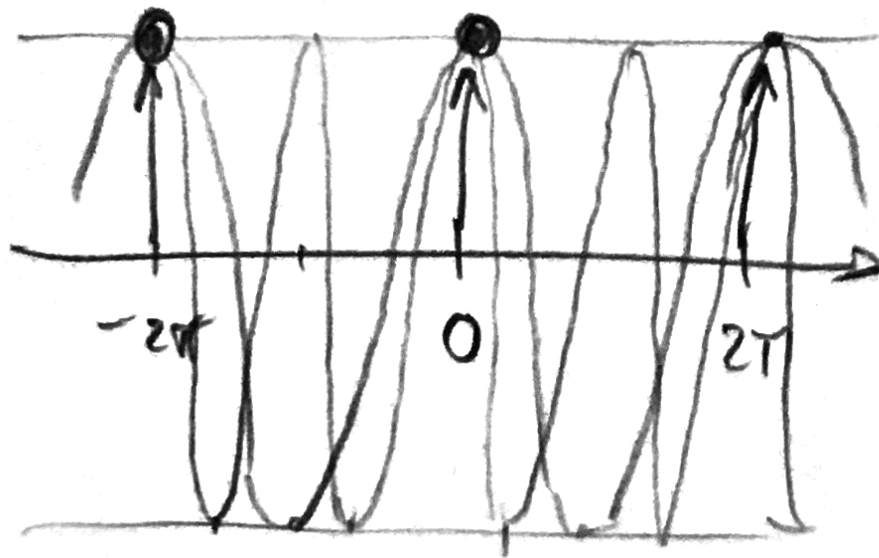
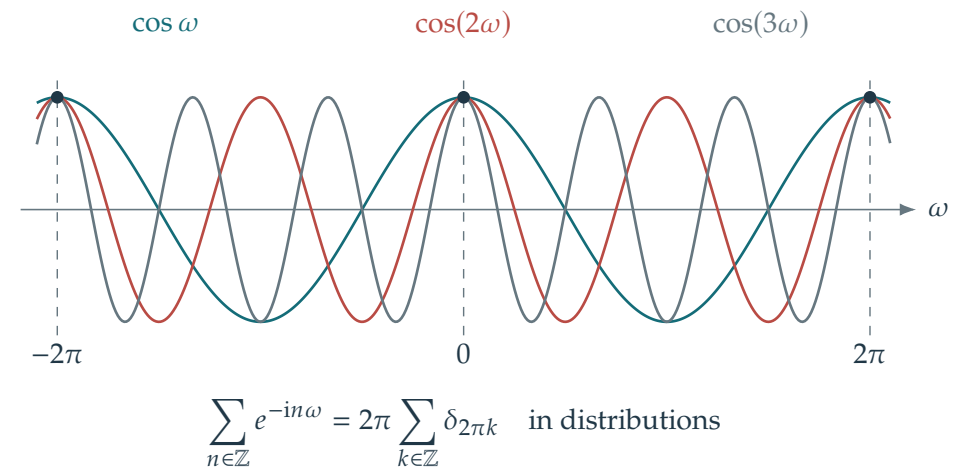
Figure 2.9

Main book, page 15

Sine waves summed in the Poisson formula.

Complete figure / Fourier modes align at the Dirac comb

Poisson formula and distributions; finite Dirichlet kernel.

Original**TikZ reconstruction**

Reading the reconstruction. Three real Fourier modes $\cos(n\omega)$, $n=1,2,3$, are shown. They all equal one at multiples of 2π . The distributional comb identity is stated separately; a finite sum is not drawn as literal Dirac masses.

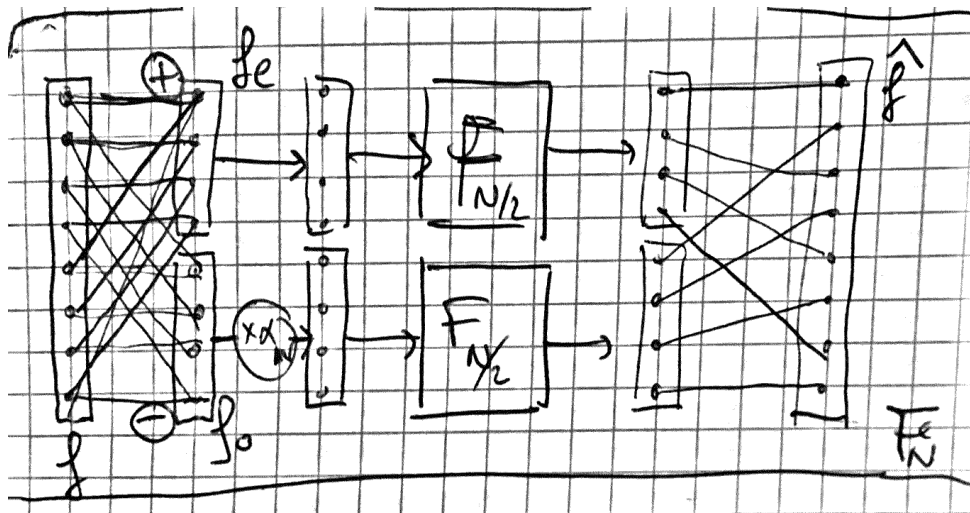
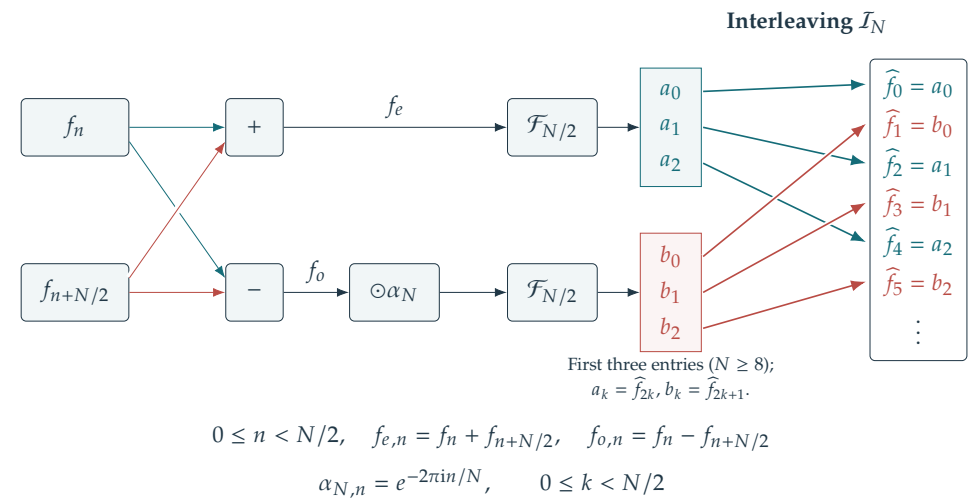
Figure 2.10

Main book, page 17

Diagram of one radix-two FFT step. Both halves of the input enter the sum and difference; the outputs of the two smaller transforms are interleaved into even and odd frequencies.

Complete figure / One radix-two FFT step

Discrete Fourier transform; recursion immediately before Figure 2.10.

Original**TikZ reconstruction**

Reading the reconstruction. Four distinct arrows connect both input halves to both the sum and difference. The difference is multiplied componentwise by the twiddle-factor vector before its half-size FFT. On the right, individual arrows map the two output streams into alternating even and odd positions; the first three entries are shown for $N \geq 8$. The zero-based indices match the DFT and the interleaving definition in the chapter. Crossings are not connections.

Figure 2.11

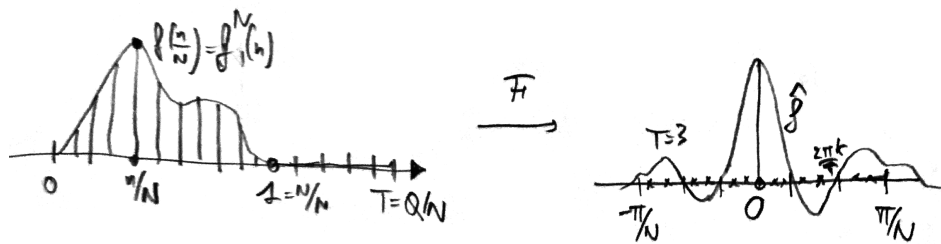
Main book, page 19

Fourier transform approximation by zero-padding in the spatial domain.

Complete figure / Spatial zero padding refines frequency sampling

Fourier approximation via spatial zero padding; Figure 2.11.

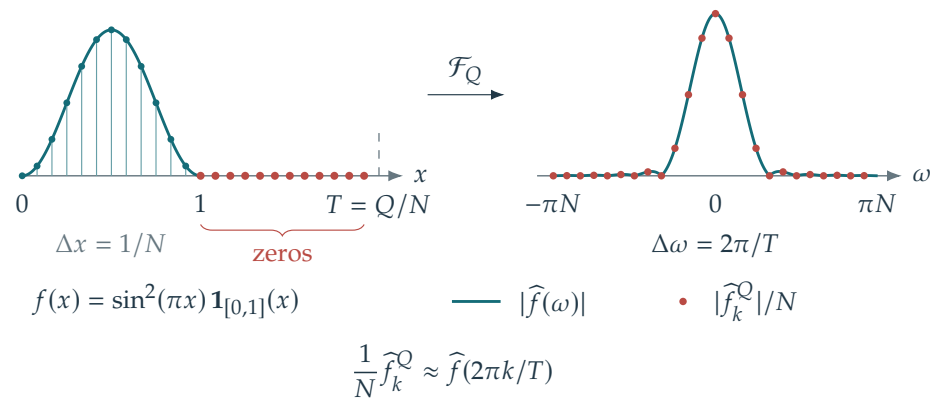
Original



TikZ reconstruction

$$f_n^Q = f(n/N), \quad N = 12, Q = 24$$

continuous transform and DFT samples



Reading the reconstruction. Interpretation to check: An exact analytic example uses $N=12$, $Q=24$ and $f(x)=\sin^2(\pi x)$ on $[0,1]$, zero elsewhere. The Q spatial samples stop at $T-1/N$, where $T=Q/N$; the final point T is excluded. The continuous curve is the magnitude of the actual Fourier transform, and the Q markers are normalized DFT magnitudes at signed indices $-Q/2$ through $Q/2-1$. Thus the unchanged Nyquist bound is πN and the frequency step is $2\pi/T$, correcting the handwritten reciprocal bounds. The DFT approximates the continuous transform by a Riemann sum; it does not equal it exactly. These are analytic values, not measurements.

Figure 2.12

Main book, page 20

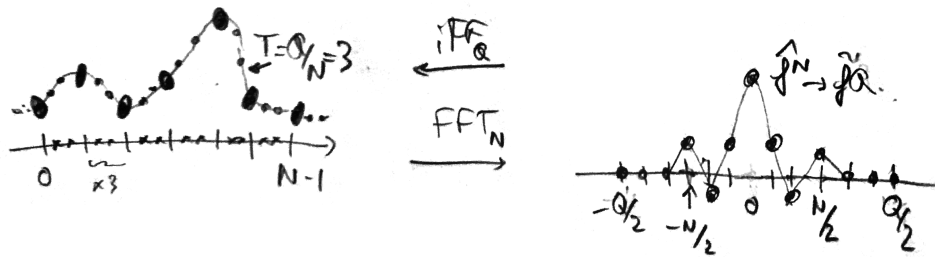
Interpolation by zero-padding in the frequency domain.

Complete figure / Spectral zero padding interpolates the samples

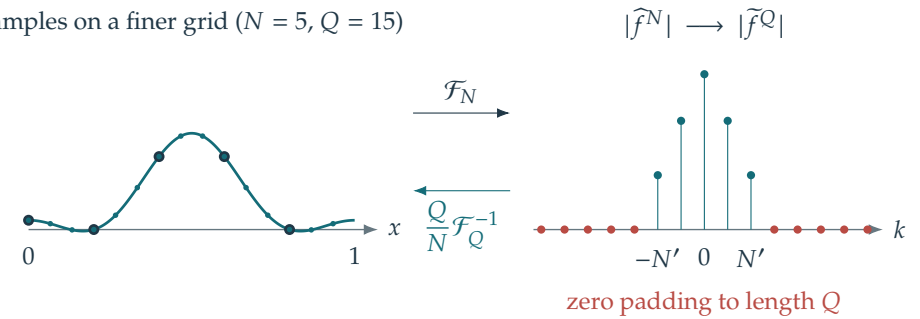
Fourier interpolation via spectral zero padding; Figure 2.12.

Original

TikZ reconstruction



samples on a finer grid ($N = 5, Q = 15$)



$$N = 2N' + 1, \quad \tilde{f}_k^Q = \hat{f}_k^N (|k| \leq N'), \quad 0 \text{ otherwise}$$

Reading the reconstruction. Interpretation to check: The retained signed-frequency coefficients are padded with zeros, and the inverse transform is multiplied by Q/N . The illustrative trigonometric polynomial and coefficient magnitudes are consistent for $N=5$ and $Q=15$; dark markers identify the five original samples and smaller colored markers the finer grid. These are analytic example values, not measurements. Odd N avoids the even-length Nyquist splitting convention.

Figure 2.13

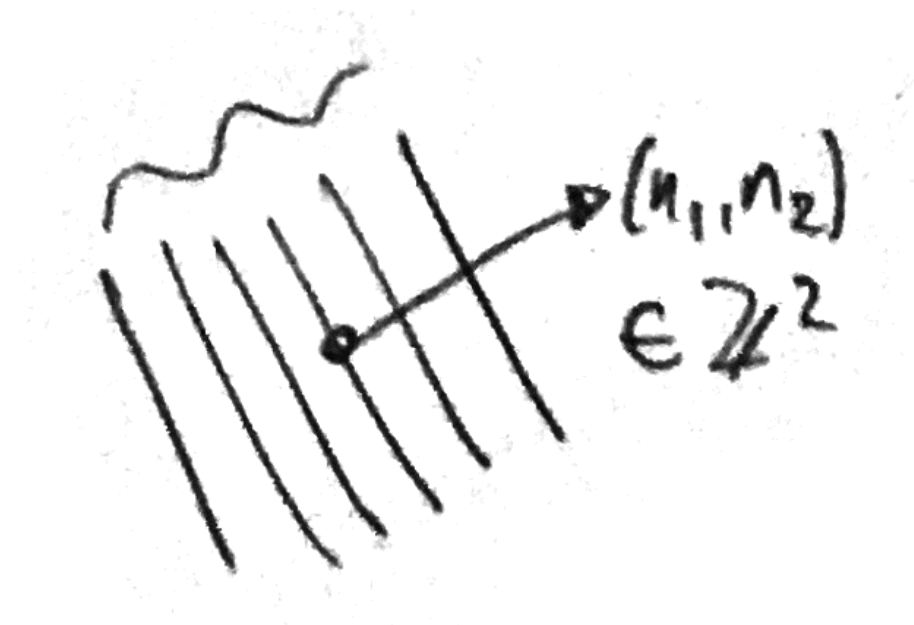
Main book, page 21

2-D sine wave.

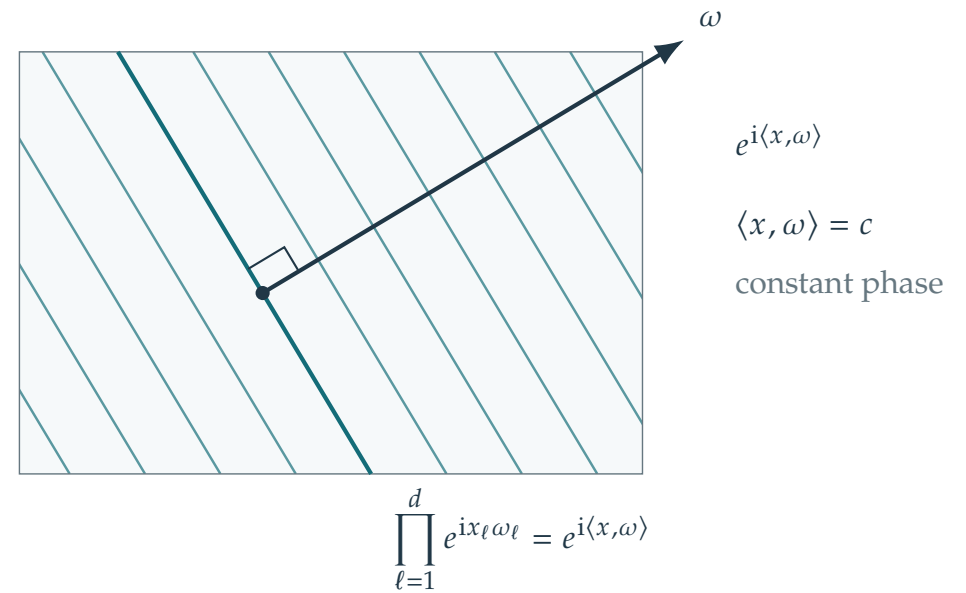
Complete figure / A plane wave and its normal direction

Fourier in multiple dimensions; tensor products of complex exponentials.

Original



TikZ reconstruction



Reading the reconstruction. The parallel lines are constant-phase sets, normal to the wave vector ω . The marked point lies on the emphasized phase line, and the right-angle marker joins that line to its normal. This indicates geometry rather than a propagation direction. Integer wave vectors apply on the periodic domain; the general figure uses ω in $\mathbb{R}^d$.

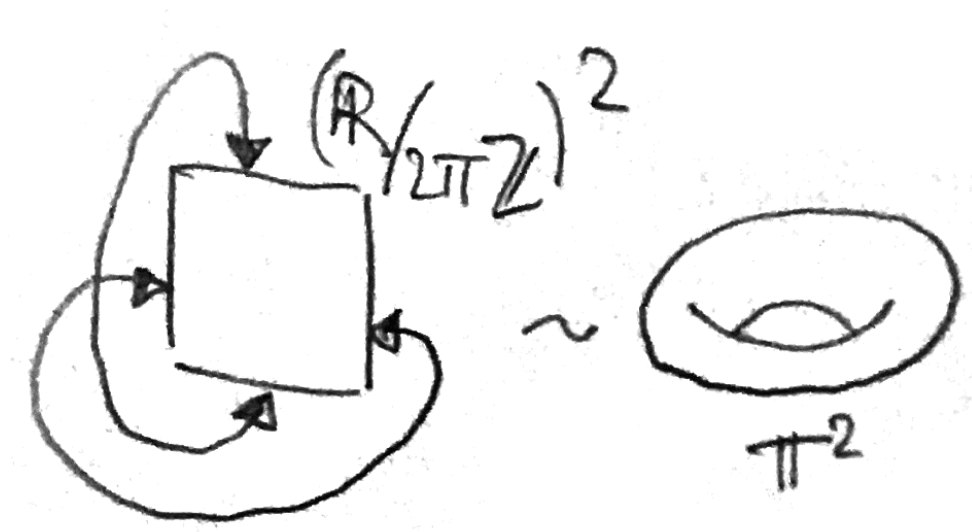
Figure 2.15

The two-dimensional torus $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$.

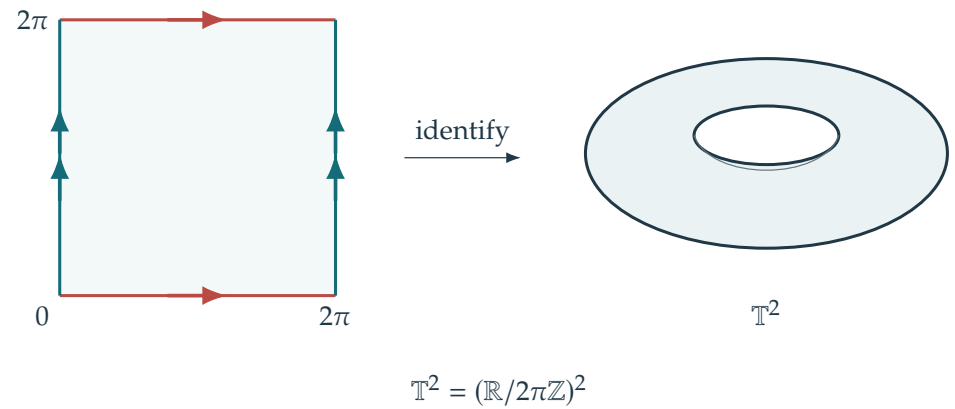
Complete figure / A square with opposite edges identified

Multidimensional torus; $(\mathbb{R} / 2\pi\mathbb{Z})$ squared.

Original



TikZ reconstruction



Reading the reconstruction. Opposite sides of the square carry matching oriented arrows. Identifying each pair produces the two-dimensional torus. The pictured doughnut is a topological visualization, not an isometric embedding of the flat quotient.

Figure 2.16

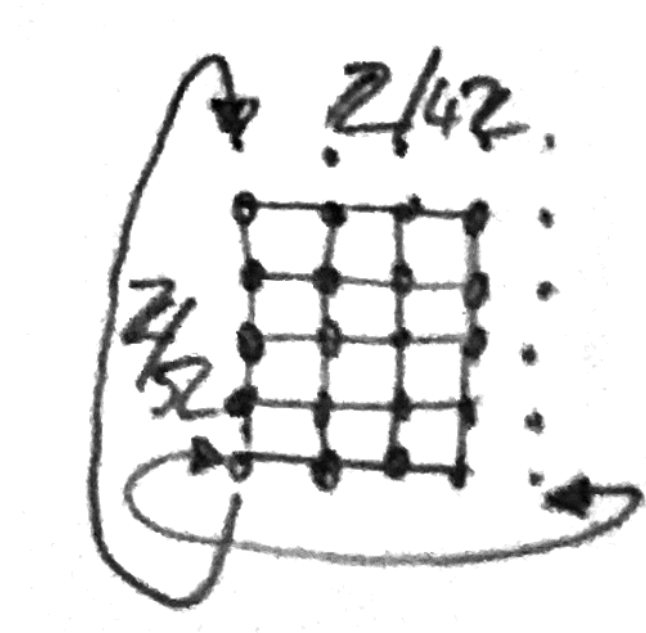
Main book, page 22

Discrete 2-D torus.

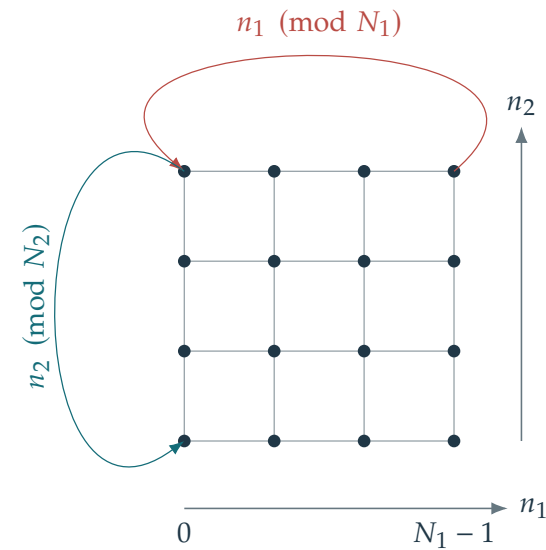
Complete figure / A periodic two-dimensional grid

Discrete multidimensional Fourier basis; product of cyclic index sets.

Original



TikZ reconstruction



$$(n_1, n_2) \in (\mathbb{Z}/N_1\mathbb{Z}) \times (\mathbb{Z}/N_2\mathbb{Z})$$

$$(n_1 + N_1, n_2) \sim (n_1, n_2)$$

$$(n_1, n_2 + N_2) \sim (n_1, n_2)$$

Reading the reconstruction. A four-by-four representative grid illustrates the two periodic coordinate directions. A step beyond the last displayed index wraps back to zero; the first and last displayed vertices remain distinct. The identifications occur after N_1 or N_2 steps, so the rectangle has no physical boundary. The grid size is illustrative.

Figure 2.18

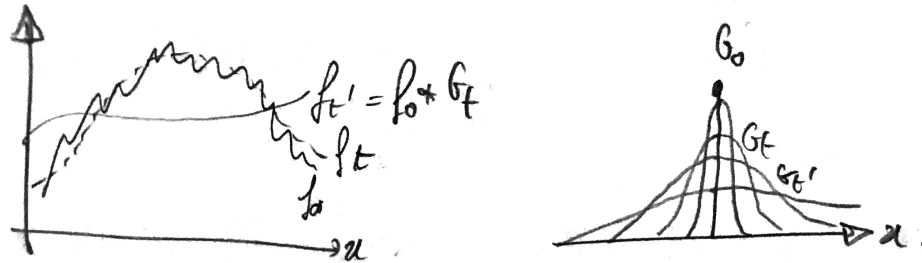
Main book, page 24

Heat diffusion as a convolution.

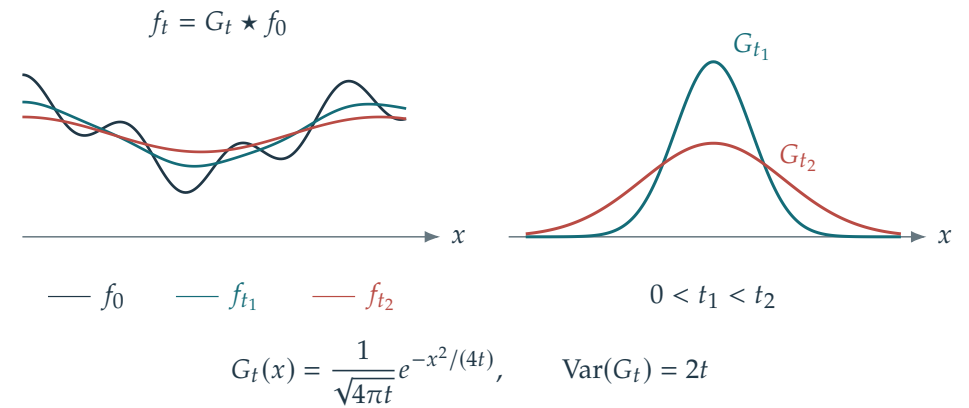
Complete figure / Heat diffusion is Gaussian convolution

Fourier multipliers and differential equations; Figure 2.18.

Original



TikZ reconstruction



Reading the reconstruction. The left curves show an illustrative signal with two Fourier modes at time zero and after positive diffusion times. The right curves are Gaussian heat kernels for the same two times: increasing time broadens the kernel and lowers its peak. The normalization and variance follow the chapter heat equation.

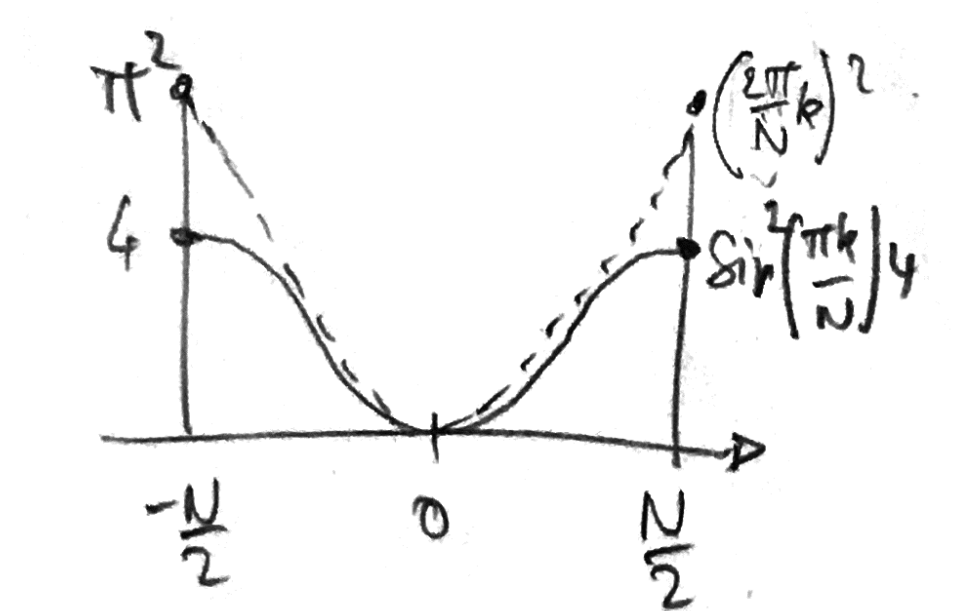
Figure 2.19

Main book, page 25

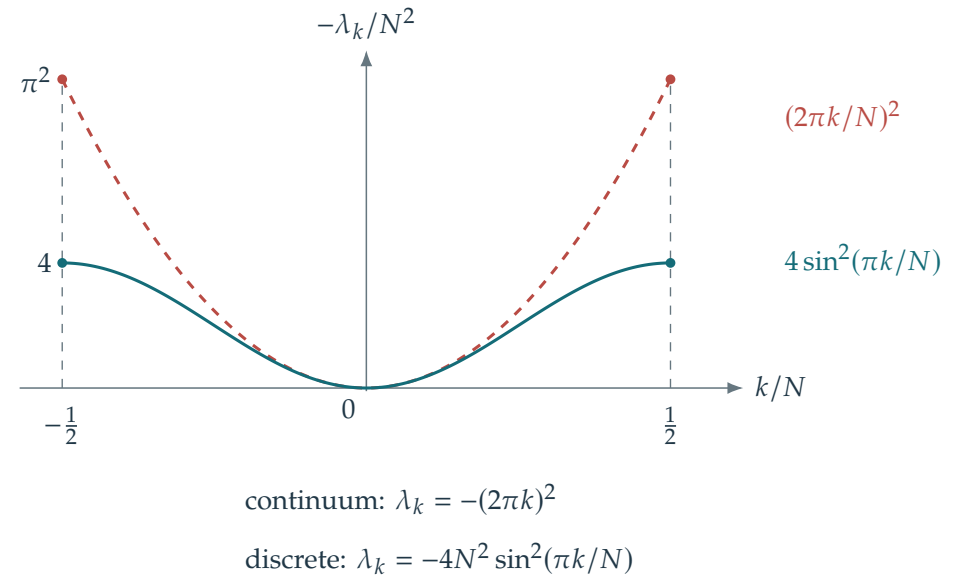
Comparison of the spectra of Δ and D_2 .**Complete figure / Continuum and discrete Laplacian spectra**

Finite domain and discretization; equations (2.17) and (2.18).

Original



TikZ reconstruction



Reading the reconstruction. The original positive curves are explicitly labeled as negative eigenvalues divided by N squared. The continuum curve is $(2\pi k/N)^2$ squared, reaching π squared at Nyquist; the discrete curve is $4 \sin^2(\pi k/N)$, reaching four. Actual Laplacian eigenvalues are nonpositive.

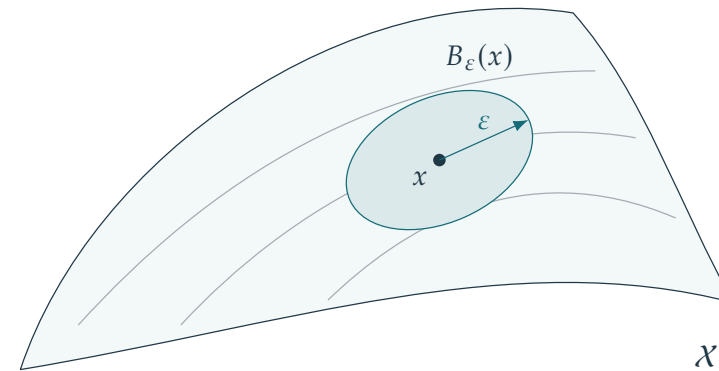
Figure 2.20

Main book, page 29

Computing the Laplacian on a surface.

Complete figure / A small geodesic ball on a surface

Laplace–Beltrami operator and the small-ball mean-value expansion.

Original**TikZ reconstruction**geodesic ball
 $\dim X = d$

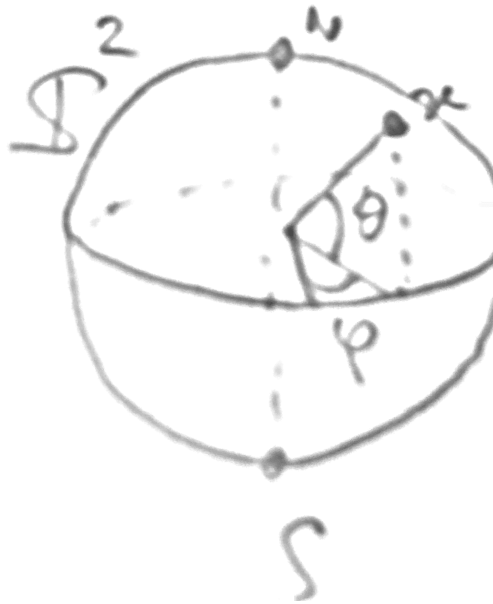
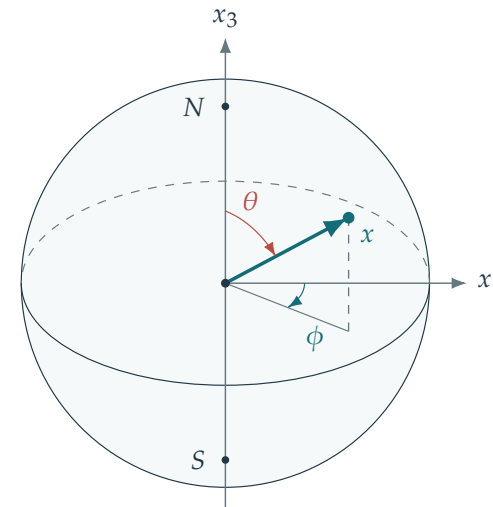
$$\frac{1}{\text{Vol}(B_\varepsilon(x))} \int_{B_\varepsilon(x)} f(y) \, d\mu(y) = f(x) + \frac{\varepsilon^2}{2(d+2)} \Delta f(x) + o(\varepsilon^2)$$

Reading the reconstruction. The sketch is a local surface patch containing a geodesic ball centered at x . The ellipse represents the projected ball, not a claim that geodesic balls are Euclidean circles. The displayed expansion retains the necessary epsilon-squared scaling and dimension-dependent constant.

Figure 2.21

Main book, page 29

Spherical coordinates.

Complete figure / Colatitude and azimuth on the sphereSpherical harmonics; $x = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$.**Original****TikZ reconstruction**

$$x \in \mathbb{S}^2$$

$$x = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

$$0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi$$

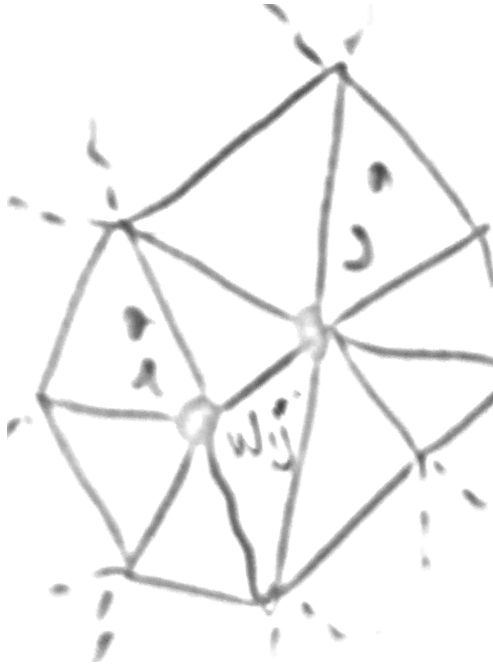
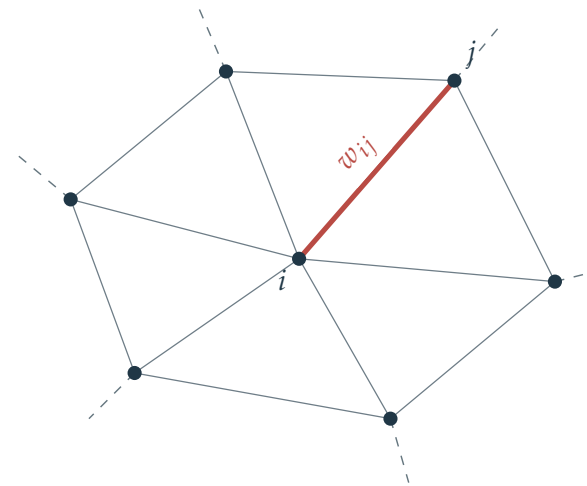
orthographic view

Reading the reconstruction. Interpretation to check: Theta is colatitude from the positive north axis, matching the chapter formula; the original angle is ambiguous and may instead suggest latitude. Phi is equatorial azimuth. The point, its equatorial projection, both angle arcs, the equator and the poles are projected from one consistent unit sphere. A viewing elevation of 30 degrees separates the arcs. Screen angles need not equal spatial angles, and the projected poles lie inside the circular silhouette. The illustrative point has colatitude 50 degrees and azimuth 38 degrees.

Figure 2.22

Main book, page 30

Weighted graph.

Complete figure / A weighted graph and its local LaplacianGraph Laplacian; $\Delta = W - D$.**Original****TikZ reconstruction**

$$w_{ij} = w_{ji} \geq 0$$

$$(\Delta f)_i = \sum_{j \sim i} w_{ij}(f_j - f_i)$$

$$\Delta = W - D$$

Reading the reconstruction. Interpretation to check: The graph patch preserves the intended local adjacency picture, highlighting an edge between vertices i and j with weight w_{ij} . Unreadable numerical marks are not assigned values. The Laplacian sign matches the chapter nonpositive convention.

Figure 3.4

Main book, page 32

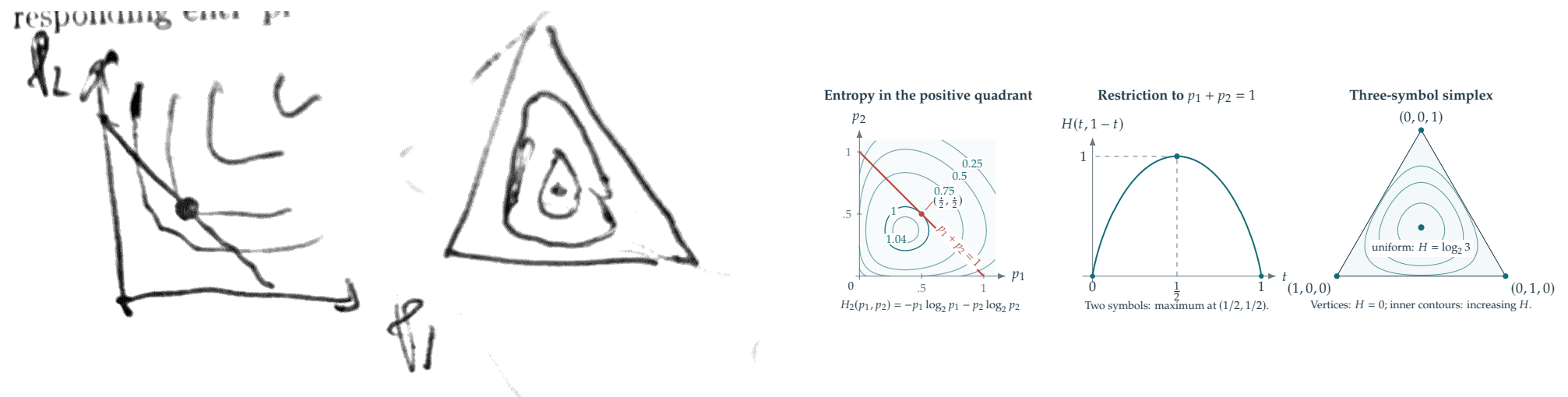
Entropy is minimized at a point mass and maximized at the uniform distribution. Left: level sets of the entropy formula in the positive quadrant and the constraint $p_1 + p_2 = 1$. Middle: its restriction to that constraint. Right: entropy level sets on the three-symbol probability simplex.

Complete figure / Entropy level sets and probability constraints

The entropy lemma states that point masses minimize entropy and the uniform distribution maximizes it.

Original

TikZ reconstruction



Reading the reconstruction. The new left panel shows actual level sets of $-p_1 \log_2(p_1) - p_2 \log_2(p_2)$ in the positive quadrant, with $0 \log_2(0) = 0$. This formula extends beyond probability vectors; the red segment $p_1 + p_2 = 1$ is the probability constraint. Its constrained maximum is at $(1/2, 1/2)$, where $H = 1$ and the contour is tangent to the segment. The unconstrained maximum of the extension is instead at $(1/e, 1/e)$, explaining why the contours are centered off the constraint. The middle panel restricts entropy to the segment. The right panel retains the three-symbol simplex, where the uniform entropy is $\log_2(3)$.

Figure 3.5

Main book, page 34

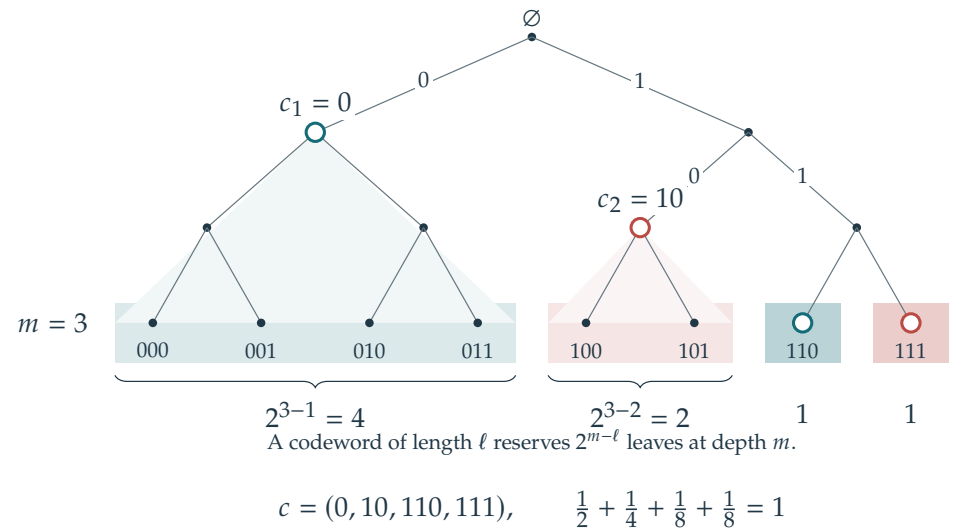
Panel 1: full binary tree obtained by completing the tree associated with the code ($c_1 = 0, c_2 = 10, c_3 = 110, c_4 = 111$). Panel 2: subtrees with the prescribed codeword lengths packed from left to right.

Panel 1 of 2 / Kraft inequality: disjoint descendant blocks

The necessity proof completes a prefix tree to depth m and counts the descendants of each codeword.

Original

TikZ reconstruction



Reading the reconstruction. The code is $(0, 10, 110, 111)$. Its codewords reserve 4, 2, 1, and 1 leaves in the full depth-three tree. Tinted subtrees connect the highlighted codeword roots to disjoint descendant blocks. Edge bits show how a path spells a codeword; the formula explains the count $2^{m-\ell}$.

Figure 3.5

Main book, page 34

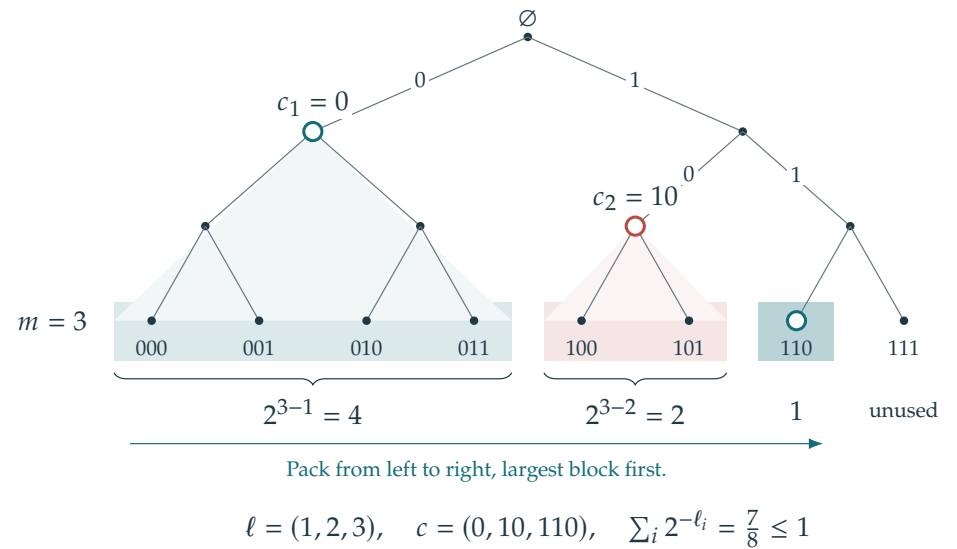
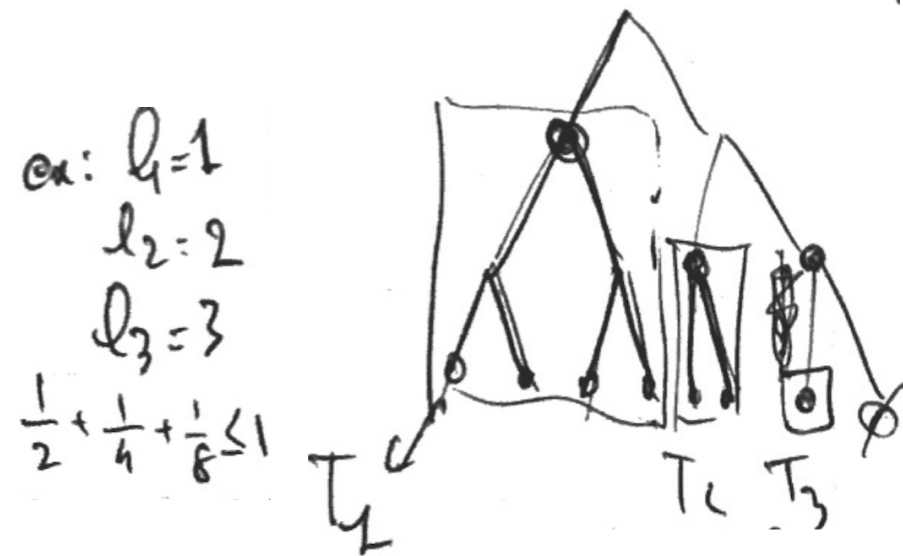
Panel 1: full binary tree obtained by completing the tree associated with the code ($c_1 = 0, c_2 = 10, c_3 = 110, c_4 = 111$). Panel 2: subtrees with the prescribed codeword lengths packed from left to right.

Panel 2 of 2 / Kraft inequality: packing prescribed lengths

The converse proof packs dyadically aligned blocks in nonincreasing order of size.

Original

TikZ reconstruction



Reading the reconstruction. For prescribed lengths $(1, 2, 3)$, the blocks contain 4, 2, and 1 leaves. The left-to-right packing arrow and tinted subtrees expose the construction: choose the largest block first, then continue at the next dyadic boundary. This gives codewords $(0, 10, 110)$, leaves 111 unused, and demonstrates $7/8 \leq 1$ without assuming equality.

Figure 4.3

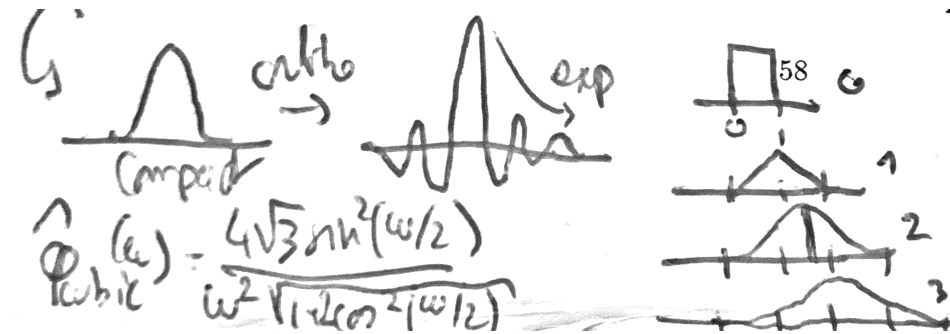
Main book, page 39

Orthogonalization of B-splines to define cardinal orthogonal spline functions.

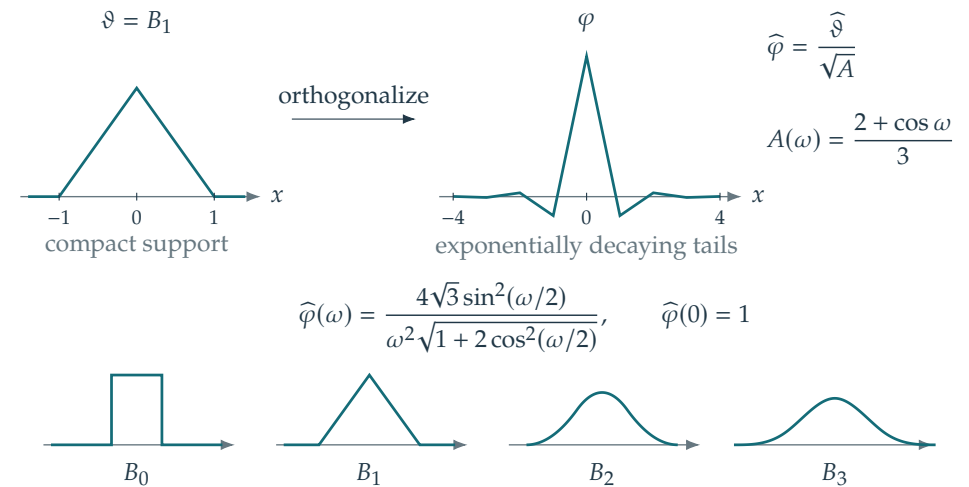
Complete figure / Orthogonalizing a cardinal spline

Spectral orthogonalization; Proposition 4.1 and Figure 4.3.

Original



TikZ reconstruction



Reading the reconstruction. The formula identifies the handwritten example as the orthogonal linear spline. Its triangular generator has compact support, while the orthogonalized function has alternating exponentially decaying tails. The replacement uses the correct Fourier multiplier $\sqrt{3/(2+\cos \omega)}$; its cosine coefficients were evaluated by 8192-point midpoint quadrature. Axis ticks distinguish the generator interval $[-1, 1]$ from the displayed orthogonalized interval $[-4, 4]$, which uses a compressed horizontal scale. Four small degree examples recover the original spline-family sketches.

Figure 4.6

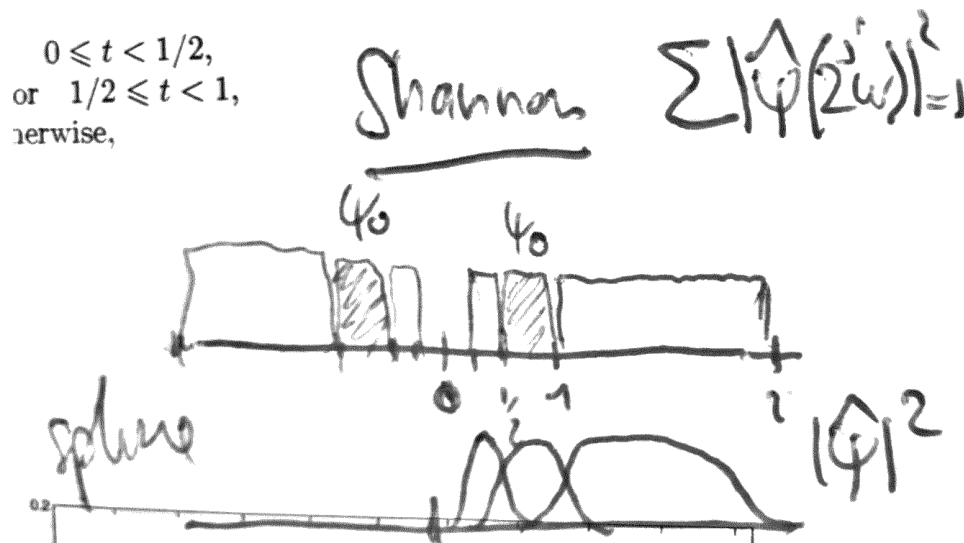
Main book, page 42

Spline and Shannon wavelet segmentation of the frequency axis.

Complete figure / Sharp and smooth frequency partitions

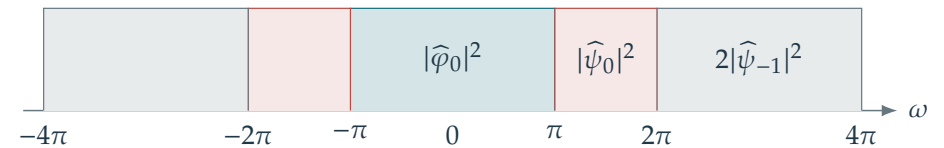
Multiresolution detail spaces; Figure 4.6.

Original

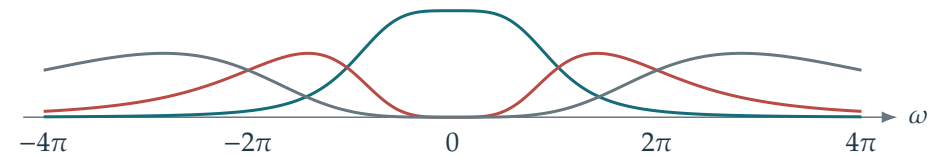


TikZ reconstruction

Shannon: sharp bands



Splines: smooth overlap (schematic)



$$2^{-j} |\hat{\psi}_{j,0}(\omega)|^2 = |\hat{\psi}(2^j \omega)|^2$$

$$\sum_{j \in \mathbb{Z}} |\hat{\psi}(2^j \omega)|^2 = 1 \quad (\text{a.e. } \omega \neq 0)$$

Reading the reconstruction. Interpretation to check: The Shannon row shows the exact low-pass band $|\omega| \leq \pi$ and the next two dyadic wavelet bands. The spline row deliberately shows schematic smooth overlap, not the spectrum of a named spline order. Only a finite frequency window is drawn; the partition identity refers to all scales. Cropped neighboring prose in the scan is omitted. The energy of a scale- j wavelet is multiplied by 2 to the power $-j$ to remove its L2 dilation factor, as the displayed identity explains; this gives the factor two on the outer scale-minus-one band.

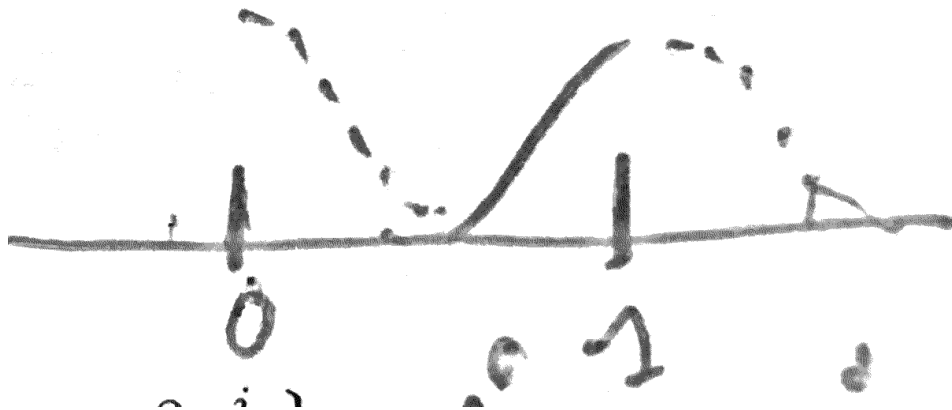
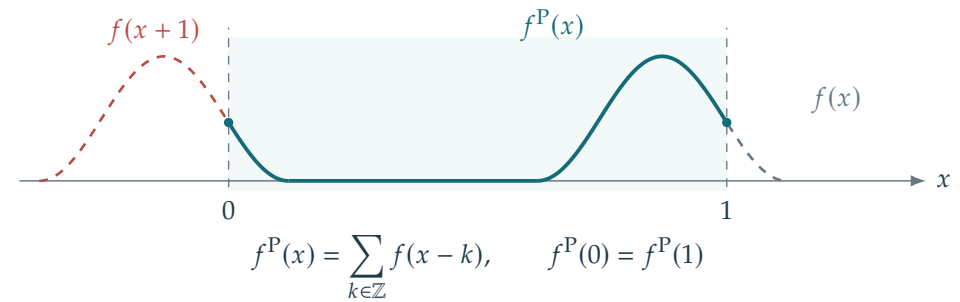
Figure 4.7

Main book, page 42

Periodized basis functions. Only the translations $0 \leq n < 2^{-j}$ are distinct, and $j_0 = 0$ is a common choice of coarsest scale.

Panel 1 of 2 / Wrapping a function around the unit interval

On bounded domains; f -periodic is the sum of integer translates.

Original**TikZ reconstruction**

Reading the reconstruction. Interpretation to check: The illustrative compact bump is $\cos^2(2\pi(x-0.87))$ for $|x-0.87| \leq 0.25$, zero elsewhere. On $[0,1]$, only $f(x)$ and its left-shifted copy $f(x+1)$ contribute, so the drawn sum and matching endpoint values are exact. The profile is not presented as a particular orthogonal wavelet; periodization applies to each basis function individually. No unknown profile or empirical values are inferred from the scan.

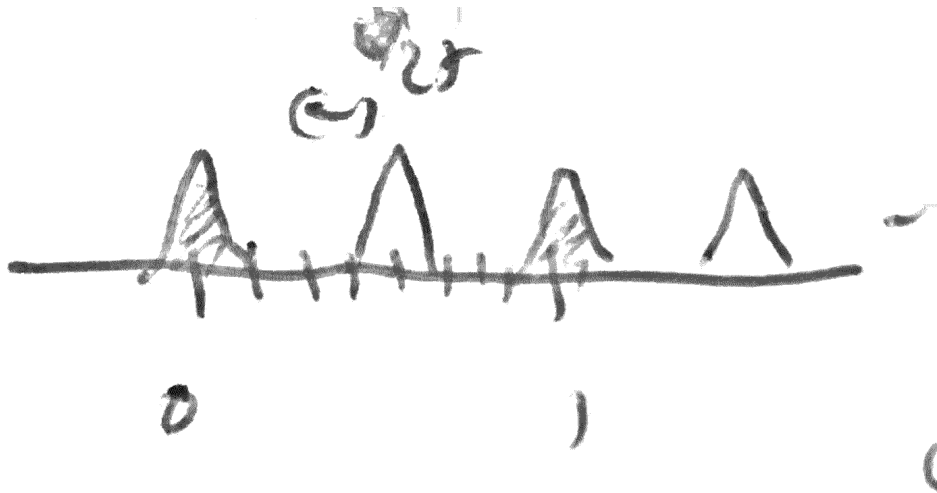
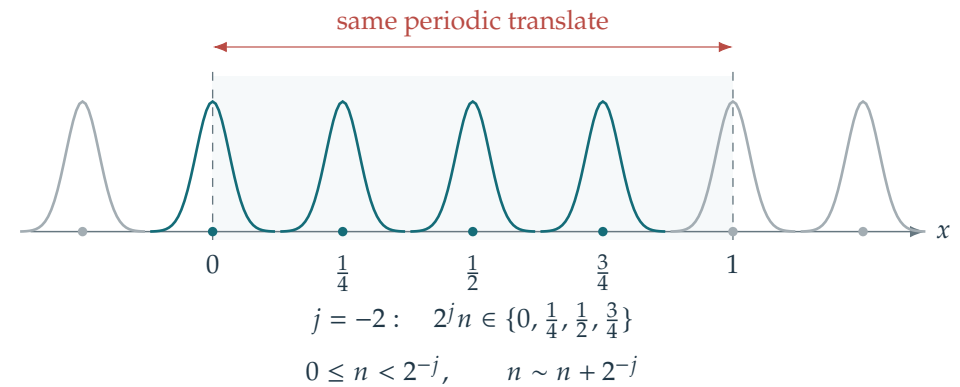
Figure 4.7

Main book, page 42

Periodized basis functions. Only the translations $0 \leq n < 2^{-j}$ are distinct, and $j_0 = 0$ is a common choice of coarsest scale.

Panel 2 of 2 / One representative of each periodic translate

On bounded domains; $0 \leq n < 2^{-j}$.

Original**TikZ reconstruction**

Reading the reconstruction. Interpretation to check: The selected translation centers lie in the half-open unit interval. The example $j=-2$ has four distinct centers; the center at one is the same periodic translate as zero. Smooth pulse profiles are schematic and do not assert that this particular shape generates an orthogonal wavelet basis.

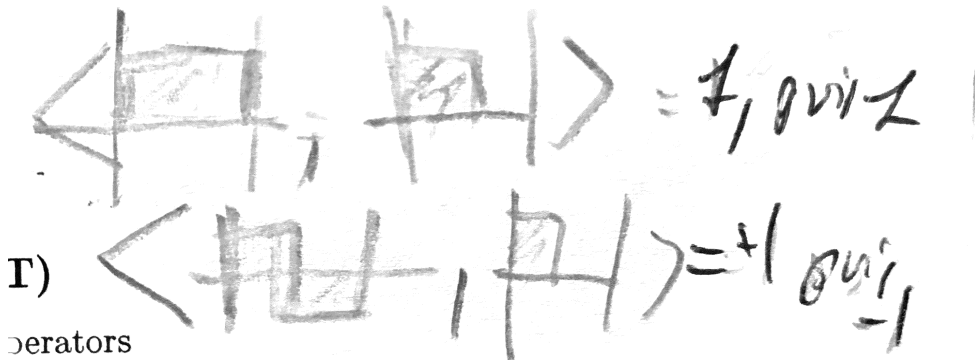
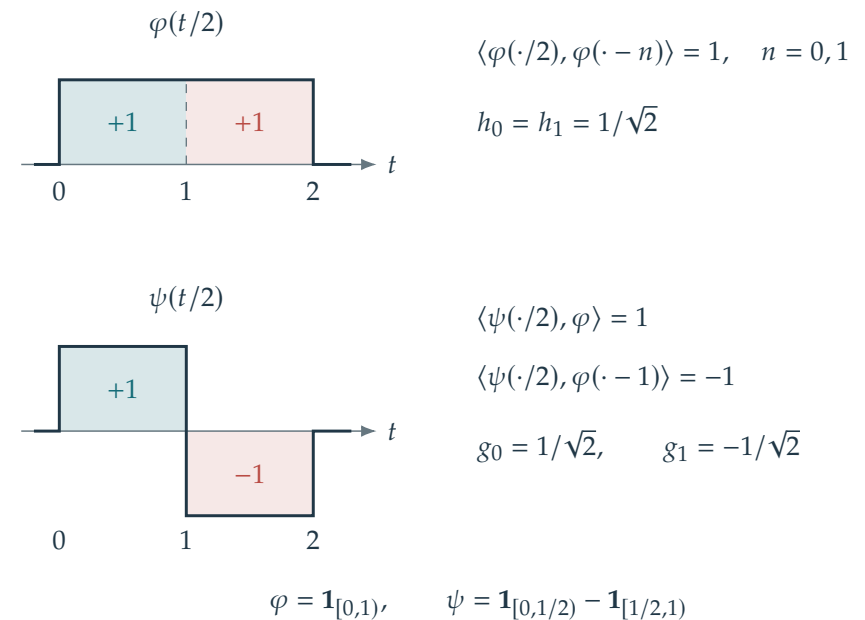
Figure 4.9

Main book, page 44

Haar filter weights as inner products.

Complete figure / Haar filter coefficients from overlap integrals

Discrete wavelet transform; equation (4.7).

Original**TikZ reconstruction**

Reading the reconstruction. The original inner products are +1 for the scaling function and +1 or -1 for the wavelet. The chapter filter definition adds a factor $1/\sqrt{2}$; the resulting $h=(1,1)/\sqrt{2}$ and $g=(1,-1)/\sqrt{2}$ are therefore explicitly distinguished from the raw overlap integrals.

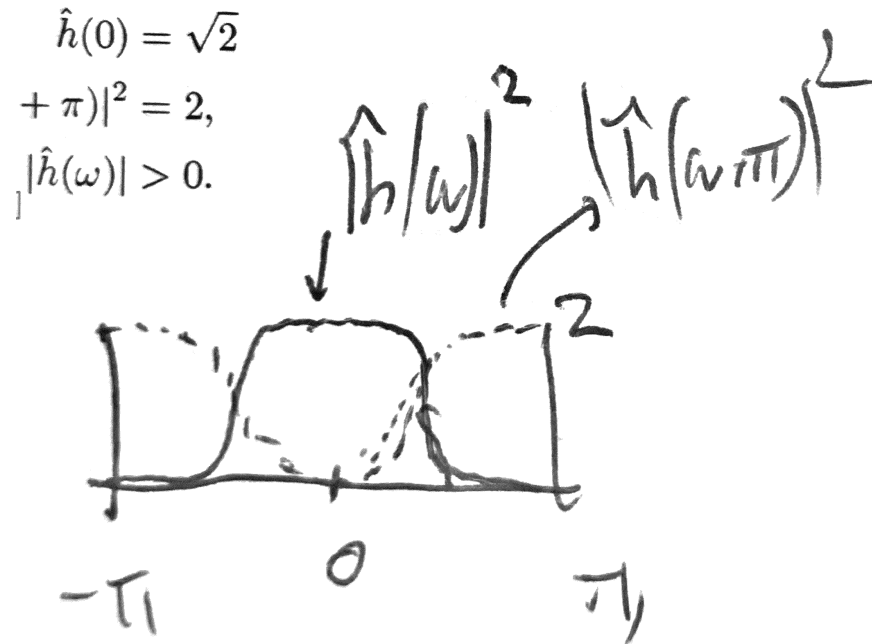
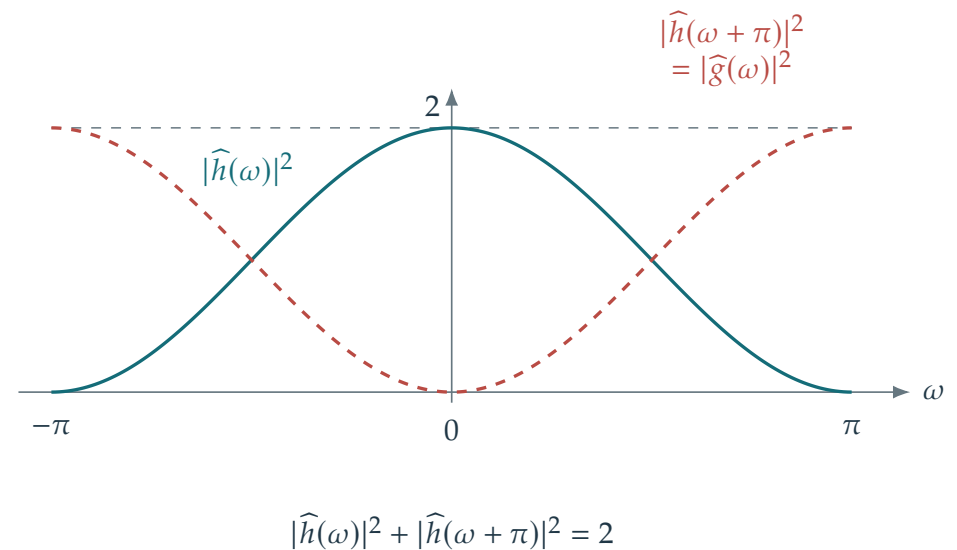
Figure 4.24

Main book, page 54

Constraints on low-pass and high-pass filters.

Complete figure / Complementary low-pass and high-pass energies

Wavelet design; low-pass and QMF constraints.

Original**TikZ reconstruction**

Reading the reconstruction. The solid and dashed curves are the exact Haar examples $|\hat{h}|^2 = 2 \cos^2(\omega/2)$ and its π shift $2 \sin^2(\omega/2)$. Their sum is two, and the shifted energy equals that of the QMF high-pass filter. The nearby cropped text in the scan is replaced by the complete relevant equality.

Figure 4.26

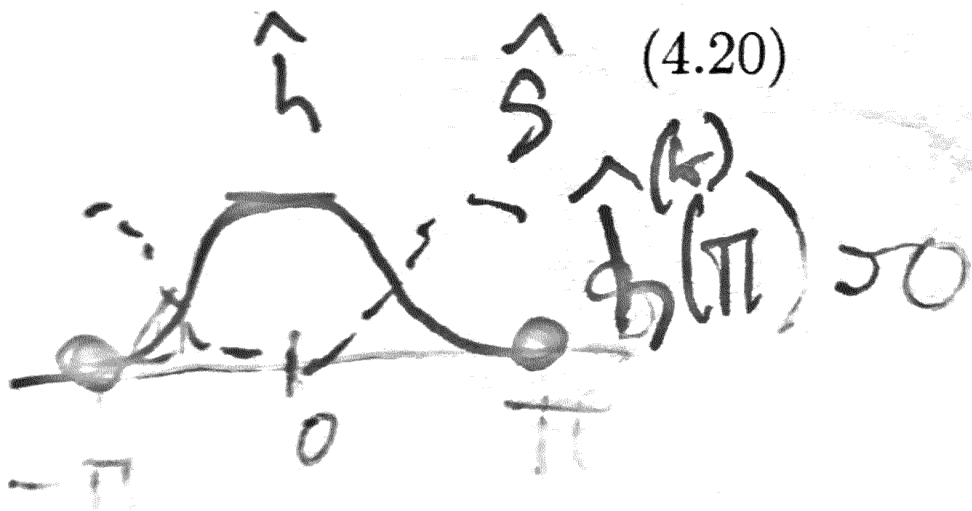
Main book, page 56

Vanishing moments correspond to vanishing derivatives of the Fourier transform.

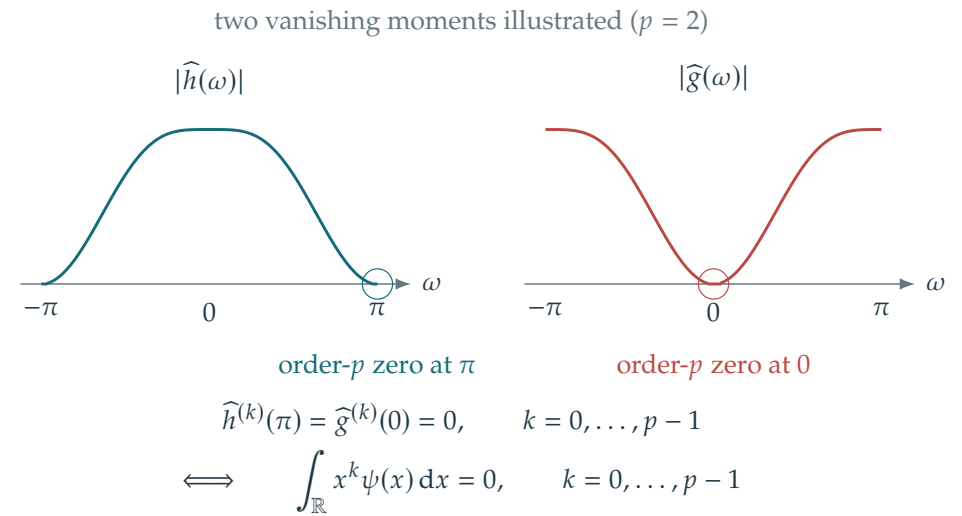
Complete figure / Vanishing moments and flat filter zeros

Wavelet design; vanishing-moment condition (4.22).

Original



TikZ reconstruction



Curves show magnitudes; derivatives refer to the filter symbols.

Reading the reconstruction. The drawing separates the low-pass zero at π from the high-pass zero at zero, avoiding the ambiguous overlaid labels in the scan. The curves are magnitudes for a two-moment QMF example; the derivative statement concerns the smooth filter symbols themselves, including their phases. The displayed zeros have order two, while the formula states the general p -moment equivalence.

Figure 5.12

Main book, page 67

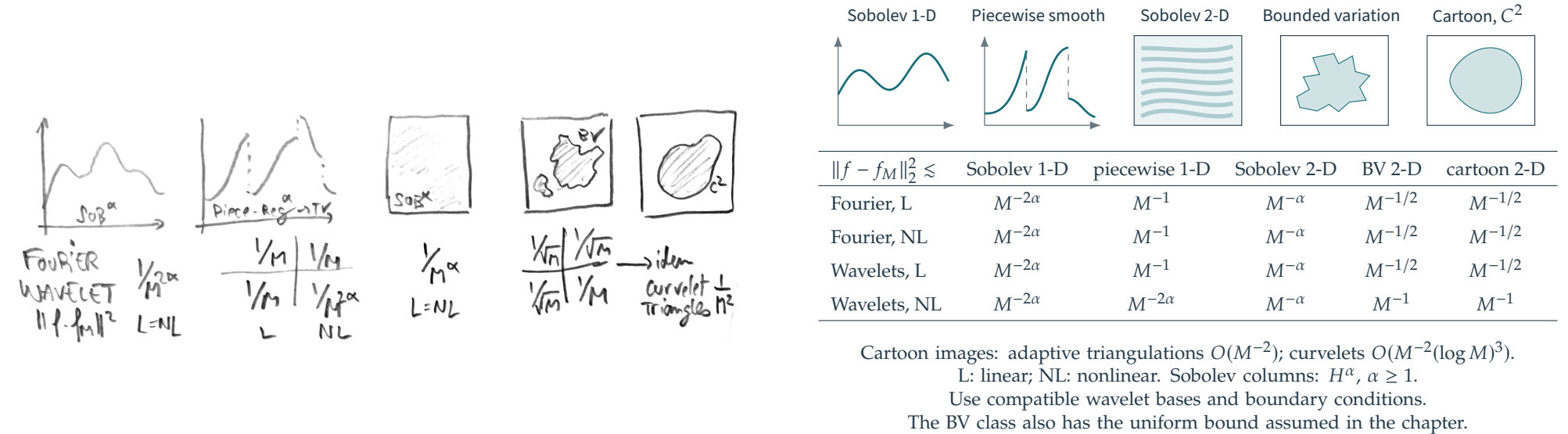
Summary of linear and nonlinear approximation rates $\|f - f_M\|^2$ for different classes of 1-D signals and images.

Complete figure / Approximation rates across signal models

Summary before Fourier approximation; squared L2 error and the smooth, piecewise-smooth, BV, and cartoon models.

Original

TikZ reconstruction



Reading the reconstruction. Interpretation to check: The sketches are reorganized into an upper-rate table. L and NL mean linear and nonlinear approximation. The Sobolev columns explicitly refer to H^α , avoiding a conflation with noninteger Holder regularity at the endpoint. The one-dimensional piecewise-smooth rates are shown for α at least one; wavelets require the stated regularity, moments, and boundary assumptions. The BV class also has the chapter uniform bound. The cartoon column assumes C^2 pieces and contours, illustrated by a smooth closed curve. The curvelet rate retains the logarithmic factor omitted by the handwritten summary. The entries are upper rates, not assertions that every signal attains them.

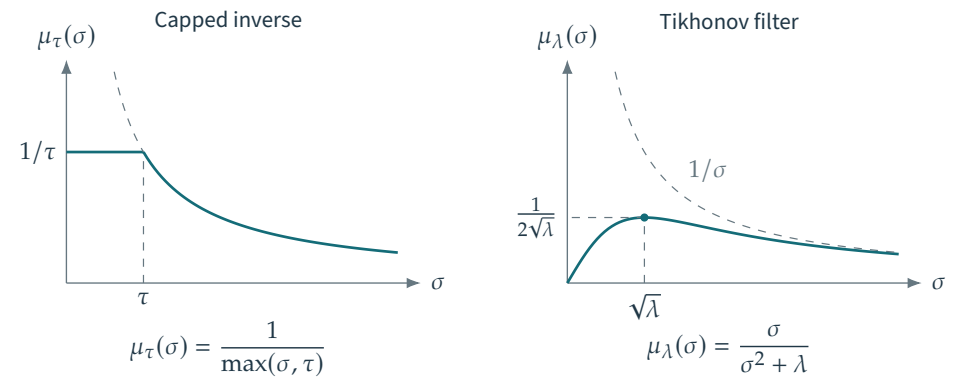
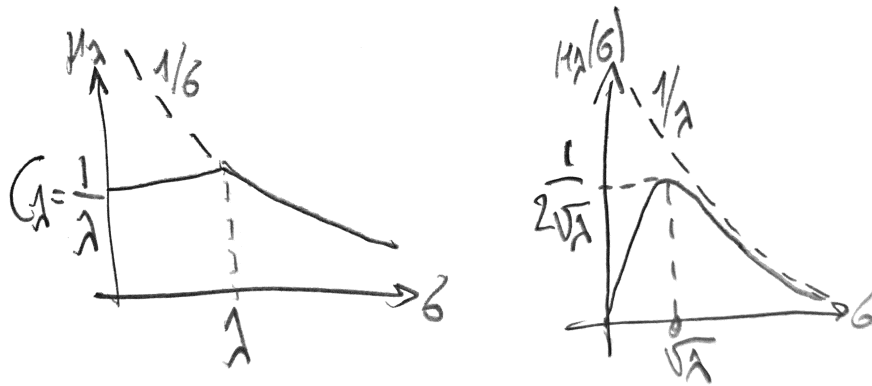
Figure 9.2

Main book, page 121

Left: capping reciprocal singular values. Right: the Tikhonov bound $\mu_\lambda(\sigma) \leq C_\lambda = \frac{1}{2\sqrt{\lambda}}$.

Complete figure / Capped inversion and Tikhonov filtering

Spectral regularization and the bound on noise amplification.

Original**TikZ reconstruction**

Common axis scales; shown with $\tau = \sqrt{\lambda} > 0$.

Reading the reconstruction. Interpretation to check: The left plateau is interpreted as the capped inverse $1/\max(\sigma, \tau)$, rather than truncated SVD. A separate cutoff τ avoids confusing its parameter with the squared-scale Tikhonov parameter λ . Both plots now use common axis scales and $\tau = \sqrt{\lambda}$: the Tikhonov maximum at $\sigma = \sqrt{\lambda}$ is half the capped gain. The dashed reciprocal curves have the same scale.

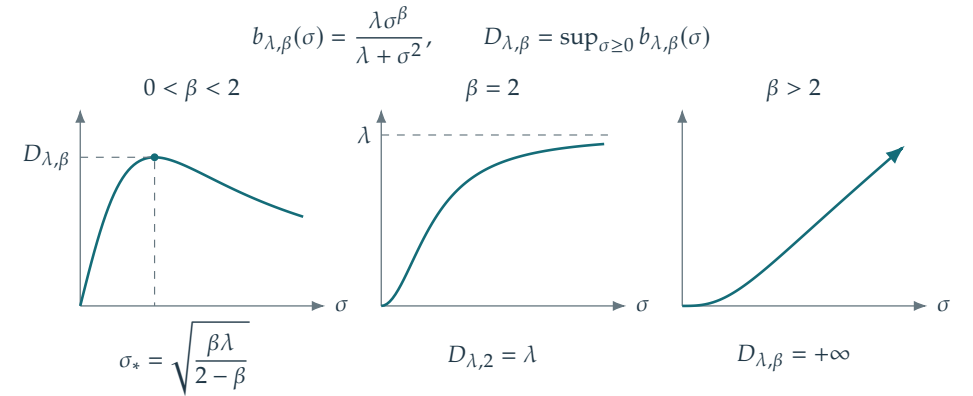
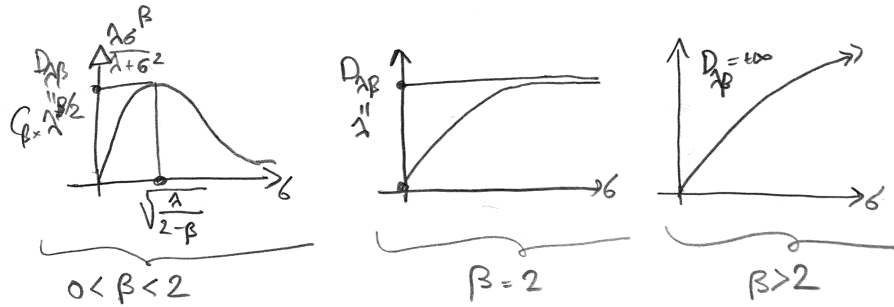
Figure 9.3

Main book, page 121

Bounding $\lambda \frac{\sigma^\beta}{\lambda + \sigma^2} \leq D_{\lambda, \beta}$.

Complete figure / The source-order dependence of the bias bound

The Tikhonov proof bounds $\lambda \sigma^\beta / (\lambda + \sigma^2)$ uniformly over $\sigma \geq 0$.

Original**TikZ reconstruction**

Here $\lambda > 0$ and the supremum ranges over all $\sigma \geq 0$.
On a fixed bounded operator, only $0 \leq \sigma \leq \|\Phi\|$ is relevant.

Reading the reconstruction. Corrects the handwritten maximizer: $\sigma^* = \sqrt{\beta \lambda / (2 - \beta)}$, including the missing factor β . For $\beta = 2$ the supremum λ is approached asymptotically. For $\beta > 2$ the supremum over all $\sigma \geq 0$ is infinite. The figure explicitly distinguishes this from the bounded spectrum of a fixed bounded operator. Curves show representative β values with rescaled axes and $\lambda > 0$.

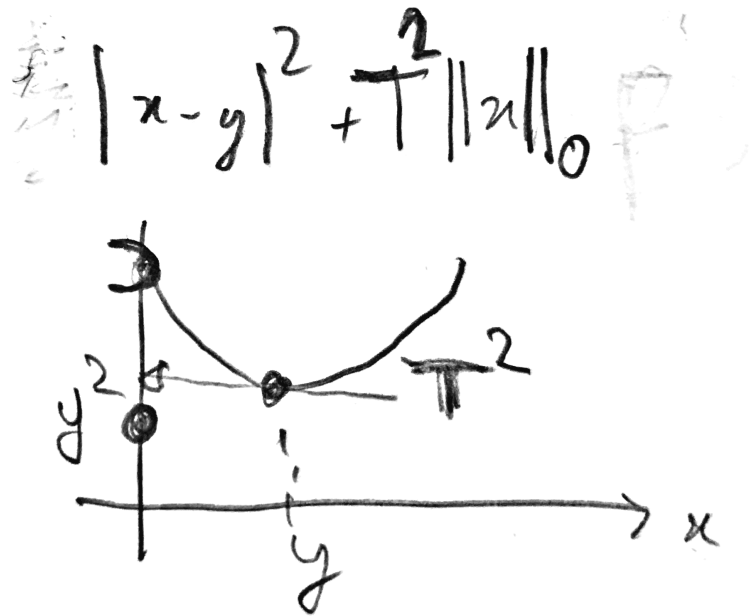
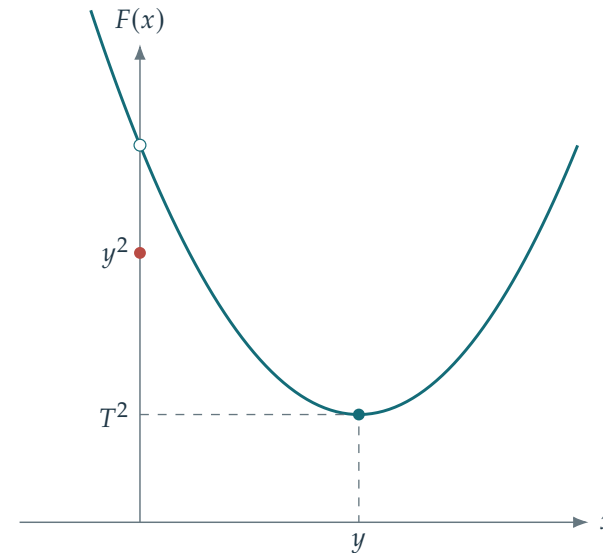
Figure 10.3

Main book, page 133

Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}|x - y|^2 + \lambda|x|$ for increasing λ .

Panel 1 of 5 / Hard thresholding as a scalar minimization

The first panel of the scalar sparsity-penalty figure uses the objective without the factor one half.

Original**TikZ reconstruction**

$$F(x) = (x - y)^2 + T^2 \mathbf{1}_{x \neq 0}$$

$$|y| < T : \quad x^* = 0$$

$$|y| > T : \quad x^* = y$$

$$|y| = T : \quad x^* \in \{0, y\}$$

Shown: $y > T > 0$.

Reading the reconstruction. Uses exactly the caption normalization $F(x) = (x - y)^2 + T^2 \mathbf{1}_{x \neq 0}$. The objective has an isolated value y^2 at zero and minimum T^2 among nonzero candidates, attained at y . The displayed example has $y > T$, and the decision rule also states the equality case.

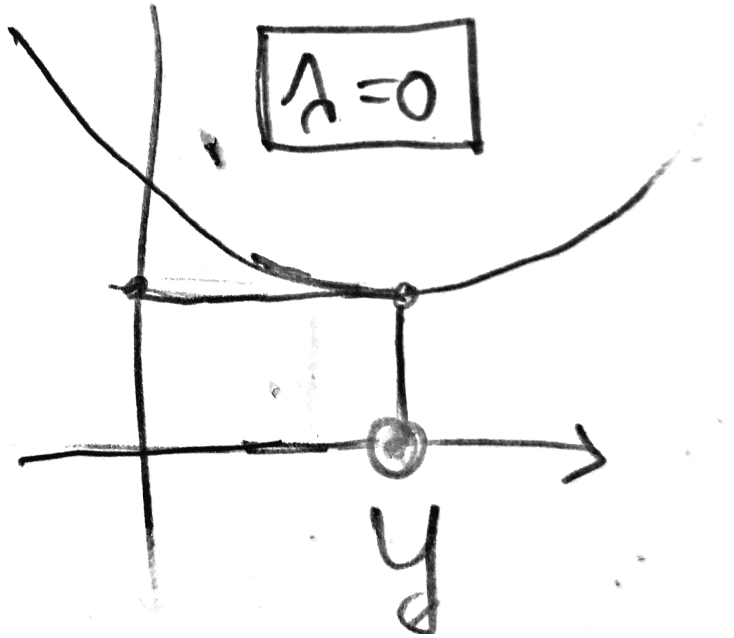
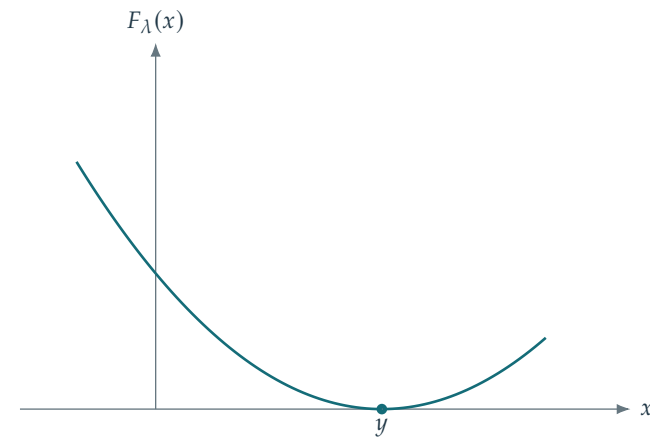
Figure 10.3

Main book, page 133

Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}|x - y|^2 + \lambda|x|$ for increasing λ .

Panel 2 of 5 / Soft thresholding: lambda=0

The four remaining panels of the scalar sparsity-penalty figure, with $y > 0$.

Original**TikZ reconstruction**

$$\lambda = 0$$

$$F_\lambda(x) = \frac{1}{2}(x - y)^2 + \lambda|x|$$

$$x^* = y$$

Shown: $y > 0$.

Reading the reconstruction. Reconstructs the exact piecewise-quadratic objective $\frac{1}{2}(x-y)^2 + \lambda|x|$. The minimizer is $\max(y-\lambda, 0)$. At $\lambda=y$ the right derivative at zero is zero; at $\lambda > y$ it is positive, while the left derivative remains negative. The curves are illustrative numerical instances of the symbolic cases.

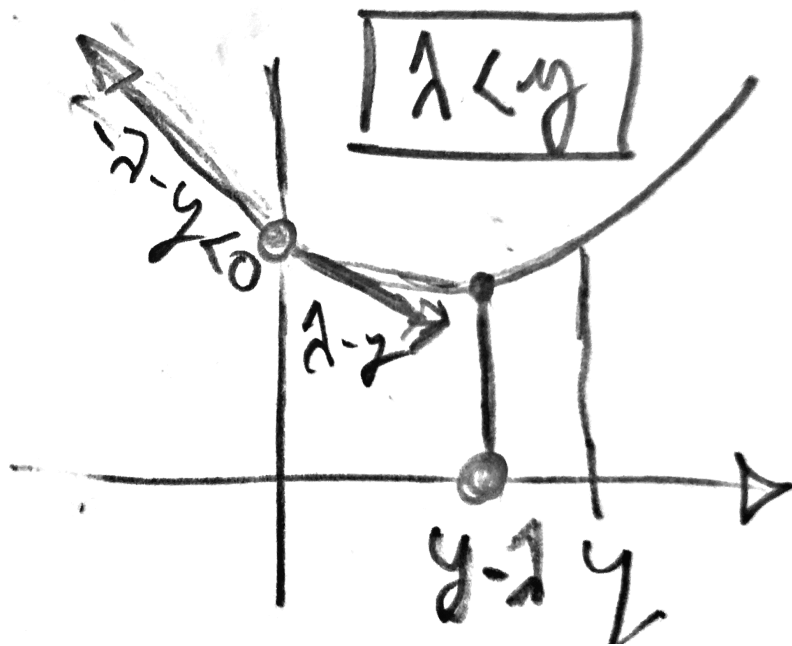
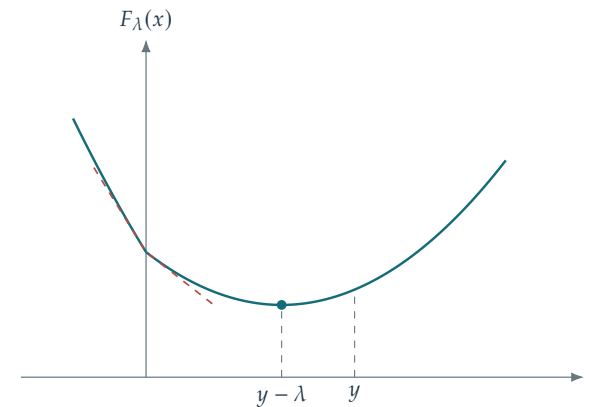
Figure 10.3

Main book, page 134

Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def}}{=} \frac{1}{2}|x - y|^2 + \lambda|x|$ for increasing λ .

Panel 3 of 5 / Soft thresholding: $0 < \lambda < y$

The four remaining panels of the scalar sparsity-penalty figure, with $y > 0$.

Original**TikZ reconstruction**

$$0 < \lambda < y$$

$$F_\lambda(x) = \frac{1}{2}(x - y)^2 + \lambda|x|$$

$$x^* = y - \lambda$$

$$\partial F_\lambda(0) = [-y - \lambda, \lambda - y]$$

Dashed: one-sided tangents at zero.

Shown: $y > 0$.

Reading the reconstruction. Reconstructs the exact piecewise-quadratic objective $\frac{1}{2}(x - y)^2 + \lambda|x|$. The minimizer is $\max(y - \lambda, 0)$. At $\lambda = y$ the right derivative at zero is zero; at $\lambda > y$ it is positive, while the left derivative remains negative. The curves are illustrative numerical instances of the symbolic cases. Dashed supporting tangents show the left slope $-y - \lambda$ and right slope $\lambda - y$ at zero; the full subdifferential interval explains when zero is a minimizer.

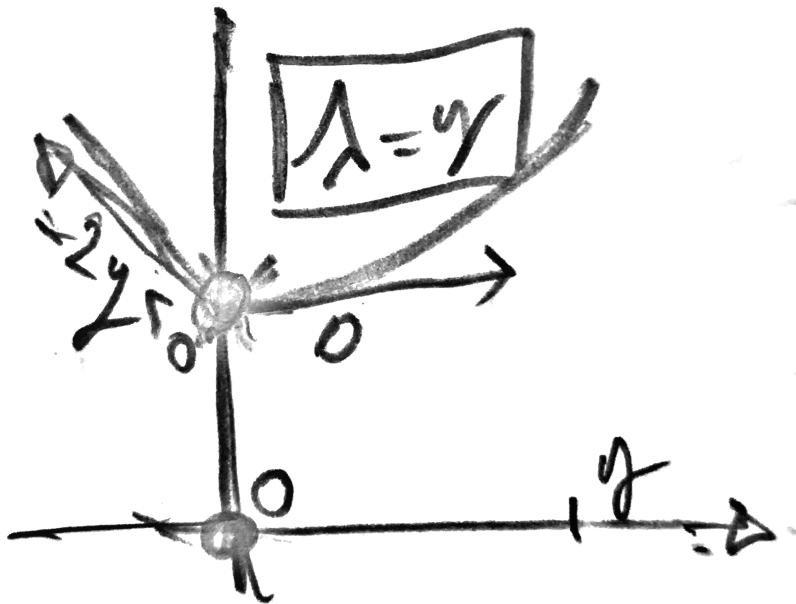
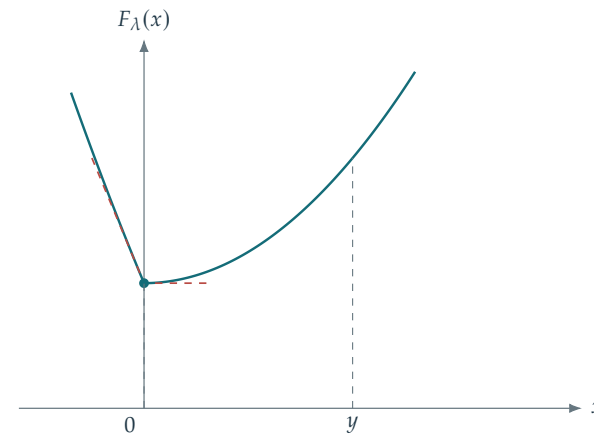
Figure 10.3

Main book, page 134

Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def}}{=} \frac{1}{2}|x - y|^2 + \lambda|x|$ for increasing λ .

Panel 4 of 5 / Soft thresholding: lambda=y

The four remaining panels of the scalar sparsity-penalty figure, with $y > 0$.

Original**TikZ reconstruction**

$$\lambda = y$$

$$F_\lambda(x) = \frac{1}{2}(x - y)^2 + \lambda|x|$$

$$x^* = 0$$

$$F'_\lambda(0^+) = \lambda - y = 0$$

$$\partial F_\lambda(0) = [-y - \lambda, \lambda - y]$$

Dashed: one-sided tangents at zero.

Shown: $y > 0$.

Reading the reconstruction. Reconstructs the exact piecewise-quadratic objective $\frac{1}{2}(x-y)^2 + \lambda|x|$. The minimizer is $\max(y-\lambda, 0)$. At $\lambda=y$ the right derivative at zero is zero; at $\lambda > y$ it is positive, while the left derivative remains negative. The curves are illustrative numerical instances of the symbolic cases. Dashed supporting tangents show the left slope $-y-\lambda$ and right slope $\lambda-y$ at zero; the full subdifferential interval explains when zero is a minimizer.

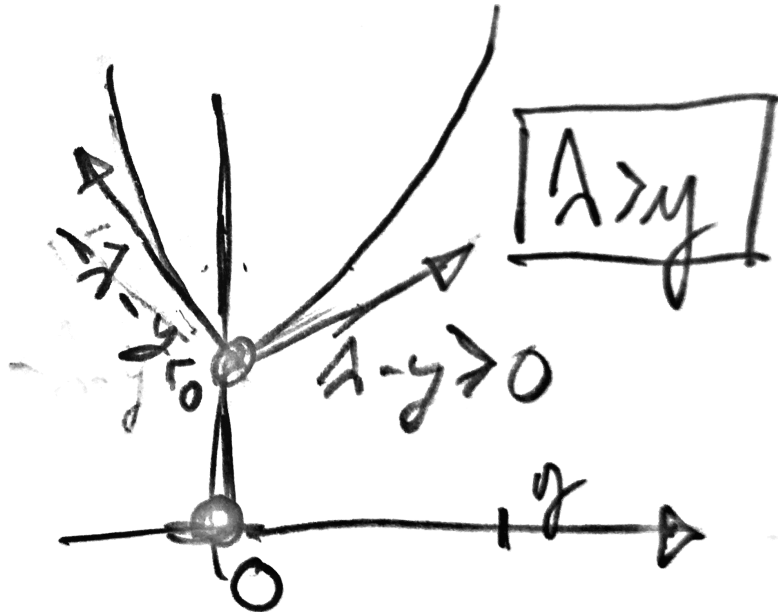
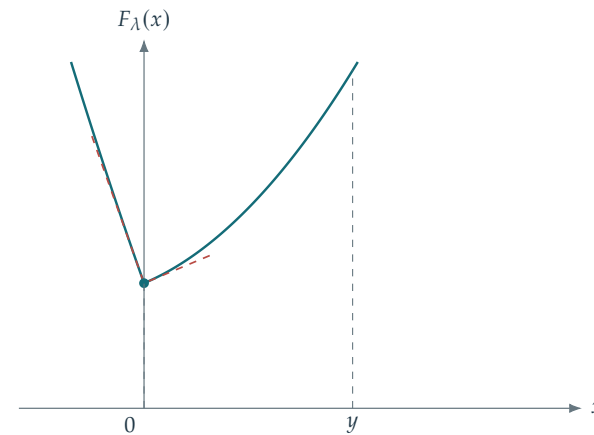
Figure 10.3

Main book, page 134

Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def}}{=} \frac{1}{2}|x - y|^2 + \lambda|x|$ for increasing λ .

Panel 5 of 5 / Soft thresholding: lambda>y

The four remaining panels of the scalar sparsity-penalty figure, with $y > 0$.

Original**TikZ reconstruction**

$$\lambda > y$$

$$F_\lambda(x) = \frac{1}{2}(x - y)^2 + \lambda|x|$$

$$x^* = 0$$

$$F'_\lambda(0^+) = \lambda - y > 0$$

$$\partial F_\lambda(0) = [-y - \lambda, \lambda - y]$$

Dashed: one-sided tangents at zero.

Shown: $y > 0$.

Reading the reconstruction. Reconstructs the exact piecewise-quadratic objective $\frac{1}{2}(x-y)^2 + \lambda|x|$. The minimizer is $\max(y-\lambda, 0)$. At $\lambda=y$ the right derivative at zero is zero; at $\lambda > y$ it is positive, while the left derivative remains negative. The curves are illustrative numerical instances of the symbolic cases. Dashed supporting tangents show the left slope $-y-\lambda$ and right slope $\lambda-y$ at zero; the full subdifferential interval explains when zero is a minimizer.

Figure 11.2

Main book, page 143

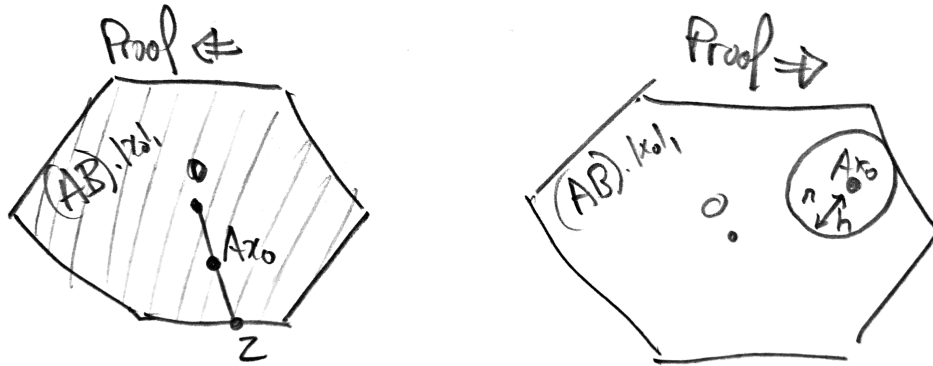
Geometry of the polytope characterization of exact recovery.

Complete figure / Interior points and failure of exact recovery

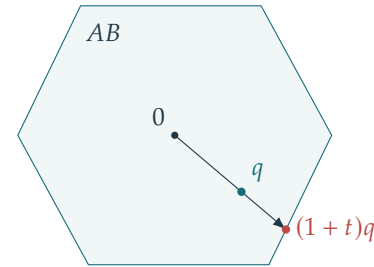
The onto-map polytope characterization: x_0 solves basis pursuit iff $Ax_0 / \|x_0\|_1$ lies on the boundary of AB .

Original

TikZ reconstruction



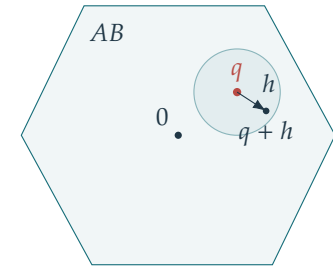
Radial extension from an interior point



$$\begin{aligned} Au &= (1+t)q, u \in B \\ z &= ru/(1+t) \text{ gives} \\ Az &= Ax_0 \text{ and } \|z\|_1 < r. \end{aligned}$$

$$r = \|x_0\|_1 > 0, \quad q = Ax_0/r, \quad A \text{ onto}$$

A better feasible vector gives an interior point



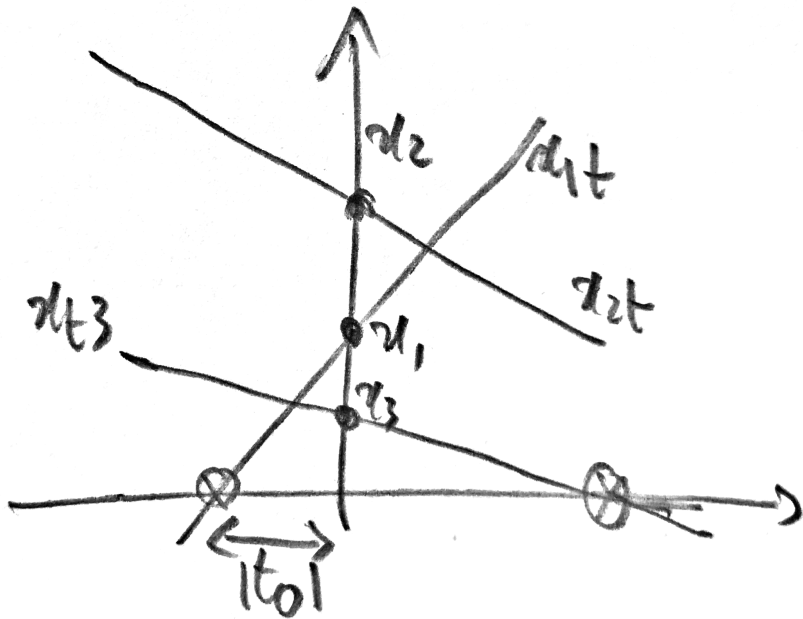
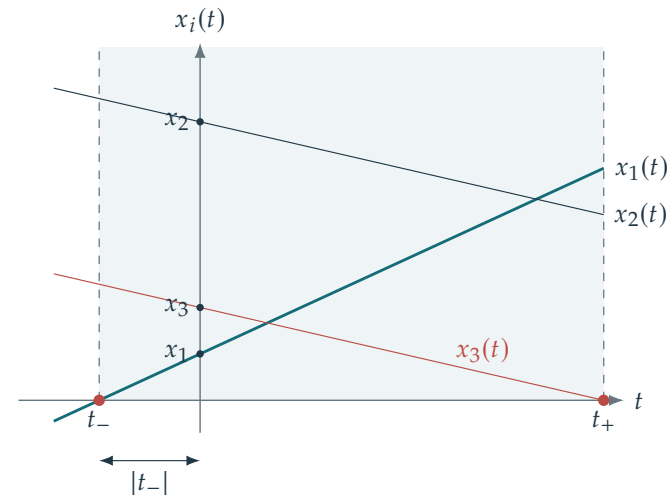
$$\begin{aligned} Az &= Ax_0, \|z\|_1 < r \\ q+h &= A(z/r + A^+h) \in AB \\ &\text{for small } h. \end{aligned}$$

Reading the reconstruction. Uses the normalized polytope AB and $q = Ax_0/r$, $r = \|x_0\|_1$, consistently with the corrected proof. The radial extension $(1+t)q$ is constructed exactly on the ray through q and on the polygon boundary. The second panel shows h as a directed displacement to $q+h$ inside an interior ball. Both panels illustrate failure of optimality through interior membership; no uniqueness claim is implied.

Figure 11.4

Main book, page 145

Coefficient trajectories along a support-reducing direction.

Complete figure / Removing a dependent active coefficientSupport reduction along $x(t)=x+th$ when $A_I h_I=0$.**Original****TikZ reconstruction**At a minimizer x :

$$x(t) = x + th, Ah = 0$$

$$Ax(t) = Ax$$

$$\|x(t)\|_1 = \|x\|_1$$

$$\text{for } t_- \leq t \leq t_+.$$

The signs stay fixed for $t_- < t < t_+$.
At either endpoint, at least one coefficient becomes zero.

Reading the reconstruction. Three affine coefficient trajectories have a marked interval with both endpoints shown. The first coefficient vanishes at the left endpoint and the third at the right. The figure states that x is a minimizer: this assumption forces the affine ℓ_1 norm to have zero slope while signs stay fixed. Constant fidelity follows from $Ah=0$. The norm remains constant at the endpoints by continuity, though the sign pattern changes there.

Figure 11.5

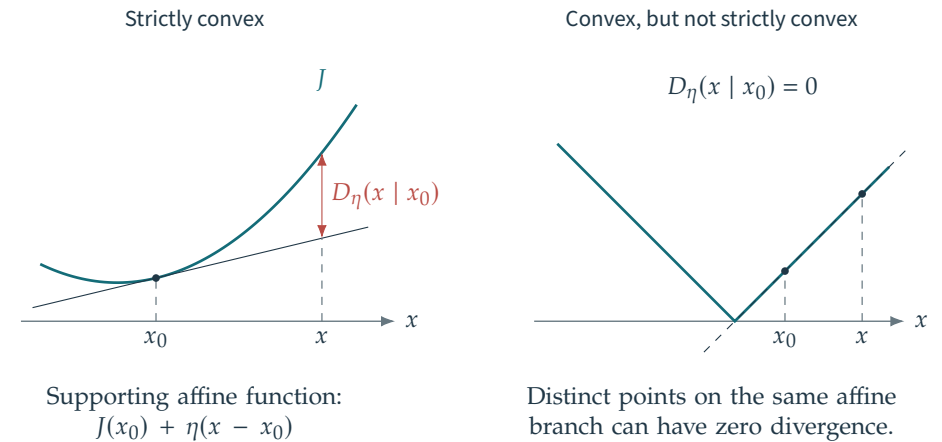
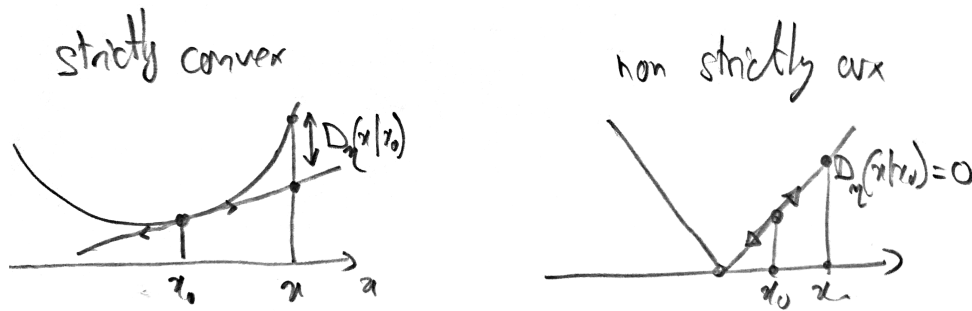
Main book, page 147

Visualization of Bregman divergences.

Complete figure / Bregman divergence as a vertical gapDefinition of $D_\eta(x | x_0)$ for η in the subdifferential of a convex J .

Original

TikZ reconstruction



Reading the reconstruction. The left panel uses a strictly convex quadratic and its tangent; the positive vertical gap is the Bregman divergence. The right panel uses $J=|x|$, with x_0 and x on the same positive affine branch, so the divergence vanishes even when x differs from x_0 .

Figure 11.6

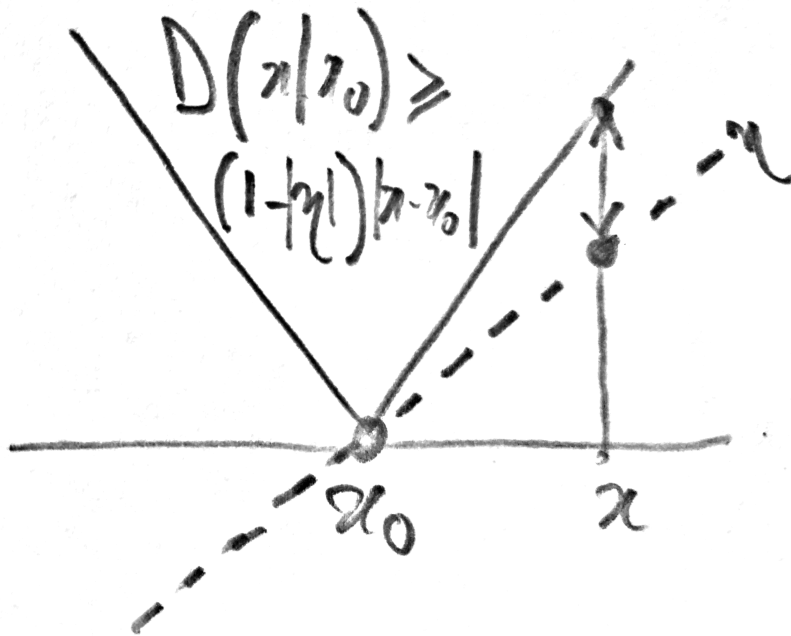
Main book, page 149

Bregman divergence controls the ℓ^1 error on coordinates where η does not saturate.

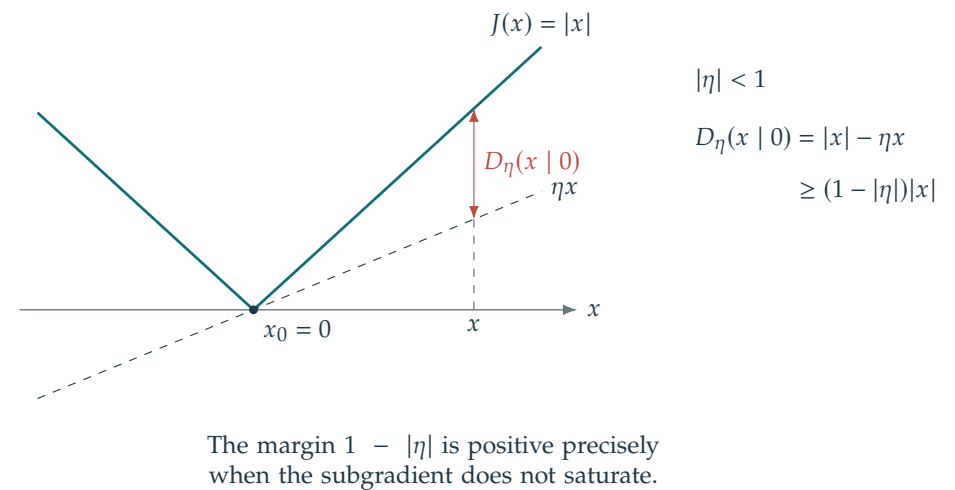
Complete figure / A strict subgradient margin controls the error

The coordinatewise estimate in the ℓ^1 Bregman lower bound.

Original



TikZ reconstruction



Reading the reconstruction. Explicitly sets $x_0=0$, as required on nonsaturated coordinates for $J=\text{abs}(x)$. For $\text{abs}(\eta)<1$ the supporting line ηx lies strictly below $\text{abs}(x)$ away from zero. The lower bound is $D_\eta(x | 0) = (1-\text{abs}(\eta))\text{abs}(x)$, not a norm bound on saturated coordinates.

Figure 11.8

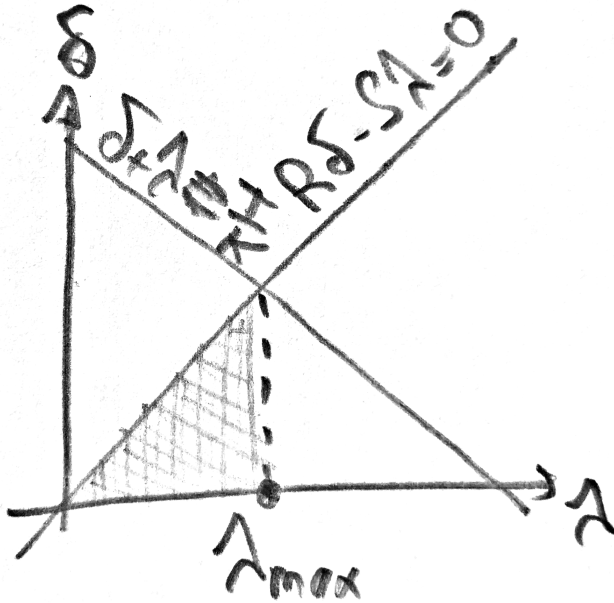
Main book, page 153

Region in the (λ, δ) plane where sign consistency holds.

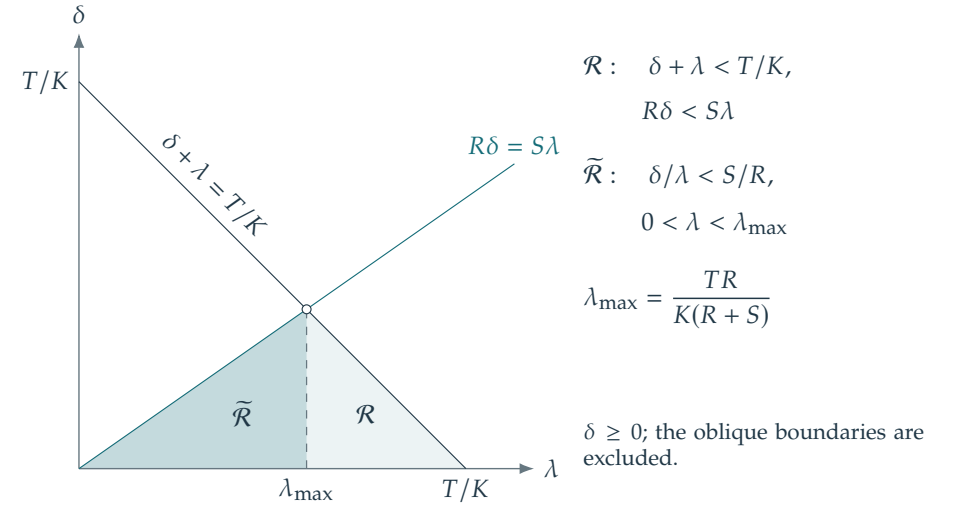
Complete figure / Parameter region for sign consistency

Intersection of $\delta + \lambda < T/K$ and $R\delta < S\lambda$ in the support-stability proof.

Original



TikZ reconstruction

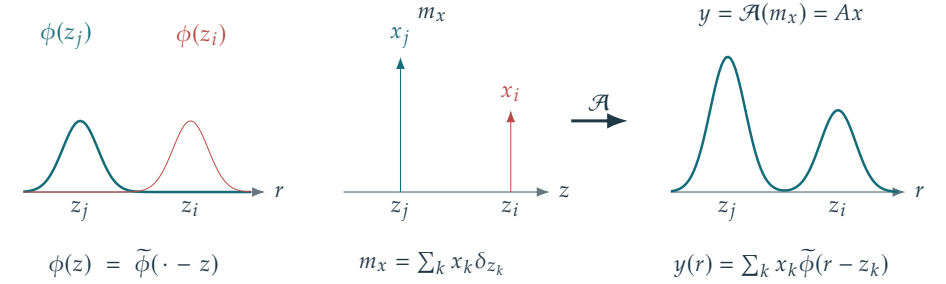
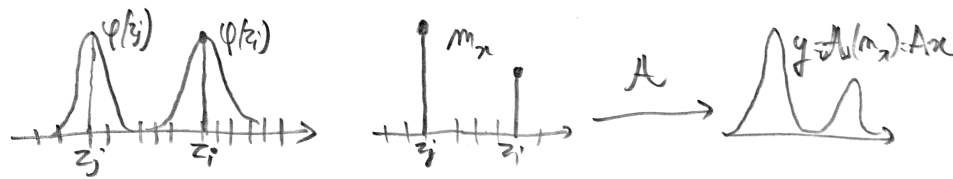


Reading the reconstruction. Reconstructs the two strict inequalities and the contained simpler triangle. The vertical cutoff is $\lambda_{\max} = TR/[K(R+S)]$, using $R = KL+1$. The oblique boundaries and the cutoff for the smaller triangle are excluded; the intersection marker is open. The zero-noise axis $\delta=0$ is allowed for positive λ satisfying the displayed strict inequalities. The horizontal axis is λ and the vertical axis is δ .

Figure 11.10

Main book, page 155

Convolution operator.

Complete figure / Convolution of a discrete measureThe dictionary map $\phi(z)$, discrete measure m_x , and measurement operator A .**Original****TikZ reconstruction**

Reading the reconstruction. Two translated Gaussian kernels have unit peak; the spike heights and the output use the same vertical scale, so the displayed output is exactly their weighted sum. Kernel centers, spike locations, and output-location labels agree across the panels. The diagram distinguishes index-space weights from physical reconstruction-grid locations, and writes the output as both the continuous operator applied to the discrete measure and the dictionary product Ax .

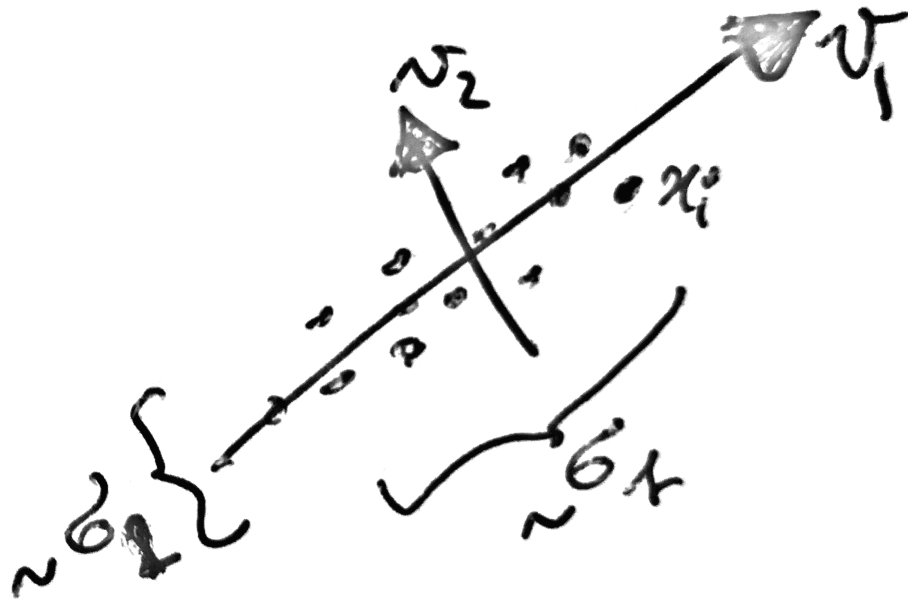
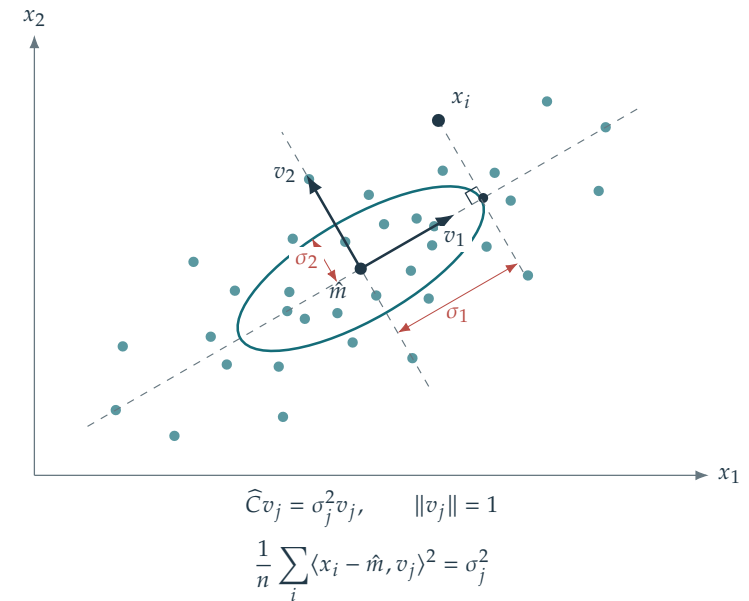
Figure 13.2

Main book, page 168

Principal directions of variation.

Complete figure / PCA directions and projected variance

Principal directions are eigenvectors of the empirical covariance; the chapter writes its eigenvalues as sigma squared.

Original**TikZ reconstruction**

Project first; the variance is the mean squared coordinate along that direction.

Reading the reconstruction. The centered illustrative cloud has exact reflection symmetries in principal coordinates. The ellipse semiaxes are empirical standard deviations with normalization $1/n$; variance is their square. A highlighted observation and its perpendicular projection show the coordinate whose squared values are averaged. Unit eigenvectors and standard-deviation lengths remain distinct.

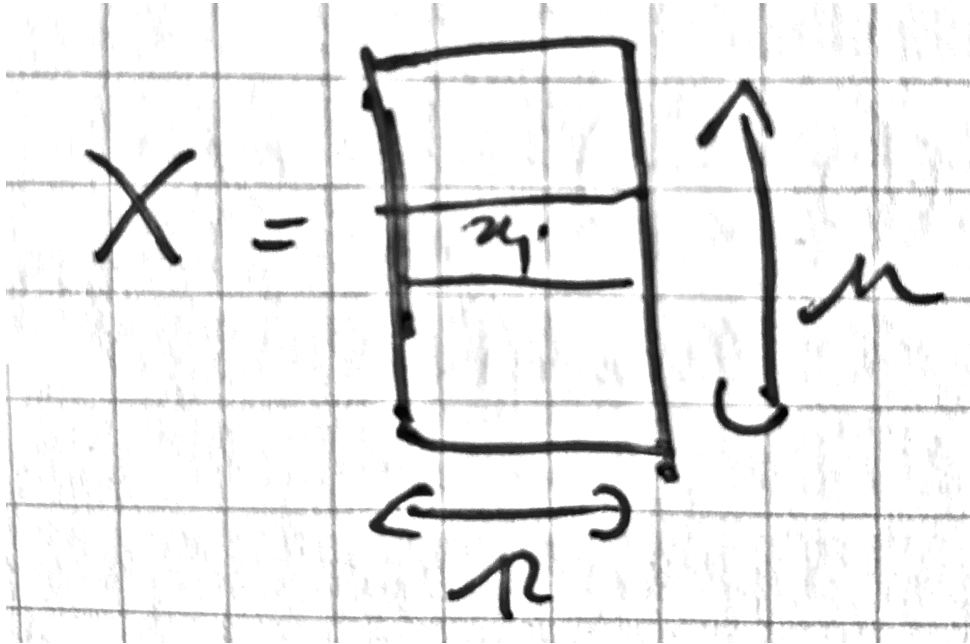
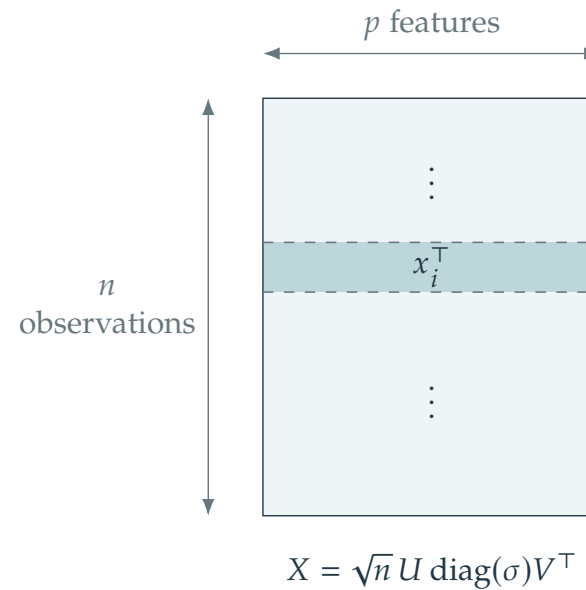
Figure 13.4

Main book, page 169

Rows of the centered data matrix are observations; columns are features.

Complete figure / Rows are observations; columns are features

The PCA proof centers the data and writes the n -by- p matrix X with observations in its rows.

Original**TikZ reconstruction**

$$X = \tilde{X} \in \mathbb{R}^{n \times p}$$

$$\hat{m} = \frac{1}{n} \sum_{i=1}^n x_i = 0$$

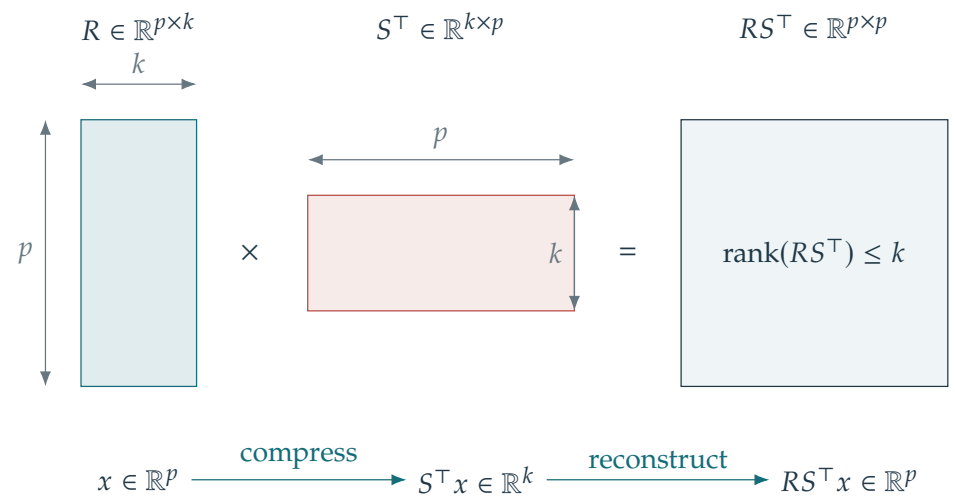
$$\hat{C} = \frac{1}{n} X^T X$$

Reading the reconstruction. The reconstruction follows the current n -observations, p -features convention. An observation x_i is a column vector in $\mathbb{R}^p$; the corresponding matrix row is its transpose. Centering is explicit, and the covariance is $X^T X$ divided by n .

Figure 13.5

Main book, page 170

Linear compression followed by reconstruction.

Complete figure / Linear compression followed by reconstructionThe variational PCA problem compares rank-at-most- k linear reconstructions RS^T .**Original****TikZ reconstruction**

Reading the reconstruction. Both R and S have p rows and k columns. Compression applies S^T ; reconstruction applies R , so the product is a p -by- p map of rank at most k . Orthogonality is not assumed for arbitrary R and S ; it is justified later when reducing to orthogonal projections.

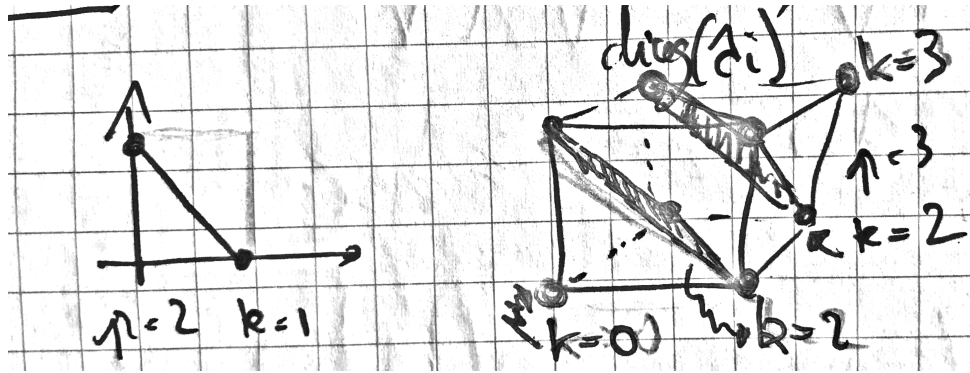
Figure 13.6

Panel 1: the linear-program bound in (13.6). Panel 2: matrix B . Panel 3: its orthogonal extension $\tilde{B}$.

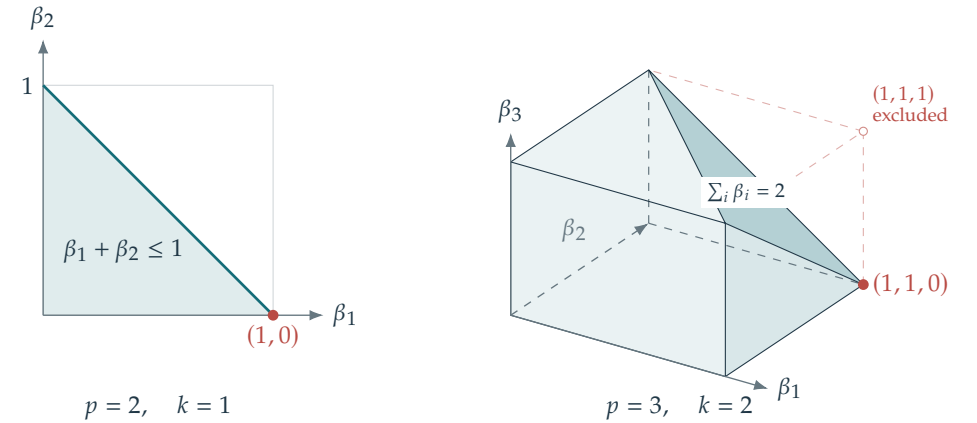
Panel 1 of 3 / The polytope bounding the PCA objective

The PCA proof bounds the trace using (13.6), a linear program maximizing a decreasingly weighted sum over $0 \leq \beta_i \leq 1$ and $\sum \beta_i \leq k$.

Original



TikZ reconstruction



Feasible region: $0 \leq \beta_i \leq 1, \sum_i \beta_i \leq k$. Red dots: a maximizing vertex.

$$\lambda_1 \geq \dots \geq \lambda_p \geq 0, \quad \max \sum_i \lambda_i \beta_i = \sum_{i=1}^k \lambda_i$$

Reading the reconstruction. The left feasible region is a triangle for $p=2, k=1$. The right is the unit cube truncated by $\sum \beta_i \leq 2$ for $p=3, k=2$. Its cutting face is shown with nonzero visible area; dashed edges distinguish the hidden cube edges from the removed corner $(1,1,1)$. Red dots mark valid maximizers $(1,0)$ and $(1,1,0)$ for ordered nonnegative eigenvalues, without asserting uniqueness.

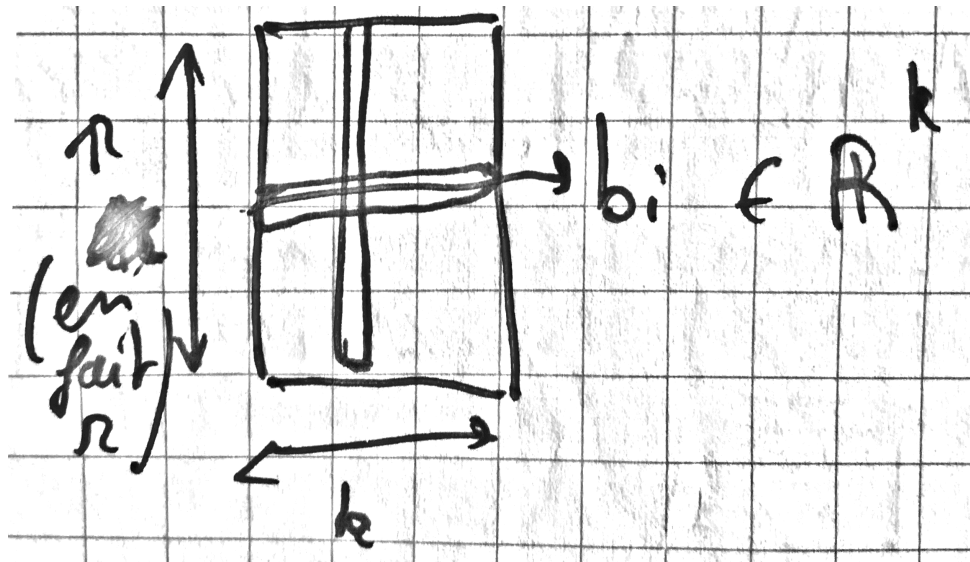
Figure 13.6

Panel 1: the linear-program bound in (13.6). Panel 2: matrix B . Panel 3: its orthogonal extension $\tilde{B}$.

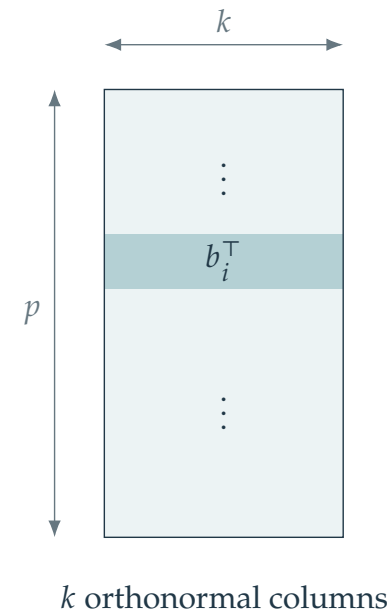
Panel 2 of 3 / Row norms of the rotated basis

The PCA proof defines $B=V^T S$ and β_i as the squared norm of its i th row.

Original



TikZ reconstruction



$$B = V^T S \in \mathbb{R}^{p \times k}$$

$$B^T B = I_k$$

$$\beta_i = \|b_i\|^2 \geq 0$$

$$\sum_{i=1}^p \beta_i = \|B\|_F^2 = k$$

Reading the reconstruction. The matrix B has p rows and k orthonormal columns. The vector b_i is in $\mathbb{R}^k$ and appears transposed as a row. Its squared row norms sum to k , the squared Frobenius norm of B . These dimensions follow the current proof rather than ambiguous letters in the scan.

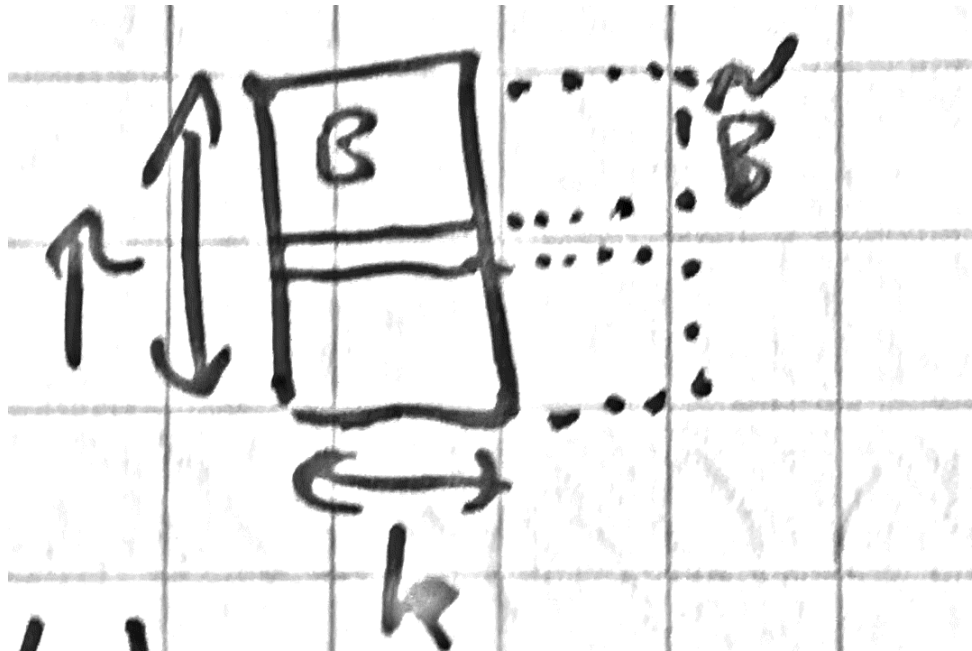
Figure 13.6

Panel 1: the linear-program bound in (13.6). Panel 2: matrix B . Panel 3: its orthogonal extension $\tilde{B}$.

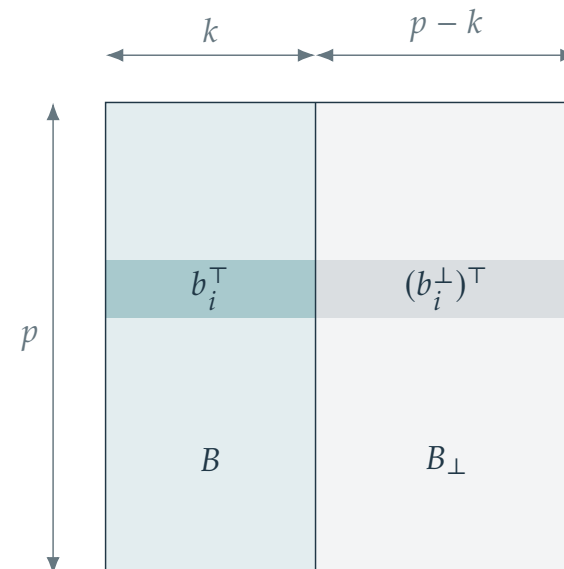
Panel 3 of 3 / Completing the columns to an orthogonal matrix

An orthogonal completion proves that every squared row norm β_i is at most one.

Original



TikZ reconstruction



$$\tilde{B} = [B \ B_{\perp}]$$

$$\tilde{B}\tilde{B}^{\top} = I_p$$

$$\|b_i\|^2 + \|b_i^{\perp}\|^2 = 1$$

$$\beta_i = \|b_i\|^2 \leq 1$$

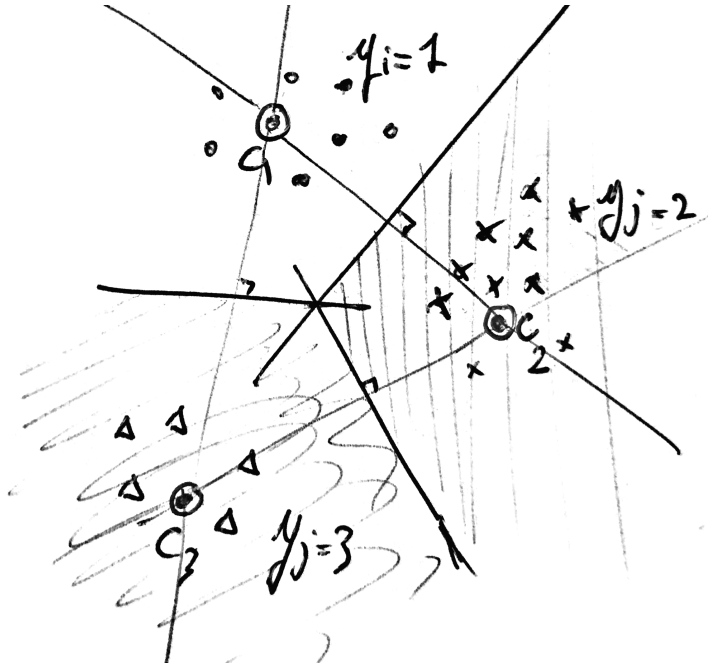
Reading the reconstruction. The first k columns are B ; the remaining $p-k$ columns form B perpendicular. Every full row of the square orthogonal matrix has norm one. Restricting a row to the first k entries cannot increase its norm. When $k=p$, the complementary block is empty.

Figure 13.7

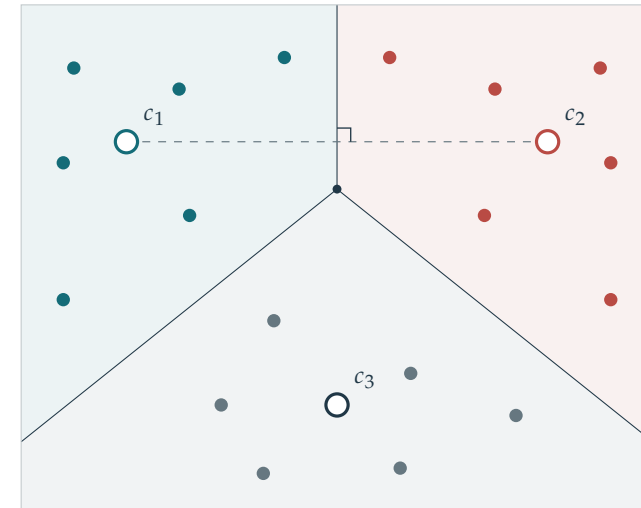
Main book, page 173

 k -means clusters according to Voronoi cells.**Complete figure / Nearest-centroid assignment and Voronoi cells**For fixed centroids, the assignment step of k -means selects the nearest centroid in Euclidean distance.

Original



TikZ reconstruction



$$V_j = \{x : \|x - c_j\| \leq \|x - c_\ell\| \text{ for every } \ell\}$$

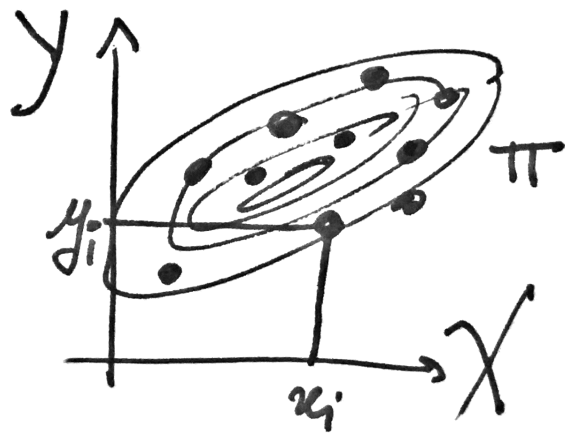
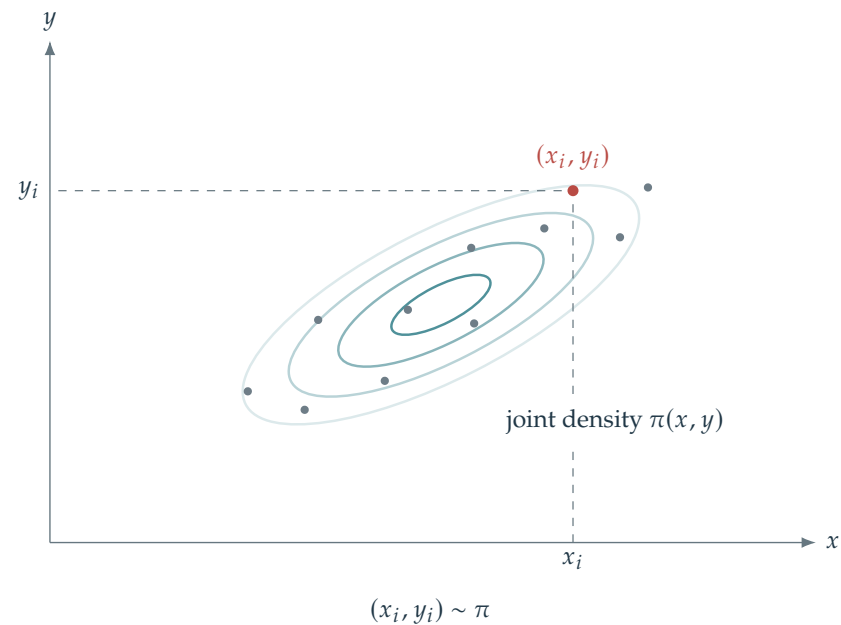
Dashed segment: two centroids. Its perpendicular bisector is their common cell edge.

Reading the reconstruction. Cell edges are the exact perpendicular bisectors of the displayed centroids. The dashed centroid segment and right-angle marker make that construction explicit. Colored observations lie in their nearest-centroid cells; the common vertex is equidistant from all three centroids. This illustrates assignment, not a claim that these centroids are already the cluster means. Boundary ties require the convention in the text.

Figure 13.10

Main book, page 177

Probabilistic modeling.

Complete figure / A joint probability model for input and outputSupervised observations are pairs (x_i, y_i) drawn from a joint law π on the product of input and output spaces.**Original****TikZ reconstruction**

Reading the reconstruction. Contours represent an illustrative bivariate Gaussian density, not an asserted fit to the original hand-drawn points. The dashed projections identify the two coordinates of one observation. The general discussion in the chapter is not restricted to Gaussian models.

Figure 13.11

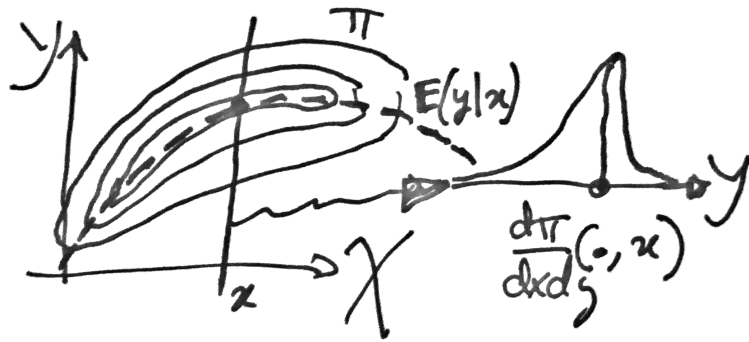
Main book, page 177

Conditional expectation.

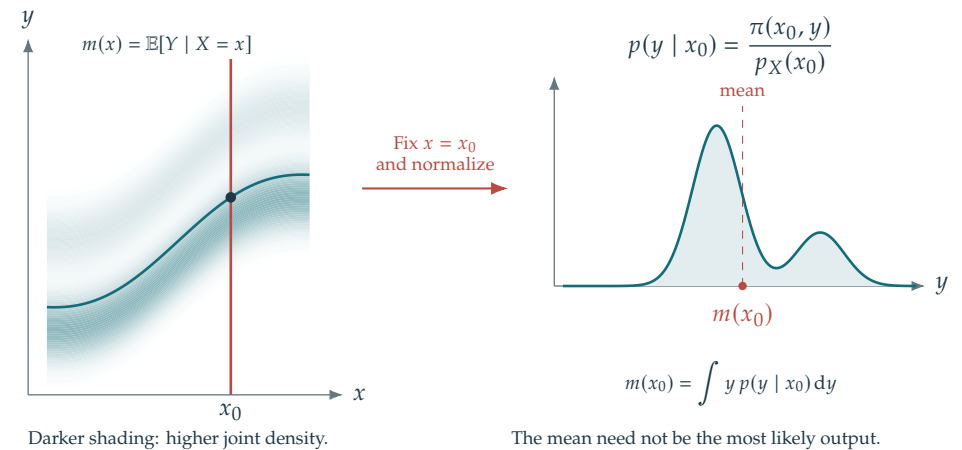
Complete figure / Conditional expectation as the mean of a slice

Under squared loss, the optimal predictor is the conditional expectation of the output given the input.

Original



TikZ reconstruction



Reading the reconstruction. Here X is uniform on $[-1.75, 1.75]$ and $Y = m(X) + \text{epsilon}$, with independent centered asymmetric mixture noise: weights $3/4$ and $1/4$, Gaussian means -0.3 and 0.9 , and standard deviation 0.28 . Shading encodes joint density; dividing its vertical slice by $p_X(x_0)$ gives the right-hand conditional density. Its mean $m(x_0)$ differs from its mode. The theorem does not assume this illustrative noise model.

Figure 13.12

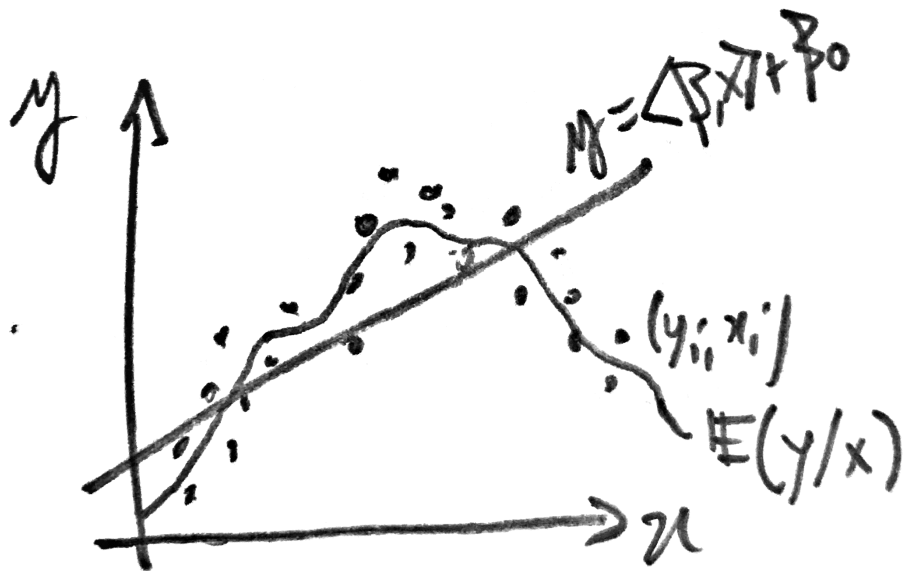
Main book, page 178

Linear regression.

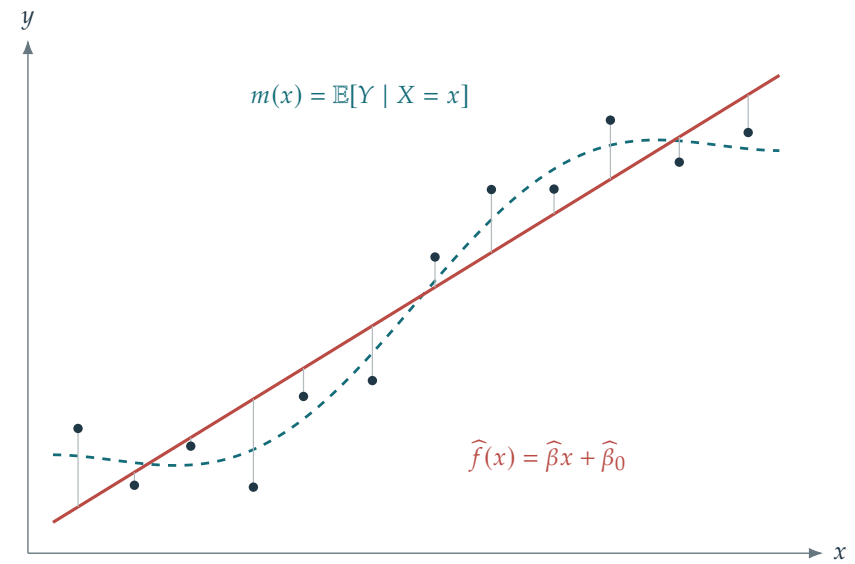
Complete figure / An affine fit and a nonlinear regression function

Linear regression restricts the predictor to a finite-dimensional model, even when the conditional expectation is nonlinear.

Original



TikZ reconstruction



Vertical segments: residuals to the fitted affine predictor.

Reading the reconstruction. The red line is the least-squares affine fit to the actual displayed illustrative points. The dashed curve is the model conditional mean used to construct those points. Residuals are vertical output errors, not orthogonal distances to the line.

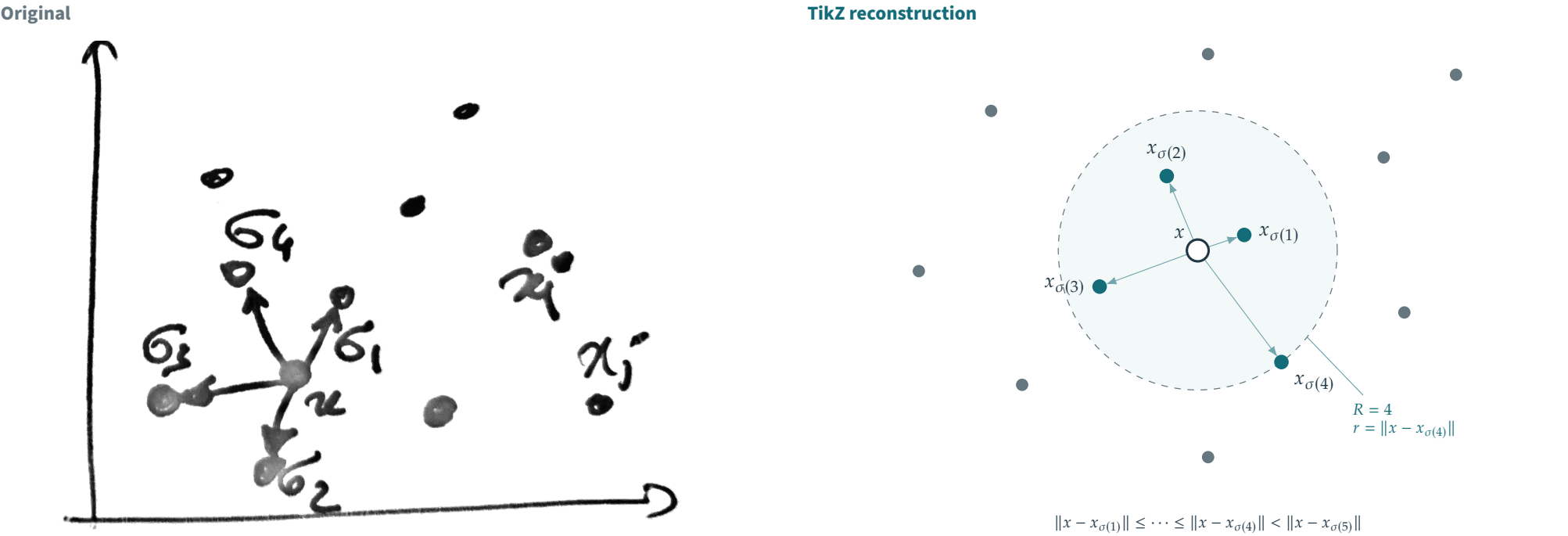
Figure 13.13

Main book, page 180

Nearest neighbors.

Complete figure / Four nearest neighbors of a query point

The R-nearest-neighbor rule sorts training points by their Euclidean distance from the query.



Reading the reconstruction. The arrows identify the four closest observations in increasing distance order. The pale disk is the selected neighborhood; its radius is exactly the distance to the fourth observation on its boundary. Every remaining point is farther away. The panel explains the distance-selection step before voting over class labels.

Figure 13.15

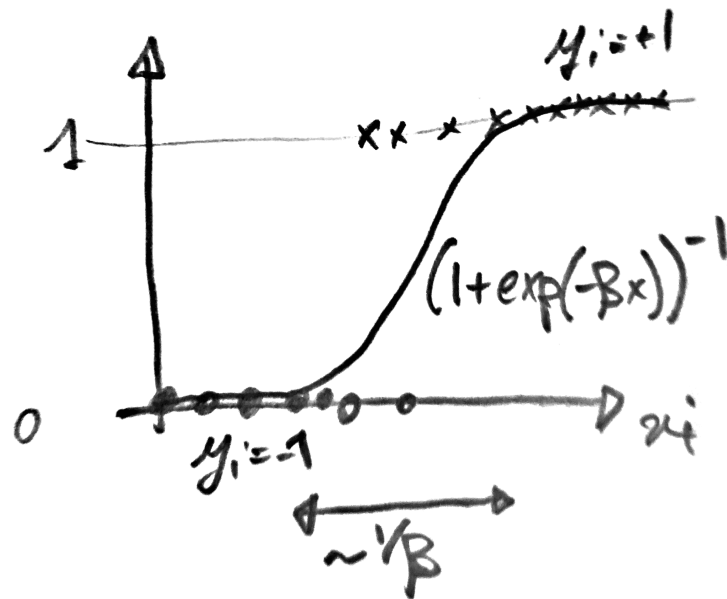
Main book, page 182

Logistic classification in one and two dimensions, showing how $\|\beta\|$ controls the width of the probability transition.

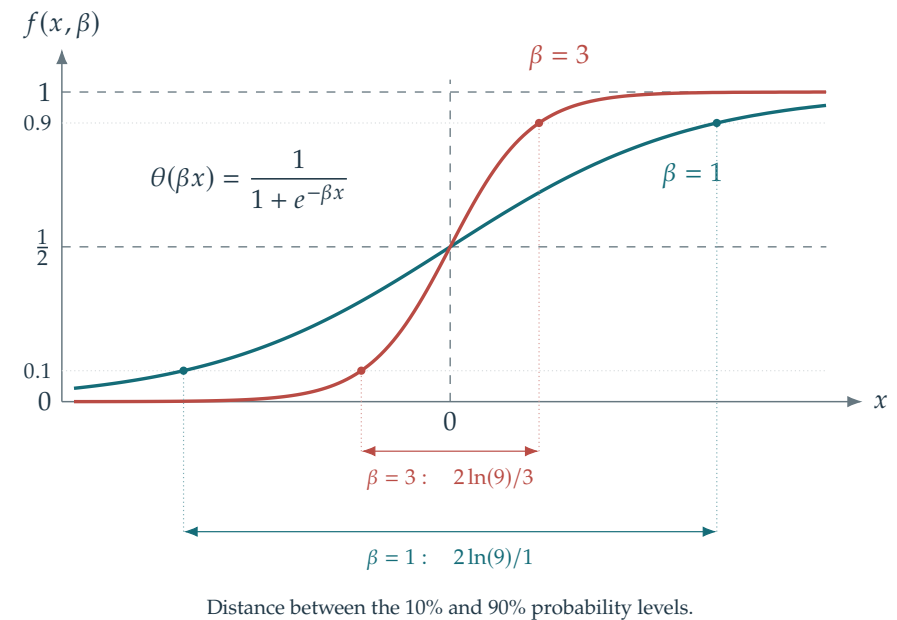
Panel 1 of 2 / Logistic probability and transition width

The probability of class +1 is $\theta(\beta x)$, with $\theta(s) = 1/(1 + \exp(-s))$.

Original



TikZ reconstruction



Reading the reconstruction. The probability is $\theta(\beta x) = 1/(1 + \exp(-\beta x))$, with positive slopes 1 and 3. Both curves pass through probability $1/2$ at $x=0$. Colored dots and guides locate the actual 0.1 and 0.9 crossings, and each dimension arrow measures the corresponding width $2 \log(9)/\beta$. The larger slope has one third of the transition width.

Figure 13.15

Main book, page 182

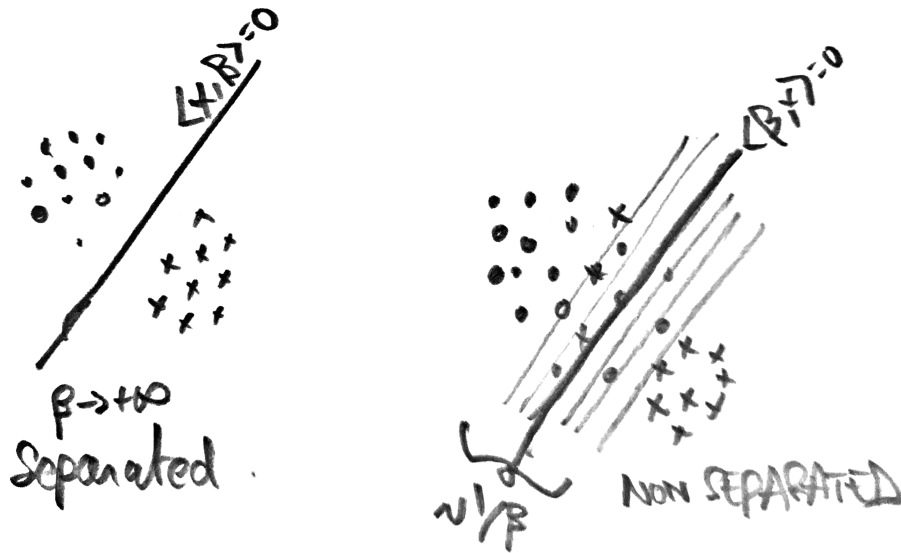
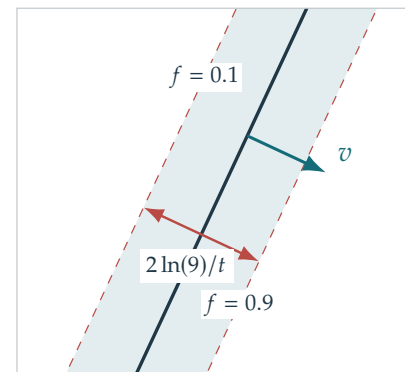
Logistic classification in one and two dimensions, showing how $\|\beta\|$ controls the width of the probability transition.

Panel 2 of 2 / The same decision boundary at two parameter norms

The parameter direction determines a hyperplane normal; its norm controls the width of the logistic probability transition.

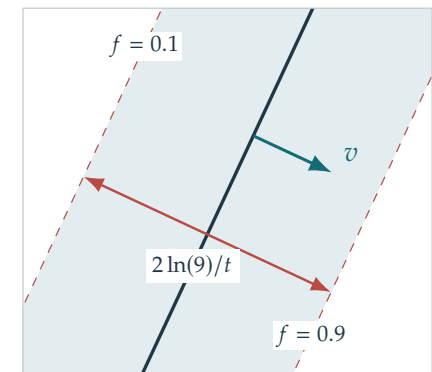
Original

TikZ reconstruction

Larger norm: $\|\beta\| = 3$ 

$$f(x, \beta) = \theta(t \langle v, x \rangle)$$

$$\text{Boundary: } \langle v, x \rangle = 0$$

Smaller norm: $\|\beta\| = 1.4$ 

$$f(x, \beta) = \theta(t \langle v, x \rangle)$$

$$\text{Boundary: } \langle v, x \rangle = 0$$

The width is measured perpendicular to the same decision boundary in both panels.

Reading the reconstruction. Both panels use $\beta = t v$ with the same unit normal v . The red dashed contours are labeled by their probabilities 0.1 and 0.9; the dimension arrows measure their normal separation, exactly $2 \log(9)/t$. The boundary at probability $1/2$ stays fixed. Parameter scaling controls probability sharpness, not whether a data set is linearly separable.

Figure 14.1

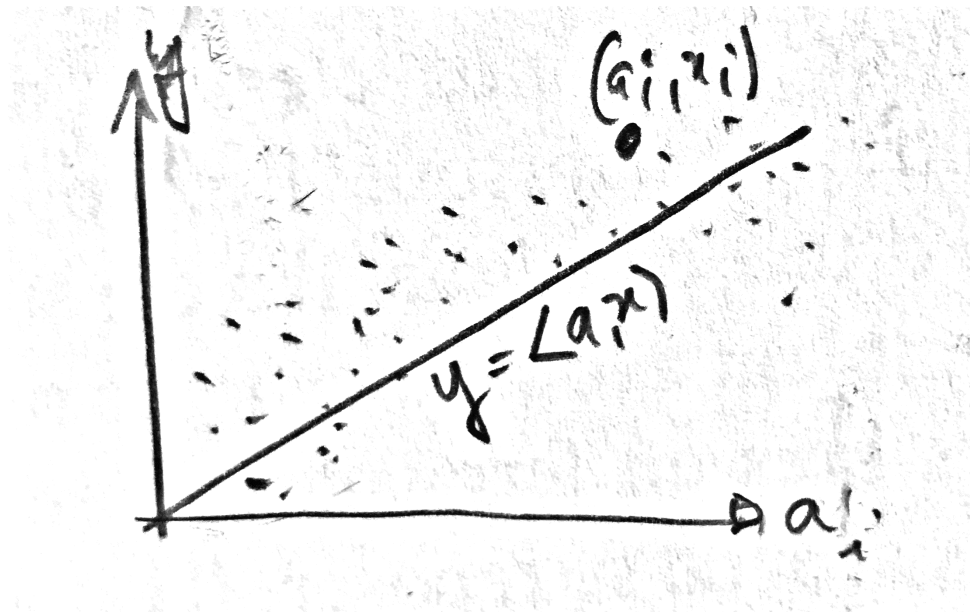
Main book, page 194

Panels 1 and 2: linear regression and a linear classifier. Panel 3: the 0-1 loss and its normalized logistic and hinge upper bounds, both tight at zero margin.

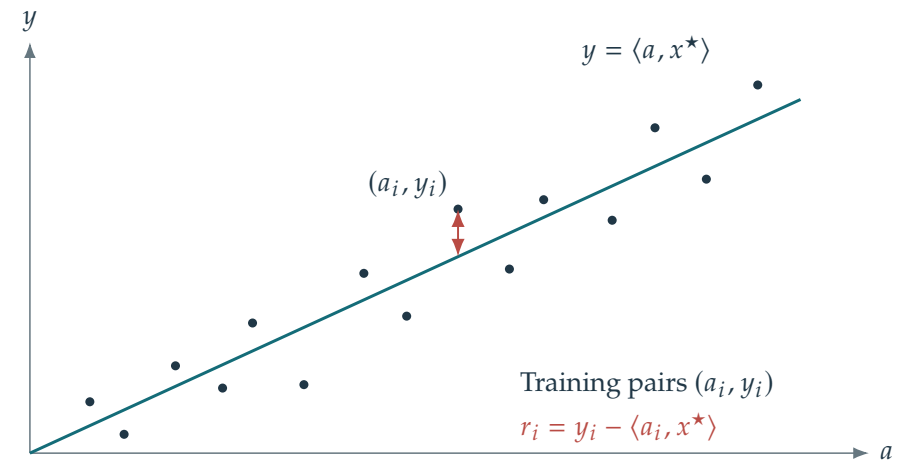
Panel 1 of 3 / Linear regression

Least-squares fitting of a linear predictor to training pairs.

Original



TikZ reconstruction



Reading the reconstruction. The fifteen training pairs form an illustrative scalar-feature dataset. The displayed slope is their exact least-squares minimizer, $x^* = \text{sum}(a_i y_i) / \text{sum}(a_i^2)$, approximately 0.4589392. The red vertical segment is the signed residual $y_i - \langle a_i, x^* \rangle$ for the marked observation; no intercept is fitted.

Figure 14.1

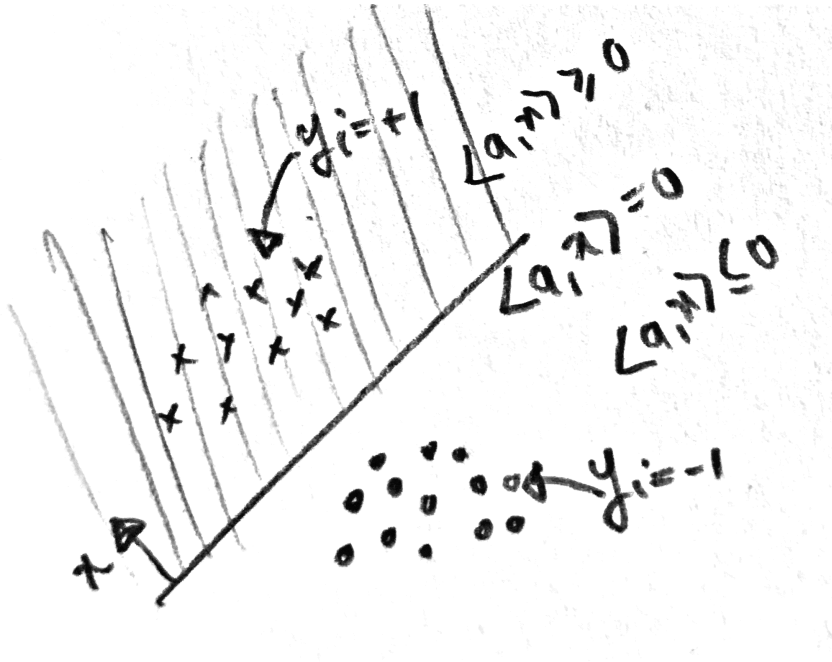
Main book, page 194

Panels 1 and 2: linear regression and a linear classifier. Panel 3: the 0-1 loss and its normalized logistic and hinge upper bounds, both tight at zero margin.

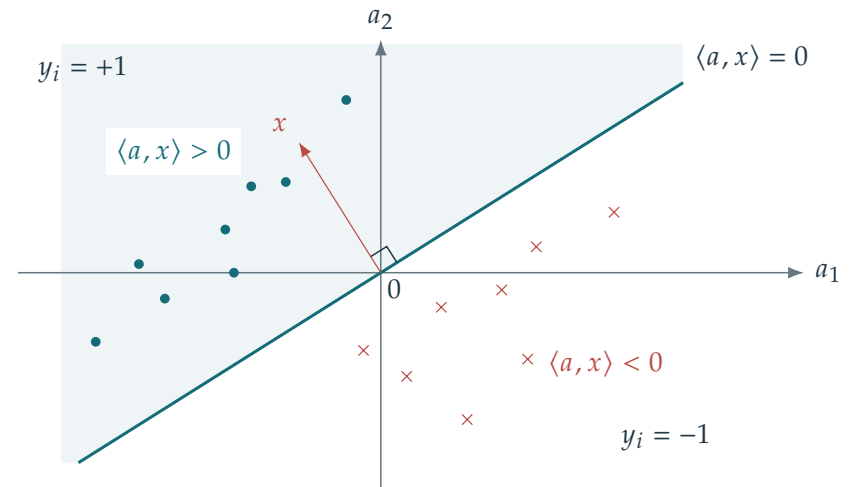
Panel 2 of 3 / A linear classifier

The parameter x is normal to the decision hyperplane in feature space.

Original



TikZ reconstruction



Reading the reconstruction. Positive labels lie on the positive-score side and negative labels on the negative-score side. The normal vector is perpendicular to the separating line, with an explicit right-angle marker. The feature-space origin is marked because this homogeneous decision hyperplane passes through it.

Figure 14.1

Main book, page 194

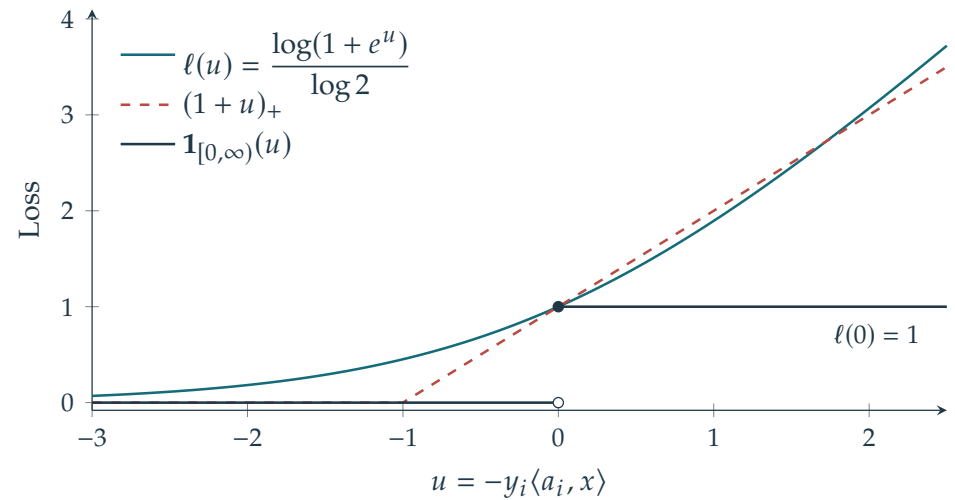
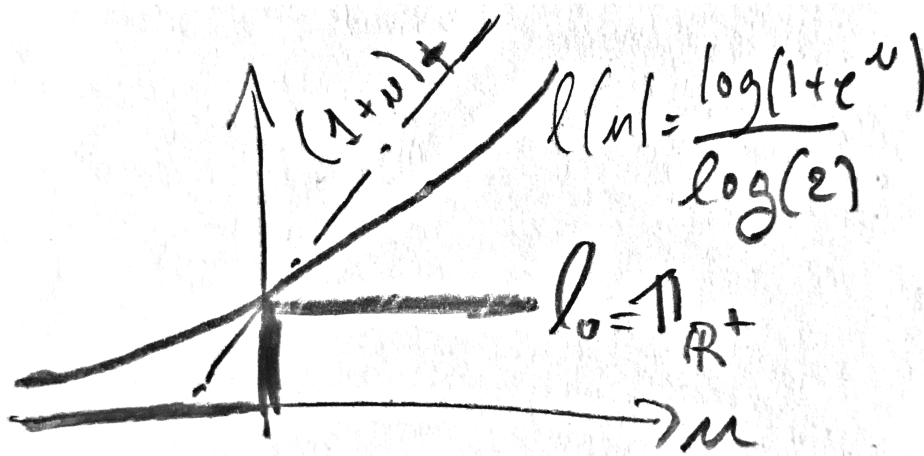
Panels 1 and 2: linear regression and a linear classifier. Panel 3: the 0-1 loss and its normalized logistic and hinge upper bounds, both tight at zero margin.

Panel 3 of 3 / Classification losses

Smooth empirical classification risk and standard margin losses.

Original

TikZ reconstruction



Reading the reconstruction. Divided the logistic loss by $\log(2)$, giving $\ell(u) = \log(1 + \exp(u)) / \log(2)$. This is an upper bound on the displayed zero-one loss and meets it at $u=0$, where both equal one. Zero margin counts as an error. The hinge loss also equals one there. Updated the chapter loss definition and its derivative consistently; the derivative is the sigmoid divided by $\log(2)$.

Figure 14.2

Main book, page 195

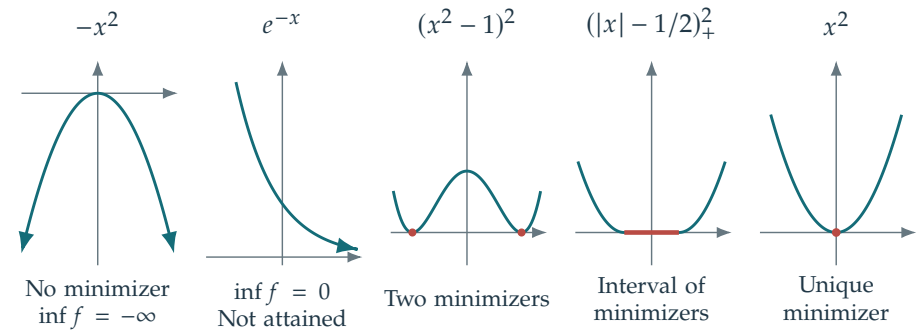
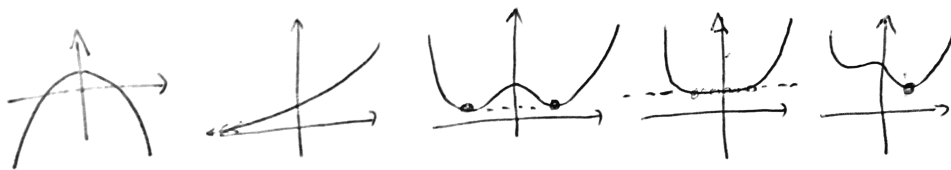
Left: no minimizer. Middle: multiple minimizers. Right: a unique minimizer.

Complete figure / Existence and uniqueness of minimizers

An objective can have no minimizer, several minimizers, or a unique minimizer.

Original

TikZ reconstruction



Reading the reconstruction. The five representative functions retain the original cases. The two no-minimizer examples explicitly distinguish an unbounded infimum, minus infinity, from the finite unattained infimum zero. The exponential curve stays strictly above its horizontal asymptote. Vertical offsets and coordinate scales are schematic.

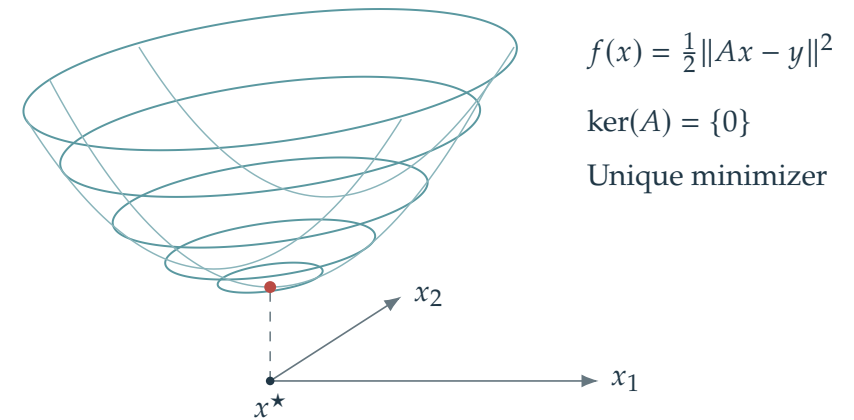
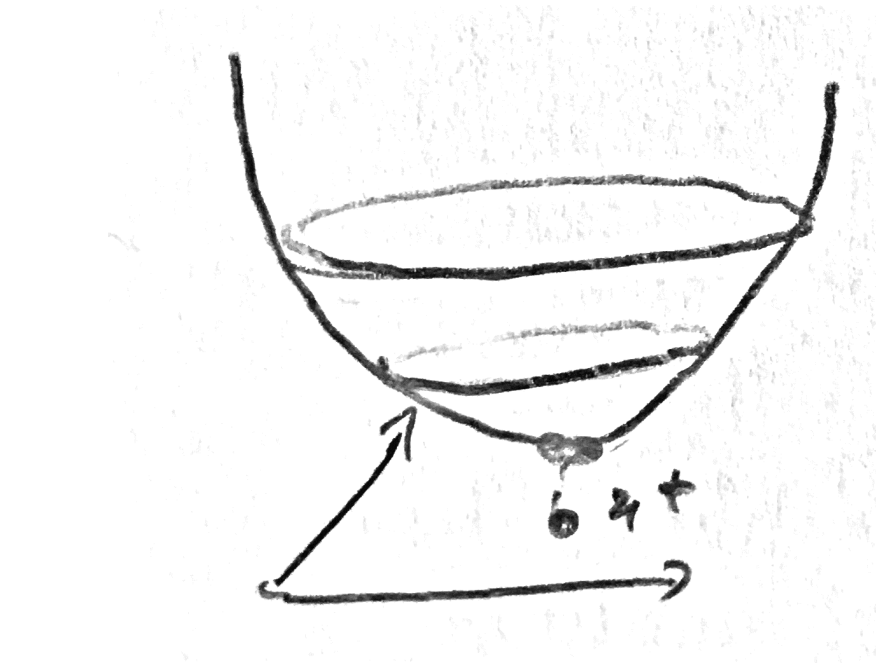
Figure 14.3

Main book, page 196

Coercivity condition for least squares.

Panel 1 of 2 / Coercive least squares

Full column rank gives a strictly convex quadratic with a unique minimizer.

Original**TikZ reconstruction**

Reading the reconstruction. The bowl depicts the full-column-rank case. Its height may include a nonzero residual at the minimizer; no exact fit is assumed.

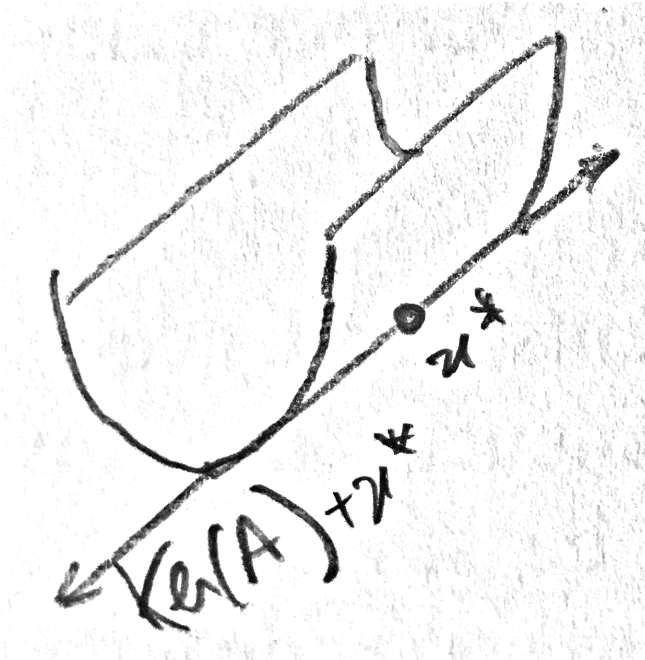
Figure 14.3

Main book, page 196

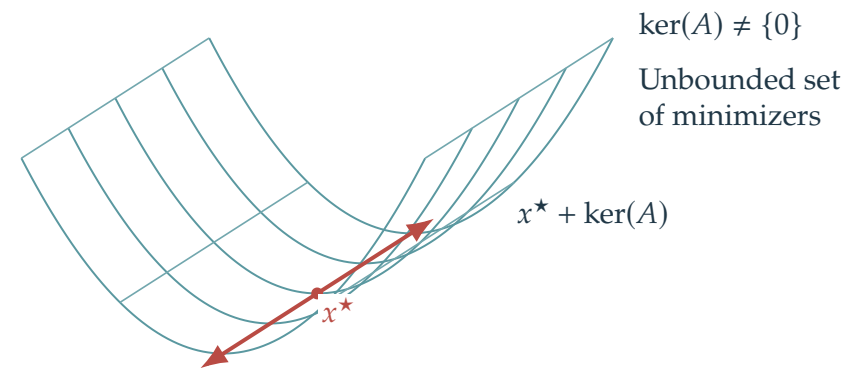
Coercivity condition for least squares.

Panel 2 of 2 / Least squares with a nontrivial kernelThe objective is constant along translations by vectors in the kernel of A .

Original



TikZ reconstruction



Reading the reconstruction. The trough is flat exactly along the kernel direction. The highlighted affine line represents $x^* + \ker(A)$; arrowheads at both ends show that this entire minimizer set is unbounded in both directions.

Figure 14.4

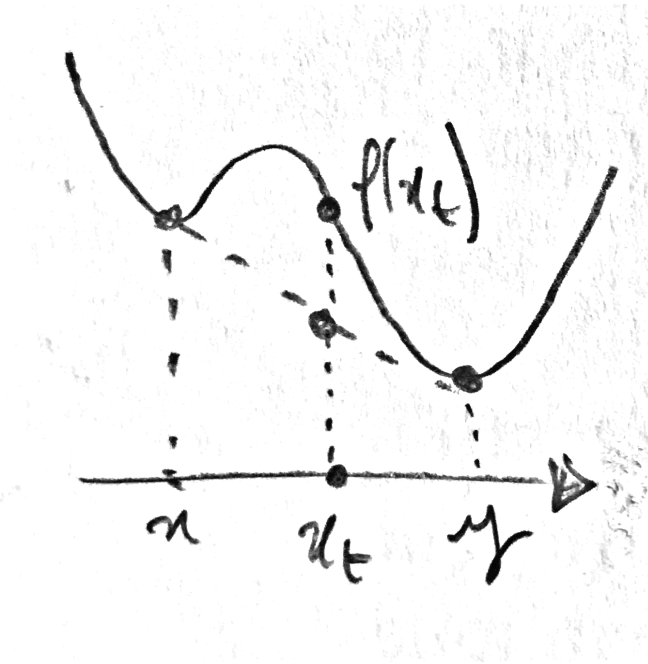
Main book, page 197

Convex and nonconvex functions; strictly convex and convex but not strictly convex functions.

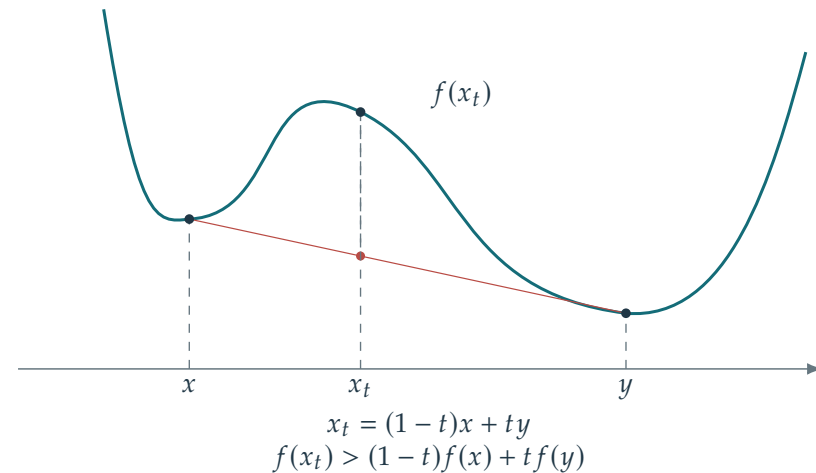
Panel 1 of 4 / Failure of the secant inequality

The graph of a convex function lies below every secant.

Original



TikZ reconstruction



Reading the reconstruction. The marked graph value is greater than the secant value at the same interior abscissa, explicitly violating convexity. The two endpoint values and the interpolated value are evaluated consistently on the drawn curve and secant.

Figure 14.4

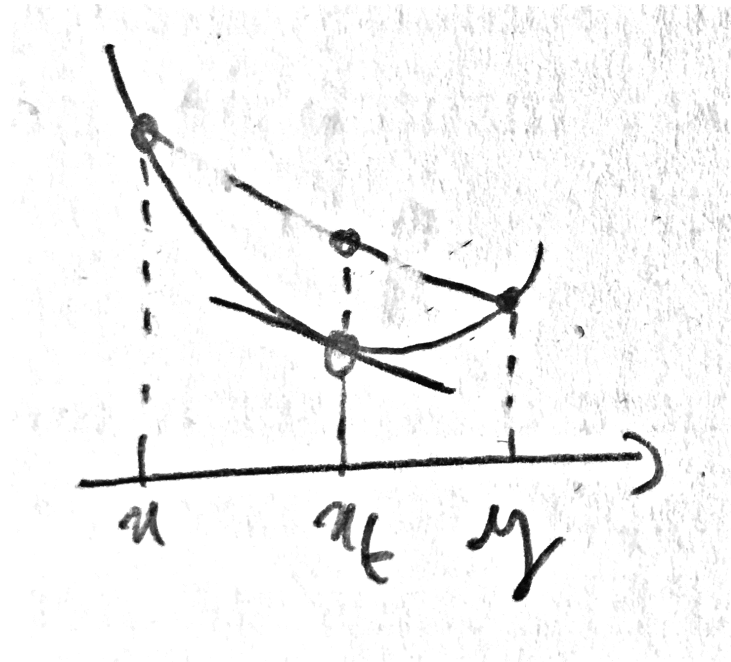
Main book, page 197

Convex and nonconvex functions; strictly convex and convex but not strictly convex functions.

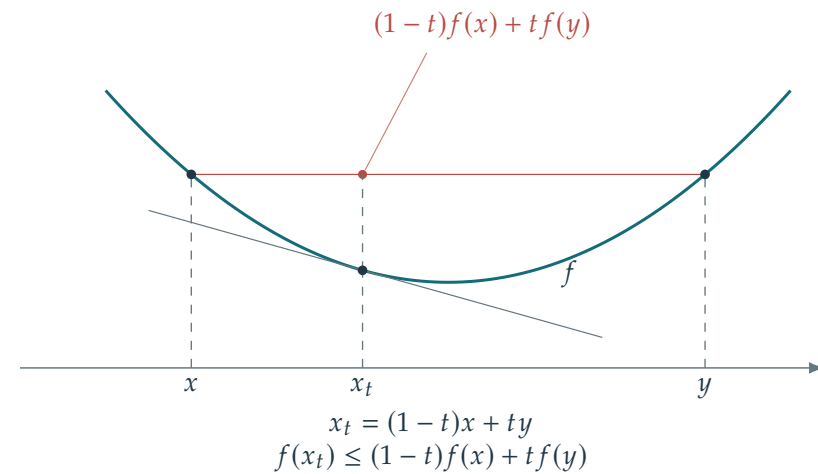
Panel 2 of 4 / Convexity and the secant inequality

The graph of a convex function lies below every secant.

Original



TikZ reconstruction



Reading the reconstruction. The marked graph value lies below the secant value at the same interior abscissa. A leader identifies the interpolated endpoint value, and the displayed inequality matches the chapter definition. The supporting line is the exact tangent to the representative quadratic at the marked graph point.

Figure 14.4

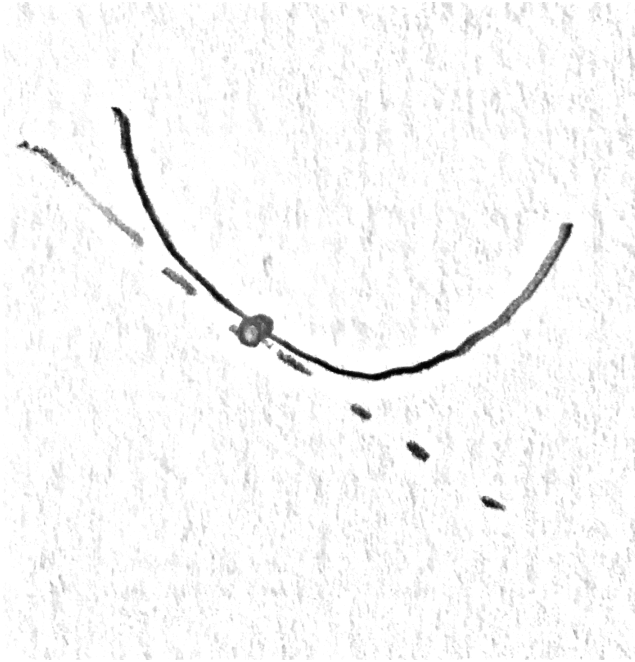
Main book, page 198

Convex and nonconvex functions; strictly convex and convex but not strictly convex functions.

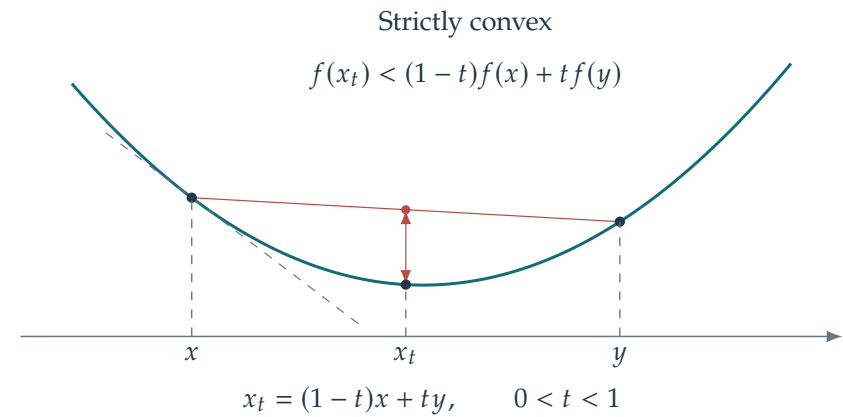
Panel 3 of 4 / Strict convexity

For distinct endpoints and an interior convex combination, the secant inequality is strict.

Original



TikZ reconstruction



Reading the reconstruction. An interior midpoint is marked on the graph and on the secant, with a positive vertical gap. For the drawn quadratic, this gap is exactly 0.875. The strict inequality and the condition $0 < t < 1$ are stated explicitly.

Figure 14.4

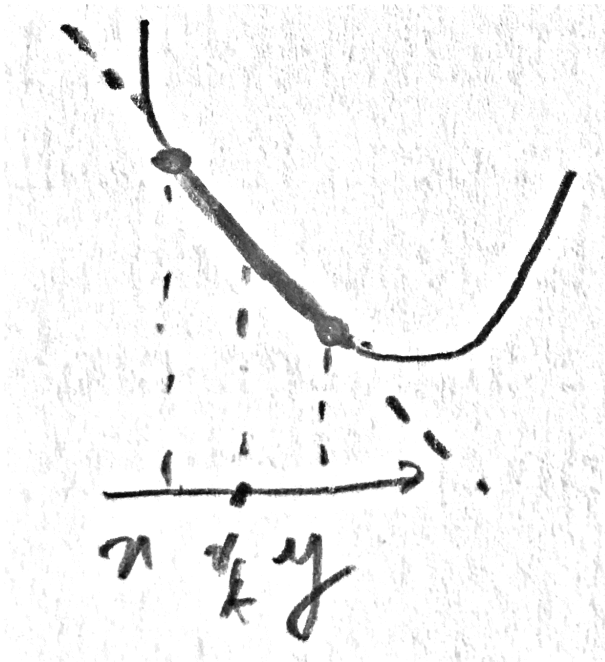
Main book, page 198

Convex and nonconvex functions; strictly convex and convex but not strictly convex functions.

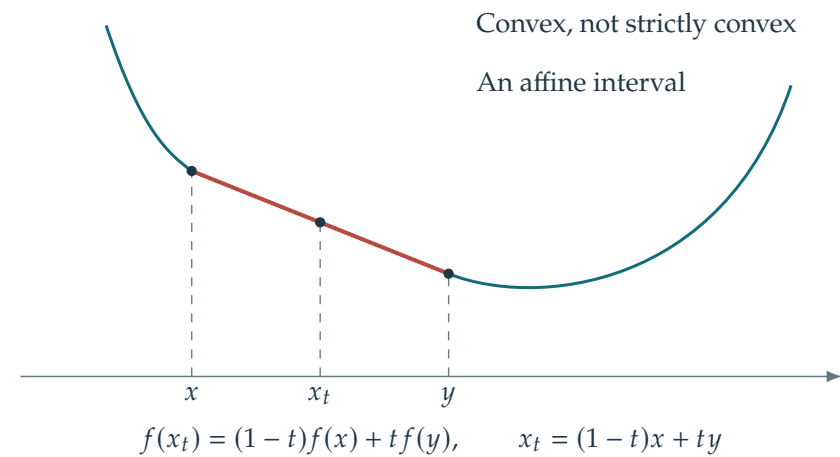
Panel 4 of 4 / Convexity with an affine interval

A convex function need not be strictly convex.

Original



TikZ reconstruction



Reading the reconstruction. The highlighted interval is exactly affine, so equality holds in the displayed secant identity. The surrounding curve has nondecreasing slope, including its joins with the affine interval, and is therefore convex.

Figure 14.5

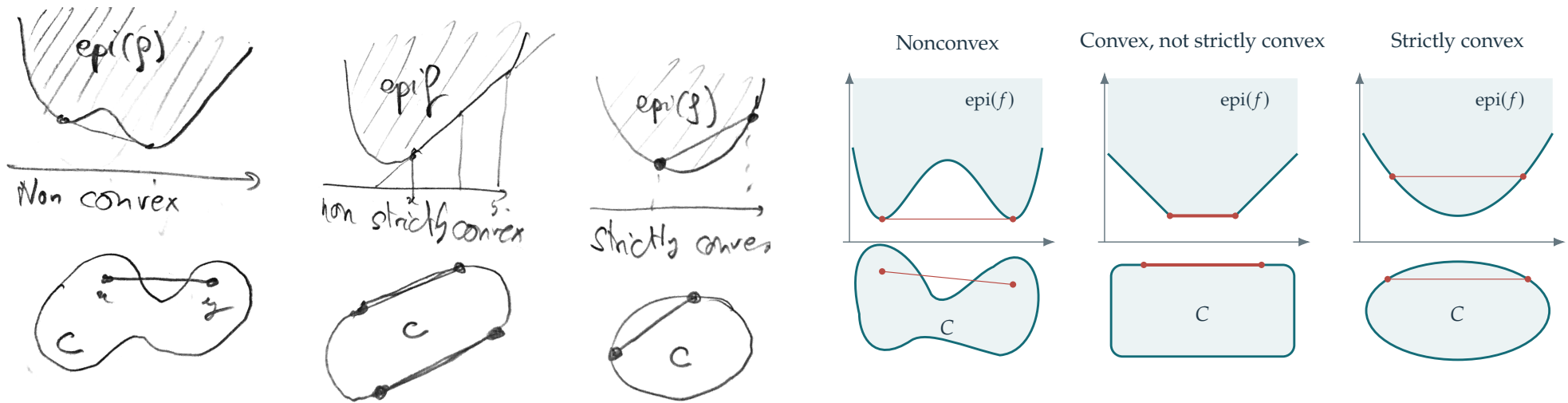
Comparison of convex functions $f : \mathbb{R}^p \rightarrow \mathbb{R}$ (for $p = 1$) and convex sets $C \subset \mathbb{R}^p$ (for $p = 2$).

Complete figure / Convexity of functions and sets

Convexity and convex sets: secants lie above a convex graph; line segments remain in a convex set.

Original

TikZ reconstruction



Reading the reconstruction. Red segments distinguish failure of convexity, non-strict convexity with boundary segments, and strict convexity. The middle function has a flat interval, and the middle set has flat boundary segments.

Figure 14.6

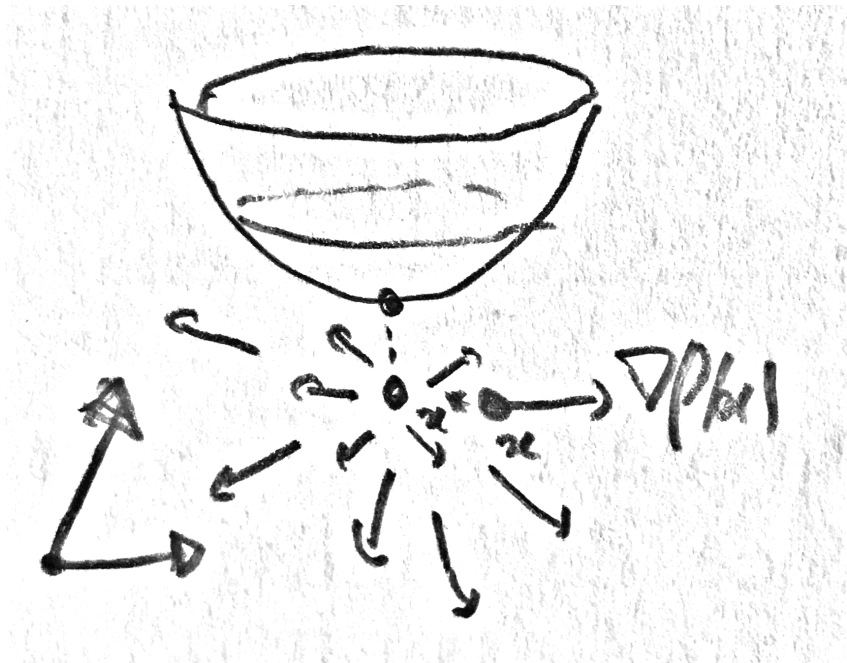
Main book, page 199

The gradient as a vector field.

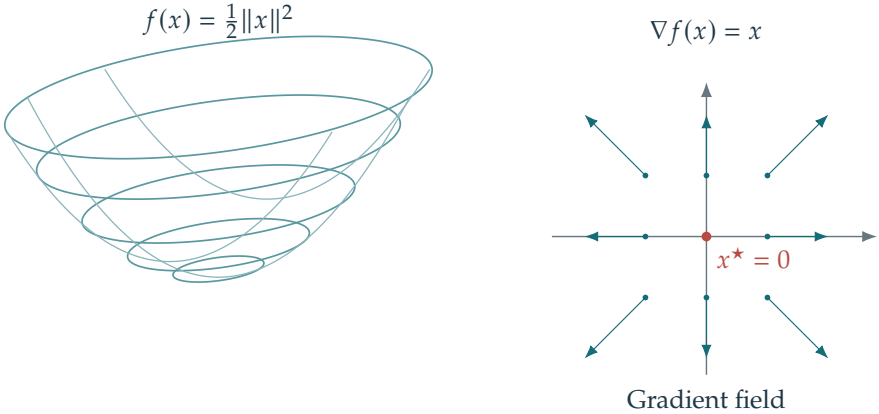
Complete figure / The gradient is a vector field

The gradient associates a vector in the domain with each evaluation point.

Original



TikZ reconstruction



Reading the reconstruction. The radial field is shown for $f(x) = \|x\|^2/2$. Each arrow based at a marked point q has displacement exactly q in the same coordinates, hence ends at $2q$. The gradient vanishes at the minimizer. The accompanying bowl uses an uncalibrated projection.

Figure 14.7

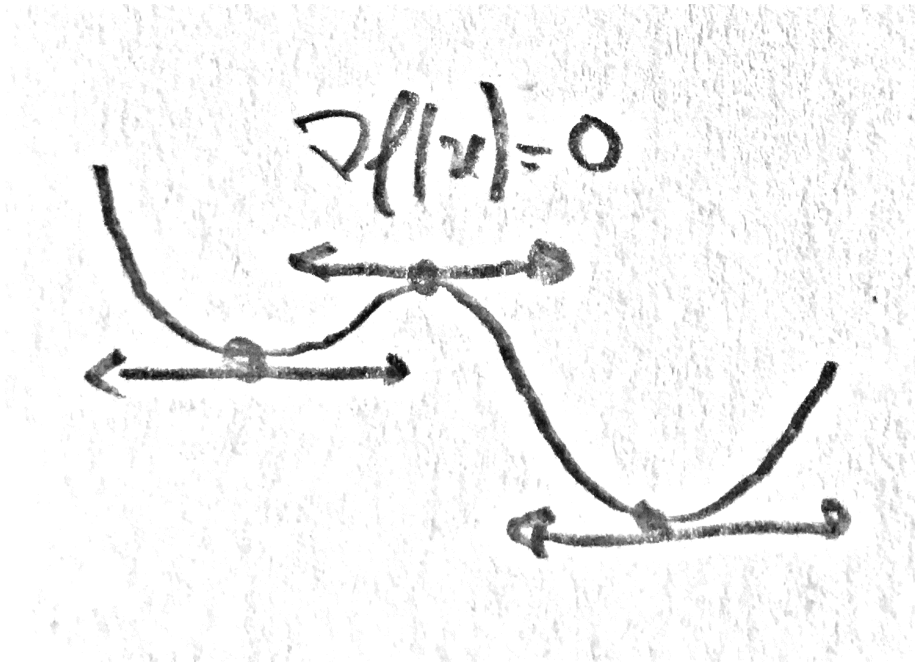
Main book, page 201

Local extrema (panel 1), a stationary inflection point (panel 2), and a global minimizer (panel 3).

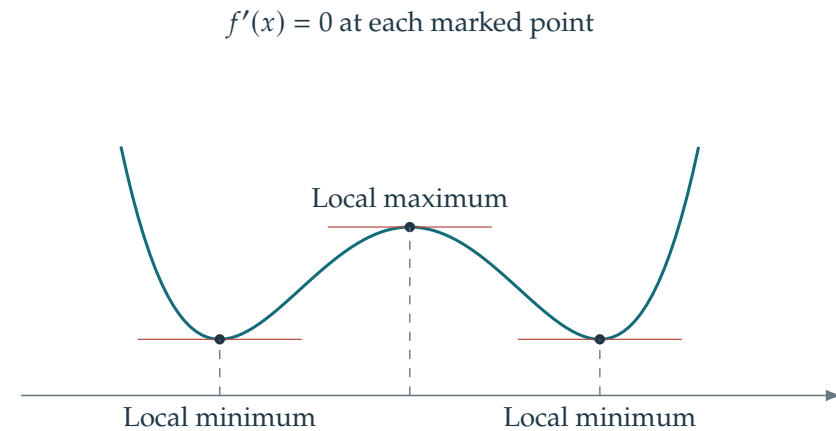
Panel 1 of 3 / Stationarity at local extrema

Vanishing derivative is necessary at differentiable local minima and maxima.

Original



TikZ reconstruction



Reading the reconstruction. The representative double-well function has two minima and a maximum, all with horizontal tangents. Stationarity alone does not distinguish these cases.

Figure 14.7

Main book, page 201

Local extrema (panel 1), a stationary inflection point (panel 2), and a global minimizer (panel 3).

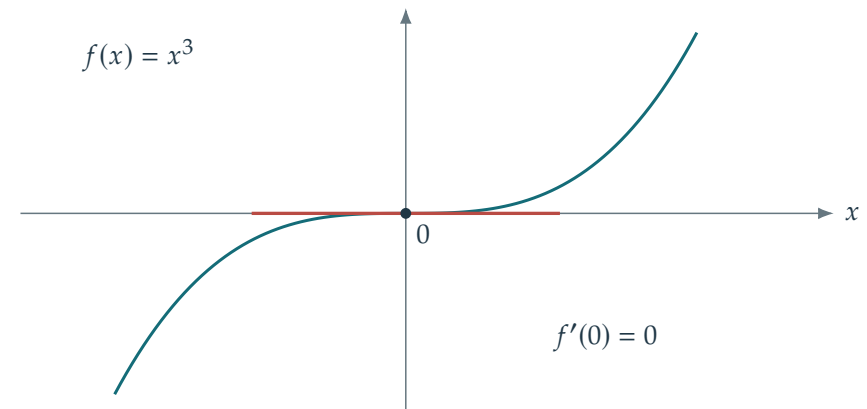
Panel 2 of 3 / A stationary point need not be an extremum

The first-order necessary condition is not sufficient without convexity.

Original



TikZ reconstruction



Stationary inflection point: neither a minimum nor a maximum

Reading the reconstruction. The reconstruction uses $f(x)=x$ cubed as in the accompanying text and identifies the origin as a stationary inflection point, which is neither a minimum nor a maximum.

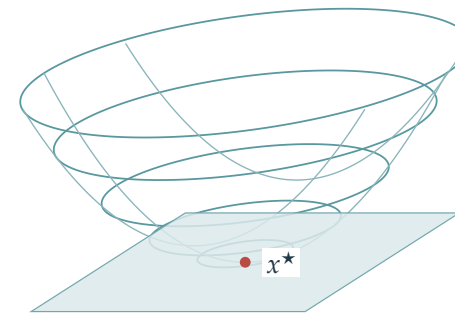
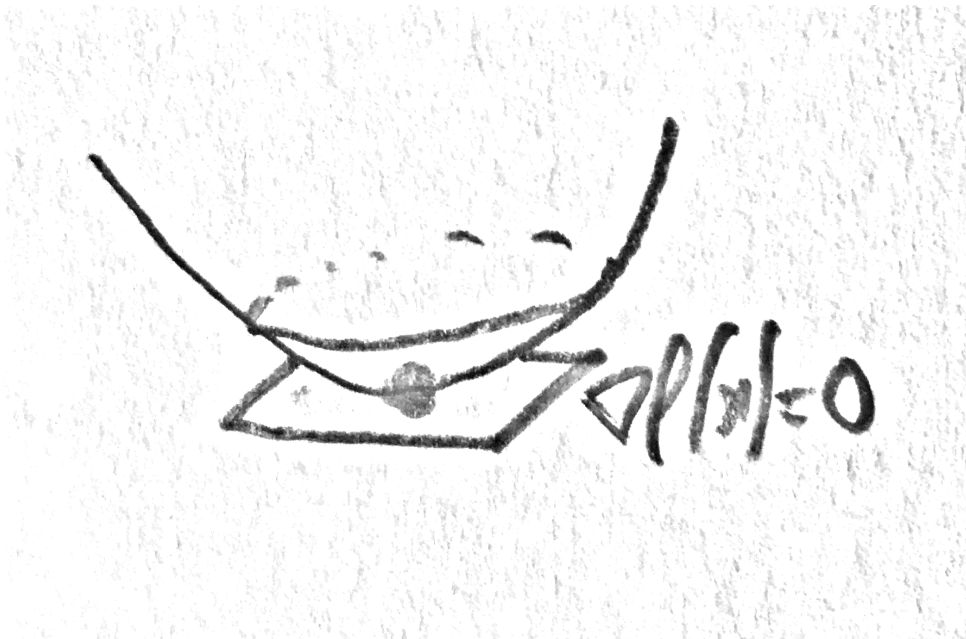
Figure 14.7

Main book, page 201

Local extrema (panel 1), a stationary inflection point (panel 2), and a global minimizer (panel 3).

Panel 3 of 3 / Stationarity for a convex objective

For differentiable convex functions, a vanishing gradient characterizes global minimizers.

Original**TikZ reconstruction**

$$\nabla f(x^*) = 0$$

A global minimizer

Horizontal supporting plane

Reading the reconstruction. The tangent plane touches the bowl at its minimum and supports the graph from below. The plane is horizontal in the depicted three-dimensional coordinates.

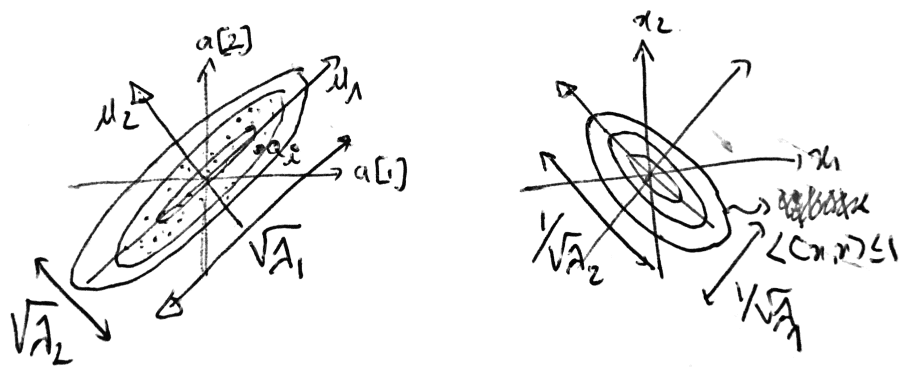
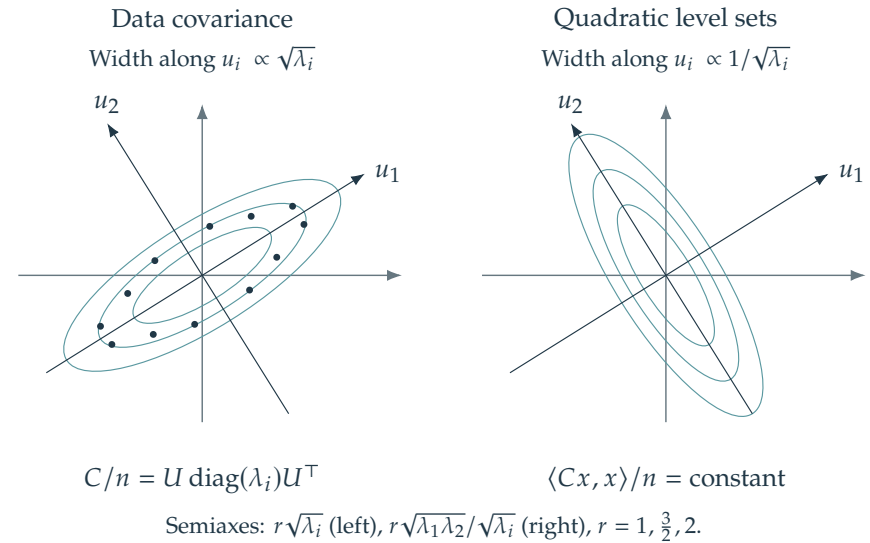
Figure 14.8

Main book, page 203

Left: point clouds $(a_i)_i$ and their principal directions. Right: the quadratic part of $f(x)$.

Complete figure / PCA and least-squares level sets

Covariance axes and quadratic level-set axes have reciprocal widths.

Original**TikZ reconstruction**

Reading the reconstruction. Both panels share the eigenvectors u_1, u_2 . The twelve reflection-symmetric points are centered and have empirical covariance eigenvalues $\lambda_1=1$ and $\lambda_2=0.325$ squared. Every covariance ellipse and quadratic level set is derived from these same eigenvalues, with reciprocal axis ratios. Width annotations sit above the panels so all data points and complete ellipses remain visible. The caption specifies the multiplier used for each family of semiaxes; C/n follows the covariance normalization in the text.

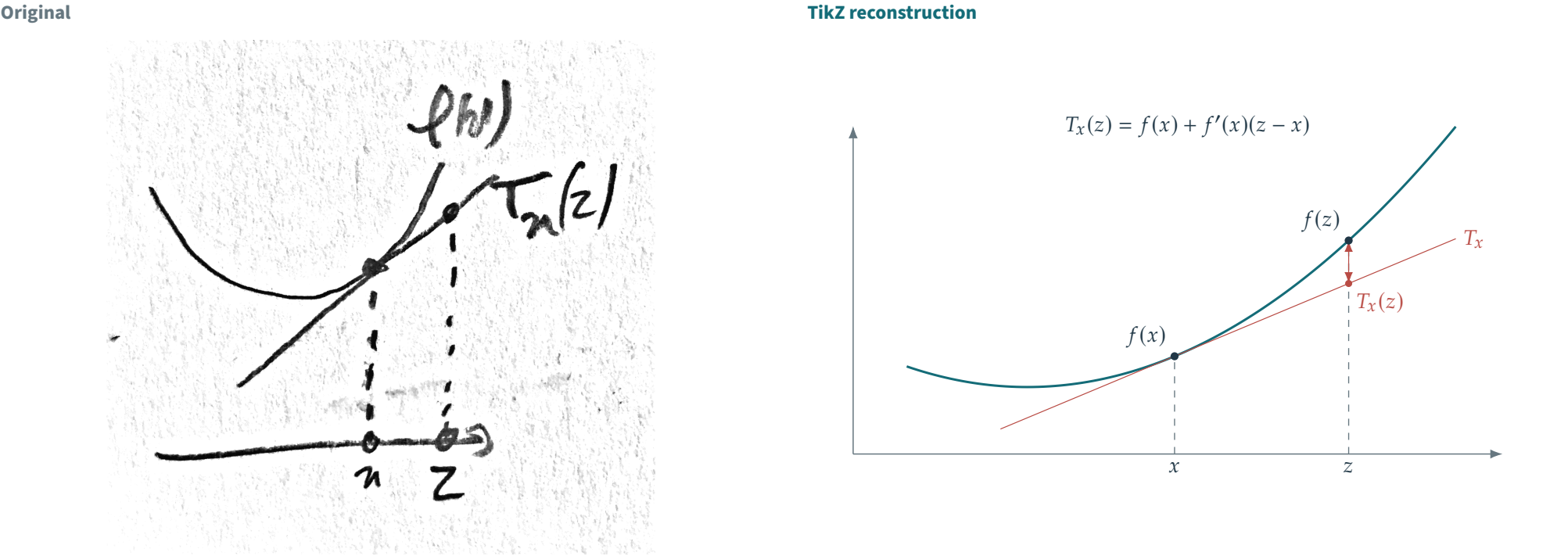
Figure 14.9

Main book, page 205

Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

Panel 1 of 4 / A first-order Taylor approximation

The affine approximation matches the function and its derivative at the expansion point.



Reading the reconstruction. The line is the exact tangent to the representative quadratic at x . Both $f(z)$ and $T_x(z)$ are labeled at their respective marked points, making the vertical Taylor remainder explicit. The tangent value at the selected z is 2.5466.

Figure 14.9

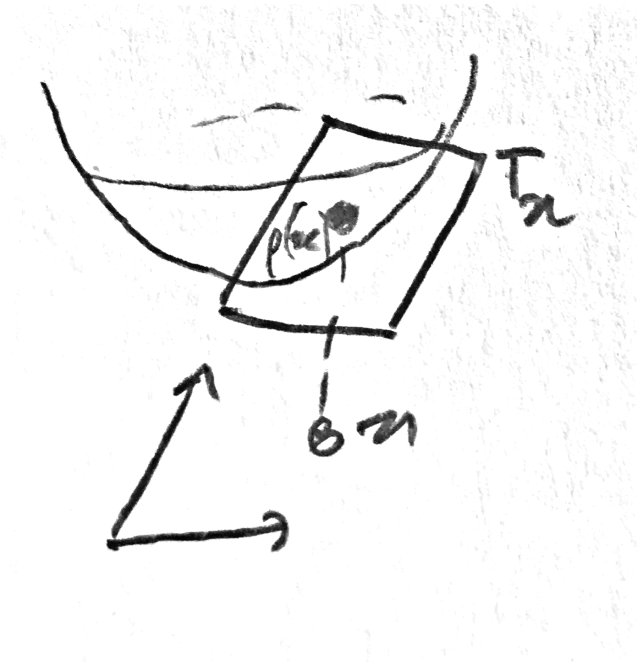
Main book, page 205

Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

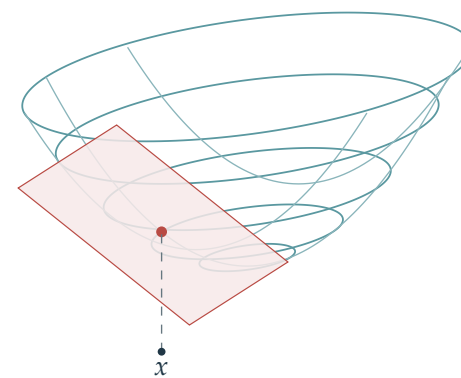
Panel 2 of 4 / A tangent plane

The first-order affine approximation of a differentiable function in two dimensions.

Original



TikZ reconstruction

 $T_x(z)$

Tangent plane

$$f(x) + \langle \nabla f(x), z - x \rangle$$

Reading the reconstruction. The tangent plane is computed for the displayed representative paraboloid and touches it at the marked point. The graph and its supporting plane are distinct objects.

Figure 14.9

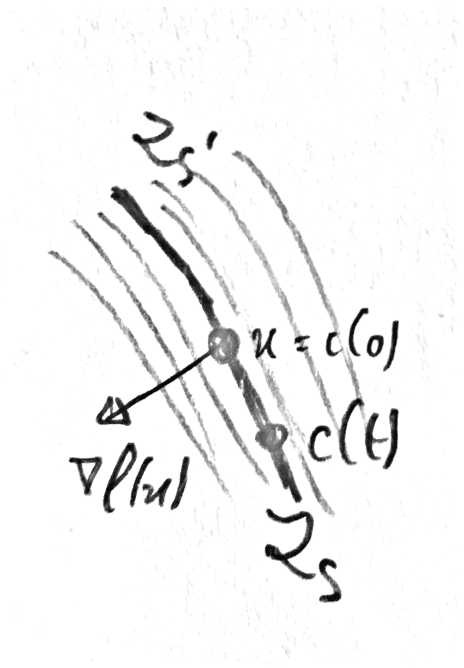
Main book, page 206

Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

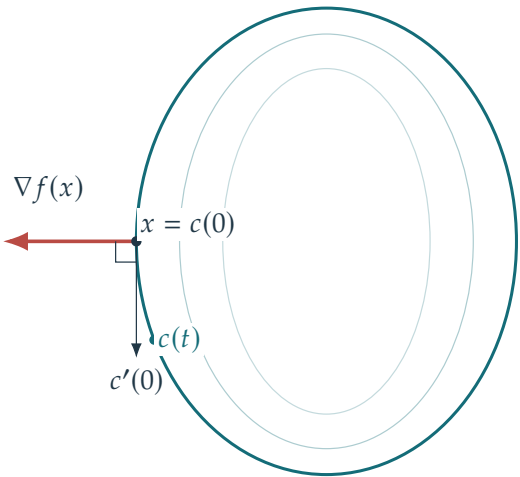
Panel 3 of 4 / The gradient is orthogonal to a level curve

Differentiating $f(c(t))=\text{constant}$ gives a zero pairing with the tangent direction.

Original



TikZ reconstruction



$$\begin{aligned} c(t) &\in \mathcal{L}_s \\ f(c(t)) &= s \\ \langle \nabla f(x), c'(0) \rangle &= 0 \end{aligned}$$

Reading the reconstruction. At the leftmost point of an elliptical level curve, the tangent is vertical and the outward gradient is horizontal. The right-angle marker records orthogonality, not a descent step.

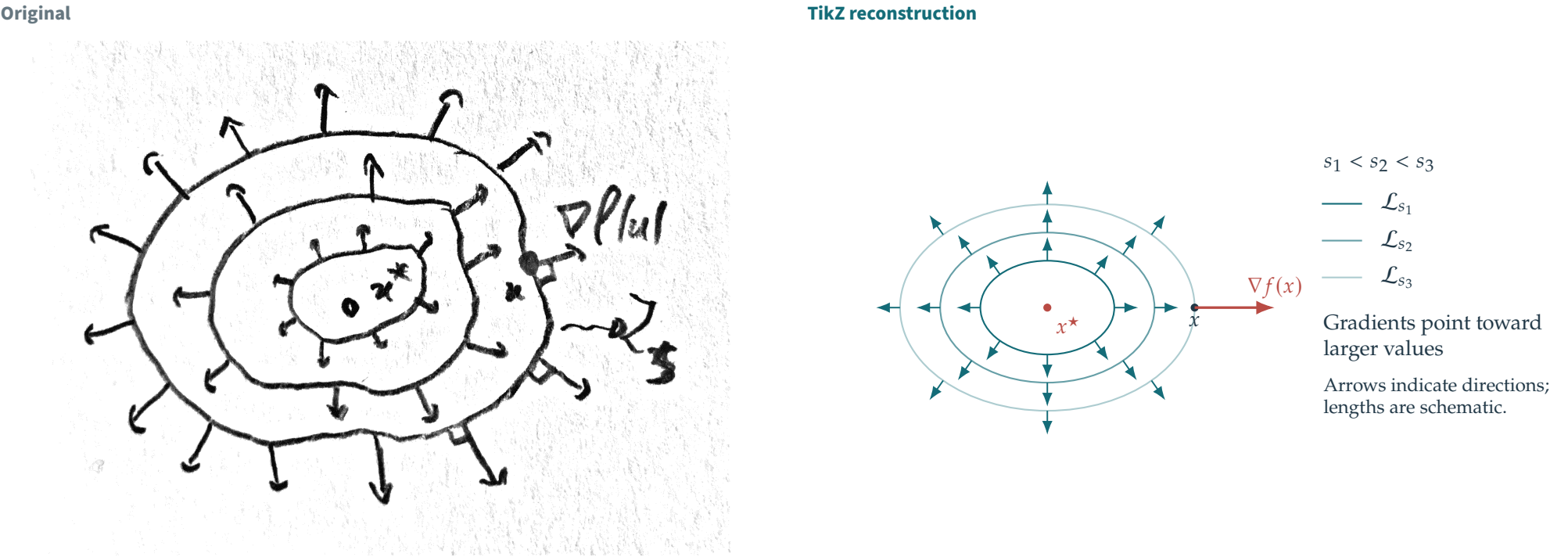
Figure 14.9

Main book, page 206

Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

Panel 4 of 4 / Gradients and nested level sets

Gradients point in the direction normal to a level set and toward larger values.



Reading the reconstruction. The arrows are normals to elliptical level curves, not radial arrows except on the principal axes. Level values increase outward. A separate key identifies the three complete ellipses, and the drawing explicitly states that arrow lengths are schematic: their directions are the mathematical content.

Figure 14.10

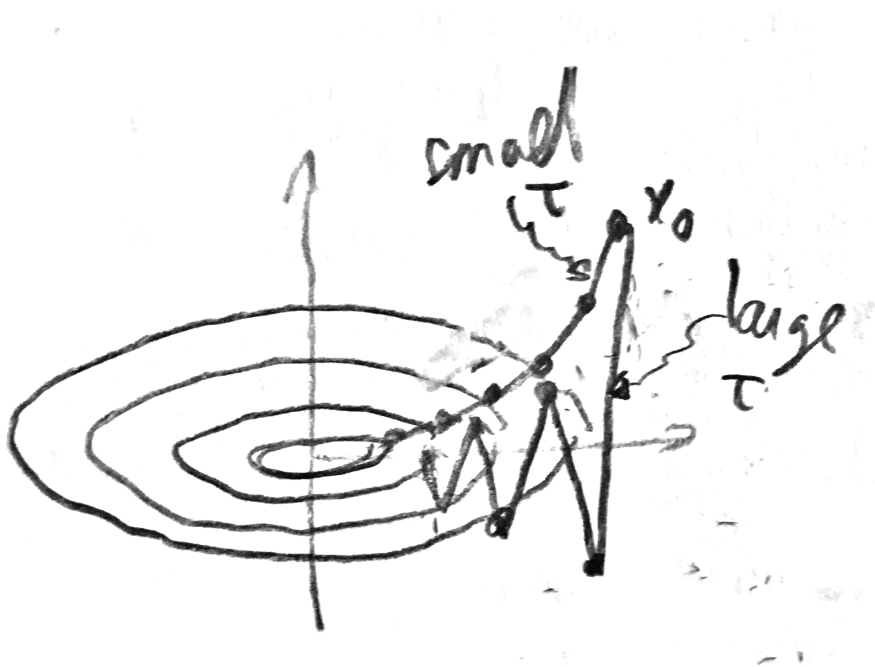
Main book, page 207

Effect of the step size τ on gradient descent (panel 1) and exact line search (panel 2).

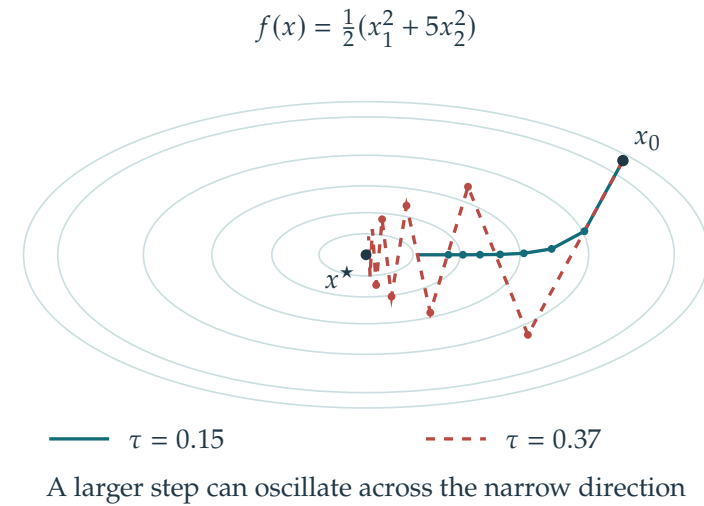
Panel 1 of 2 / Step-size effects in gradient descent

The step size controls progress and oscillation along a quadratic objective.

Original



TikZ reconstruction



Reading the reconstruction. Both trajectories are exact gradient-descent iterates for the displayed quadratic. The larger step oscillates but remains stable because both step sizes lie in $(0, 2/L)$, with $L=5$. The sketch is schematic; no numerical values were legible in it.

Figure 14.10

Main book, page 207

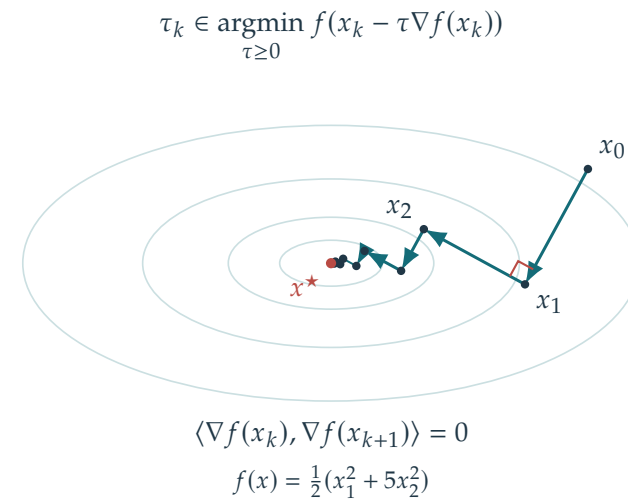
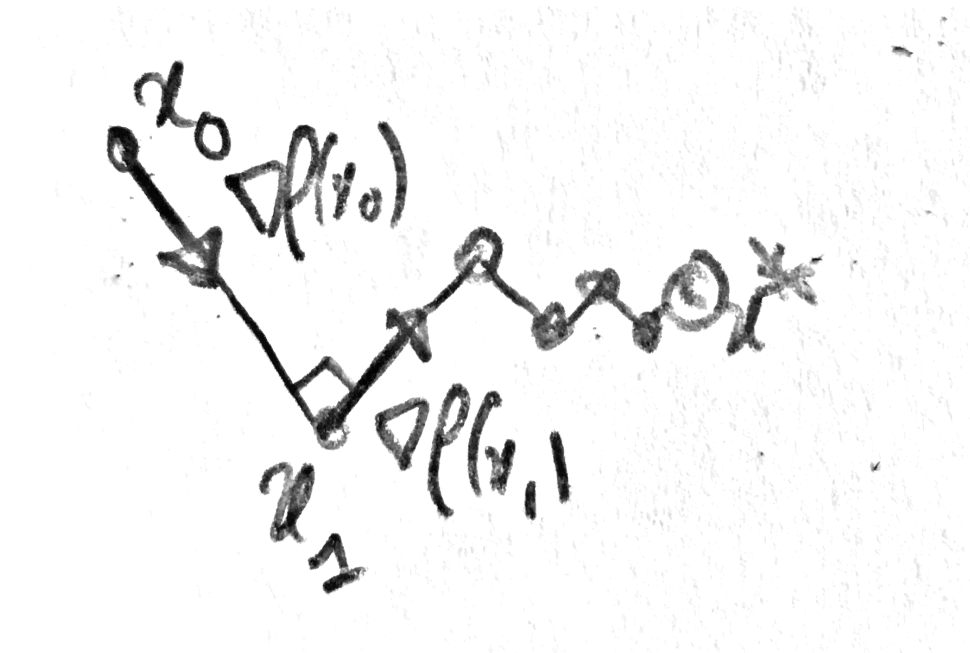
Effect of the step size τ on gradient descent (panel 1) and exact line search (panel 2).

Panel 2 of 2 / Exact line search

A differentiable interior line-search optimum makes successive gradients orthogonal.

Original

TikZ reconstruction



Reading the reconstruction. The path uses exact line search for the displayed positive definite quadratic $f(x) = (x_1^2 + 5x_2^2)/2$. The right-angle marker is computed from consecutive actual steps. The later, shorter steps remain connected as the path approaches the minimizer.

Figure 14.11

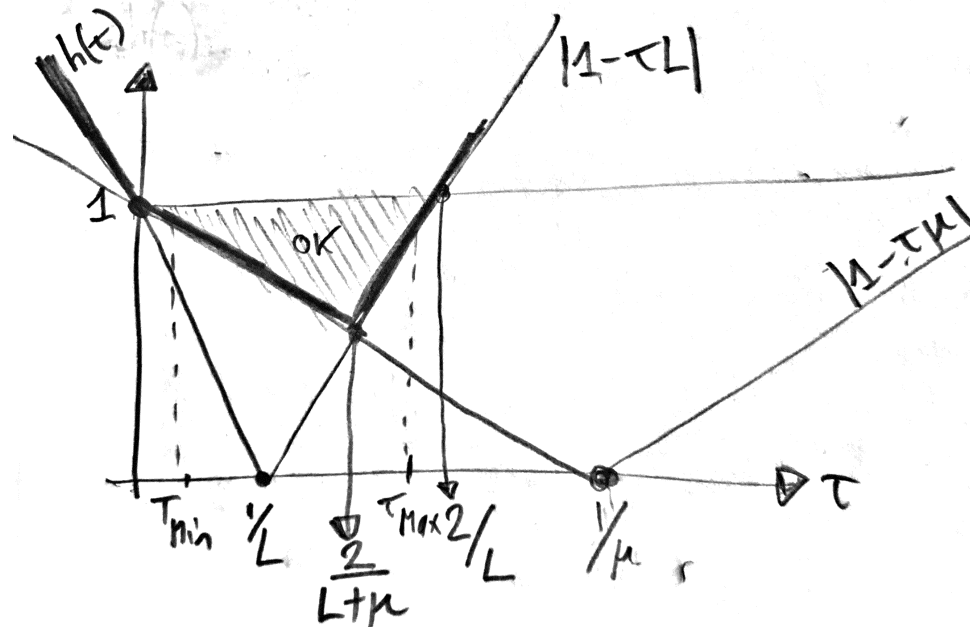
Main book, page 209

Contraction constant $h(\tau)$ for a quadratic function.

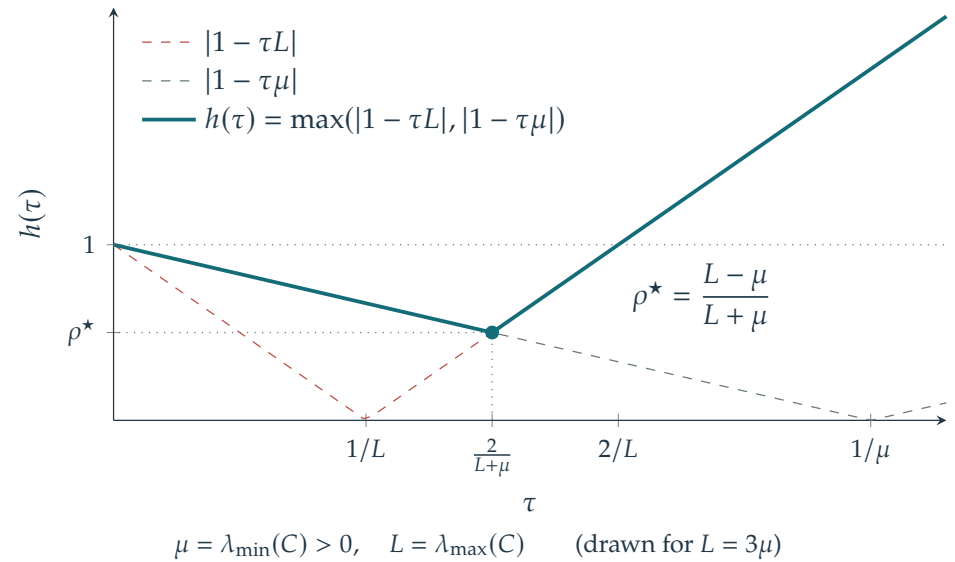
Complete figure / The optimal quadratic step size

The operator norm of $I - \tau C$ is controlled by the extreme eigenvalues.

Original



TikZ reconstruction



Reading the reconstruction. The highlighted curve is the maximum of the two absolute-value branches, with μ and L explicitly identified as the smallest and largest eigenvalues of C . Its minimum is at $2/(L+\mu)$, and $h(\tau) < 1$ exactly for $0 < \tau < 2/L$. The drawing states its representative ratio $L/\mu=3$; other labels remain symbolic.

Figure 14.12

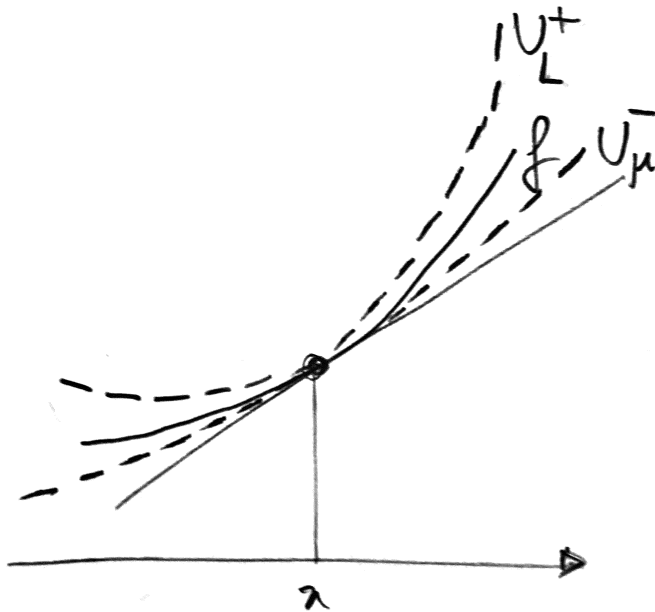
Main book, page 212

Quadratic upper and lower bounds for a smooth strongly convex function.

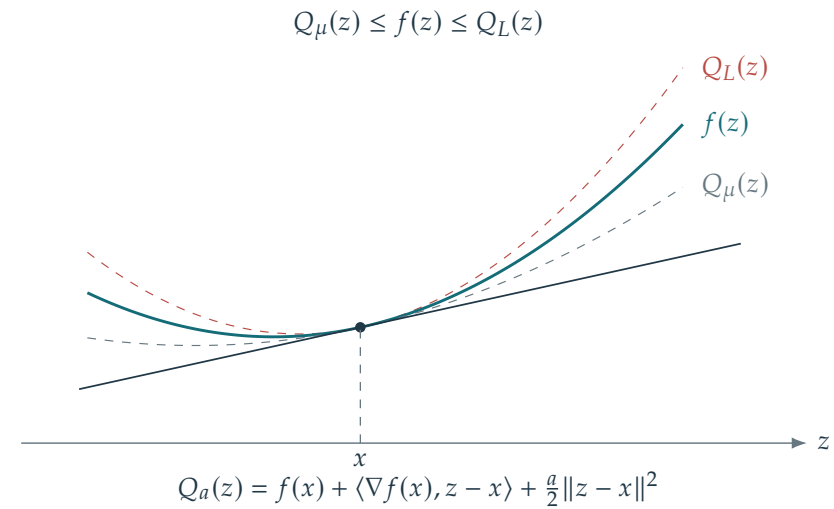
Complete figure / Quadratic upper and lower bounds

Strong convexity and Lipschitz gradient give touching lower and upper quadratic models.

Original



TikZ reconstruction



Reading the reconstruction. Both bounding quadratics have the same value and gradient as f at x . The lower curvature μ and upper curvature L are ordered correctly on both sides of the expansion point.

Figure 15.4

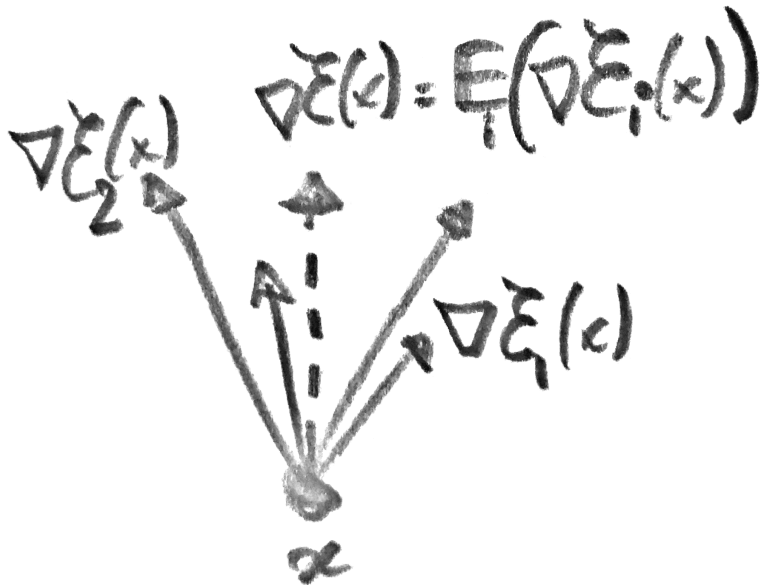
Main book, page 221

Unbiased gradient estimate

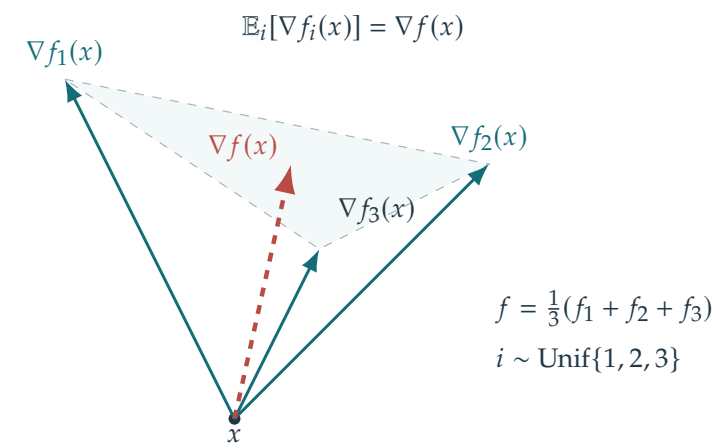
Complete figure / An unbiased stochastic gradient

Uniform sampling of a summand gives the gradient of its arithmetic mean in expectation.

Original



TikZ reconstruction



Reading the reconstruction. The red dashed vector is the exact arithmetic mean of the three individual gradient vectors. Its endpoint is the centroid of the lightly shaded triangle joining their endpoints. Uniform sampling from $\{1, 2, 3\}$ is stated explicitly, which is required for this unbiased estimate of the arithmetic-mean objective.

Figure 15.5

Main book, page 221

Schematic view of SGD iterates

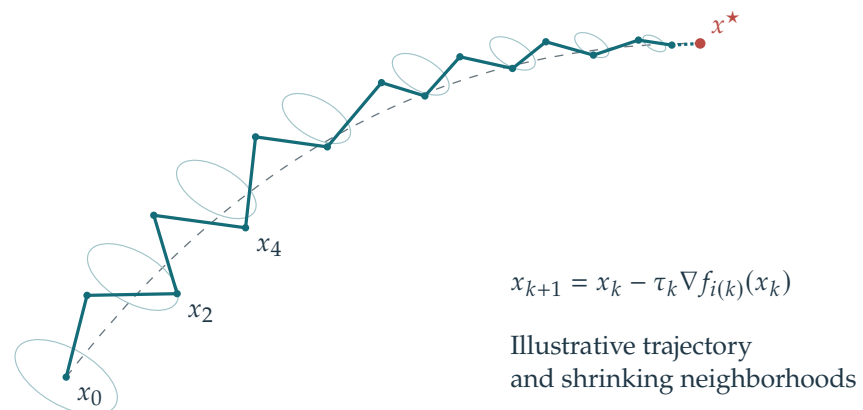
Complete figure / Stochastic gradient trajectories

Individual stochastic iterates fluctuate around the overall direction toward a minimizer.

Original



TikZ reconstruction



Reading the reconstruction. Interpretation to check: The path and shrinking ellipses preserve the qualitative intent of the sketch. They are illustrative neighborhoods, not confidence regions or a simulated dataset. A dotted continuation separates the last displayed iterate from the target minimizer. Convergence requires the objective, noise, and step-size assumptions developed in the section; step sizes alone do not guarantee it.

Figure 17.1

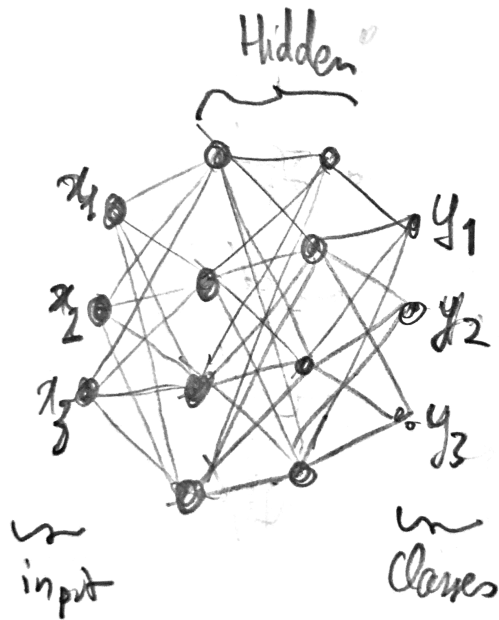
Main book, page 241

Panel 1: fully connected network. Panel 2: convolutional neural network.

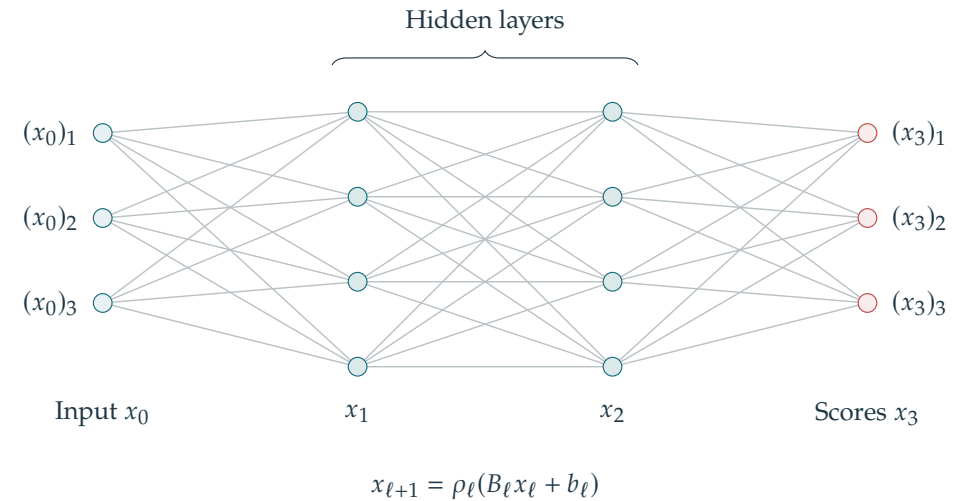
Panel 1 of 2 / A fully connected feedforward network

Deep MLP layers are affine maps followed by nonlinear activations.

Original



TikZ reconstruction



Reading the reconstruction. Retains the original three inputs, two hidden layers of four units, and three outputs. Every adjacent-layer pair is connected; line crossings are not nodes. The layer vectors are x_0, x_1, x_2, x_3 , and component labels follow that notation. The output x_3 denotes scores, avoiding confusion with the chapter target label y ; a separate logistic map converts scores into probabilities.

Figure 17.1

Main book, page 241

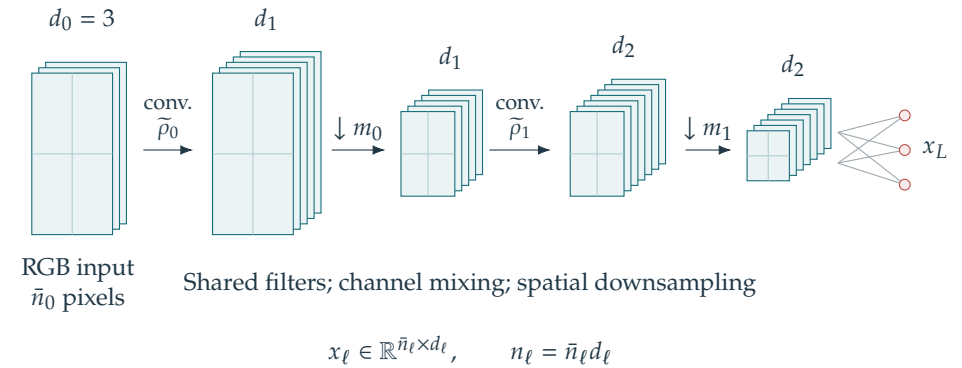
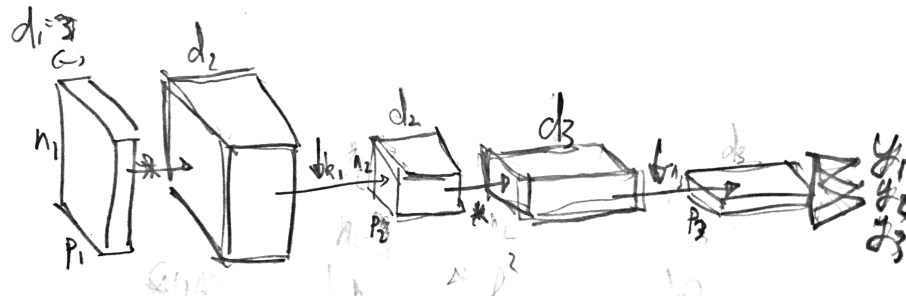
Panel 1: fully connected network. Panel 2: convolutional neural network.

Panel 2 of 2 / Feature maps in a convolutional network

Convolutional layers arrange activations by spatial location and channel; downsampling reduces spatial resolution.

Original

TikZ reconstruction



Reading the reconstruction. Interpretation to check: The low-resolution dimension labels are replaced by the chapter notation: $n_l = \bar{n}_l d_l$ and RGB input $d_0 = 3$. The five tensor blocks show an input, convolutional features, their downsampled version, a second feature stack, and a second downsampling. The activation arrows use the pointwise tilde-rho symbols because the chapter rho already includes downsampling. The output is the score vector x_L . Depth illustrates channel count and face size illustrates spatial resolution; exact numerical widths or filter sizes cannot be recovered and are not invented.

Figure 18.1

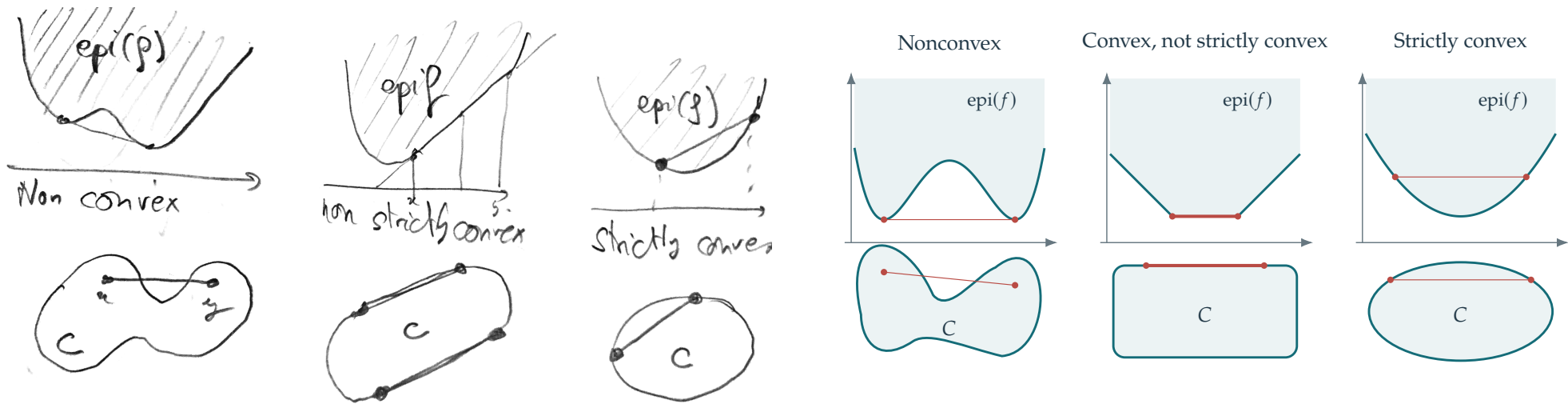
Convexity and strict convexity for functions and sets.

Complete figure / Convexity of functions and sets

The epigraph definition of convexity and the distinction between convexity and strict convexity.

Original

TikZ reconstruction



Reading the reconstruction. The same original also appears in the smooth-optimization chapter. This reconstruction is shared in design with that chapter: segments exhibit failure of convexity, an affine or flat boundary segment, and strict convexity. The epigraph is shaded above each function. The functions and sets are schematic examples rather than named formulas from the source.

Figure 18.2

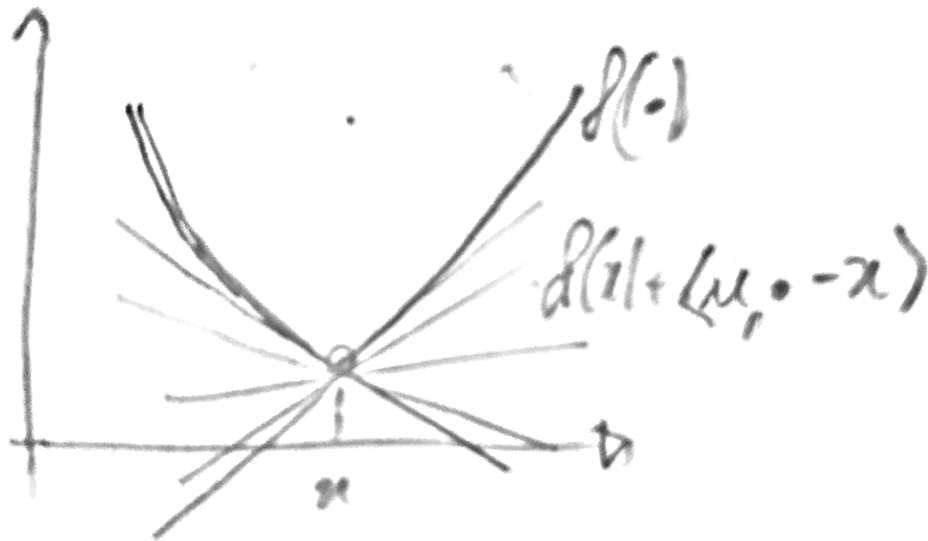
Main book, page 247

The subdifferential

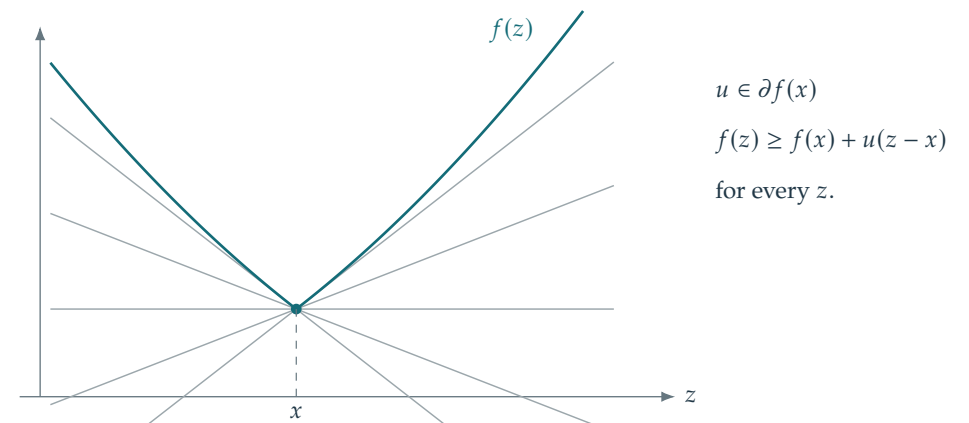
Complete figure / Supporting affine functions at a kink

Definition of the subdifferential as all supporting slopes.

Original



TikZ reconstruction



Several affine functions support the graph at this kink.

Reading the reconstruction. The original fan of supporting lines is reconstructed at a nonsmooth point of a convex function. Every drawn slope belongs to the interval between the one-sided derivatives. The diagram labels the supporting expression with a separate evaluation variable z , avoiding confusion with the base point x .

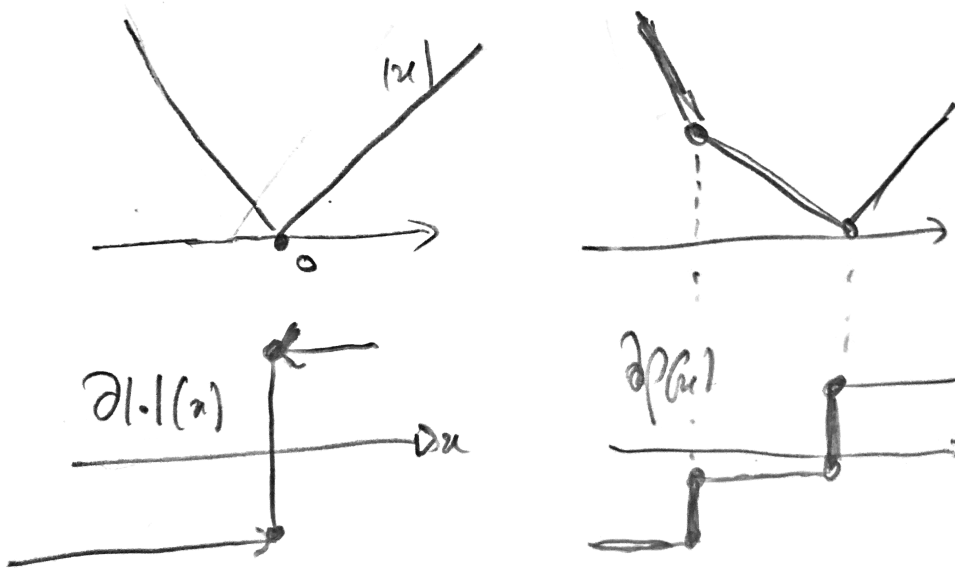
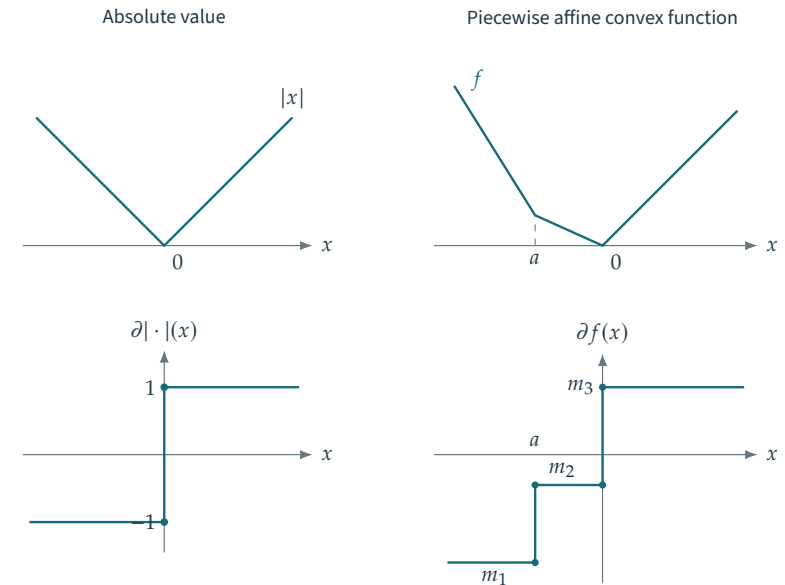
Figure 18.3

Main book, page 248

Subdifferential of the absolute value and a piecewise affine convex function.

Complete figure / Graphs of two subdifferentials

The subdifferential of absolute value and a piecewise affine convex function.

Original**TikZ reconstruction**

At each corner, the subdifferential contains every slope between the two one-sided slopes.

Reading the reconstruction. Interpretation to check: At each kink the entire closed interval of intermediate slopes is drawn, not just its endpoints. The second function uses representative slopes $m_1 < m_2 < 0 < m_3$ at knots $a < 0$ and 0 ; the original does not specify numerical slopes. The lower panels are set-valued graphs, so their vertical segments are essential.

Figure 18.4

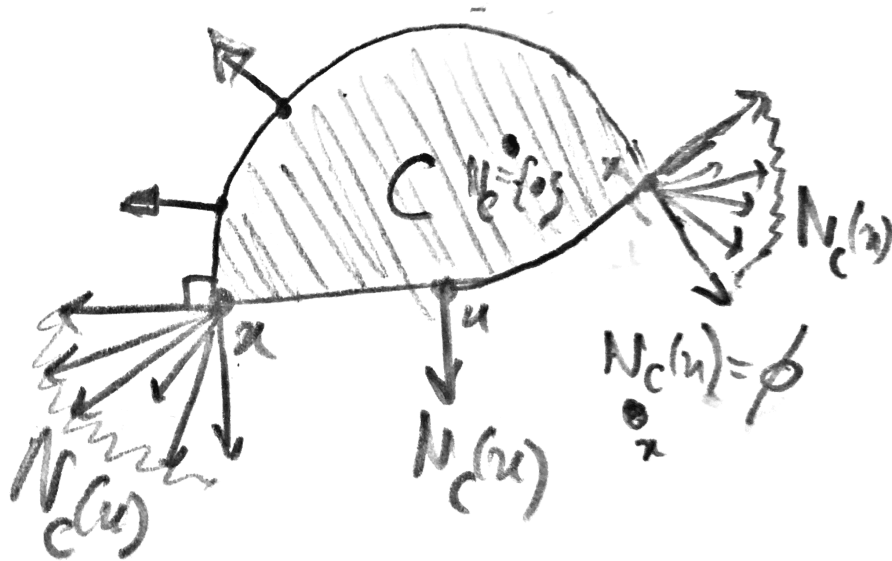
Main book, page 249

Normal cones

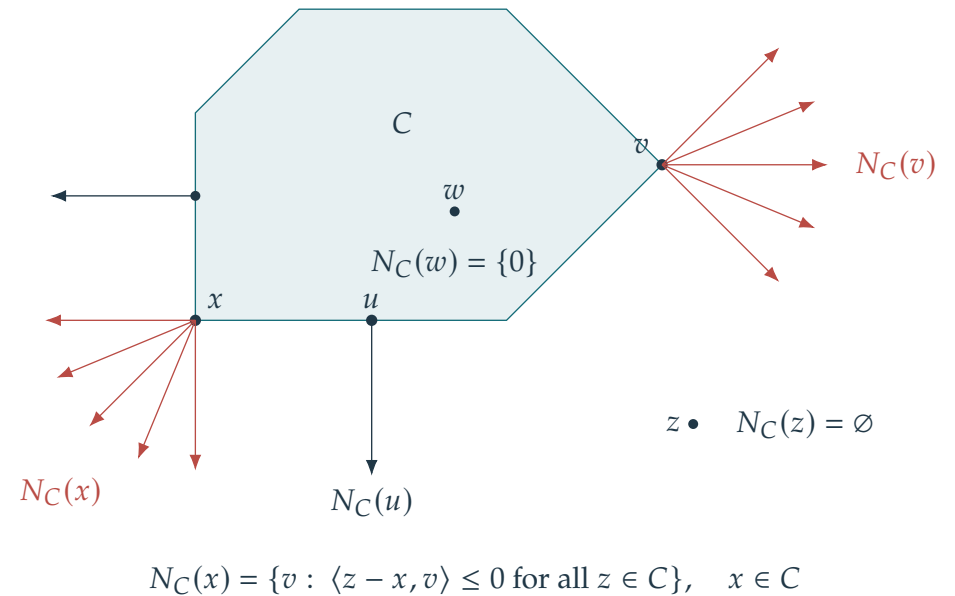
Complete figure / Normal cones at boundary, interior, and exterior points

The normal cone is the subdifferential of the indicator of a convex set.

Original



TikZ reconstruction



Reading the reconstruction. The convex body is rendered schematically as a polygon so its normal directions are exact. Edge interiors have outward normal rays, corners have cones, interior points have only the zero normal, and exterior points have an empty normal cone. The arrows depict directions and are translated to their base points; they do not bound the cone lengths.

Figure 19.1

Main book, page 257

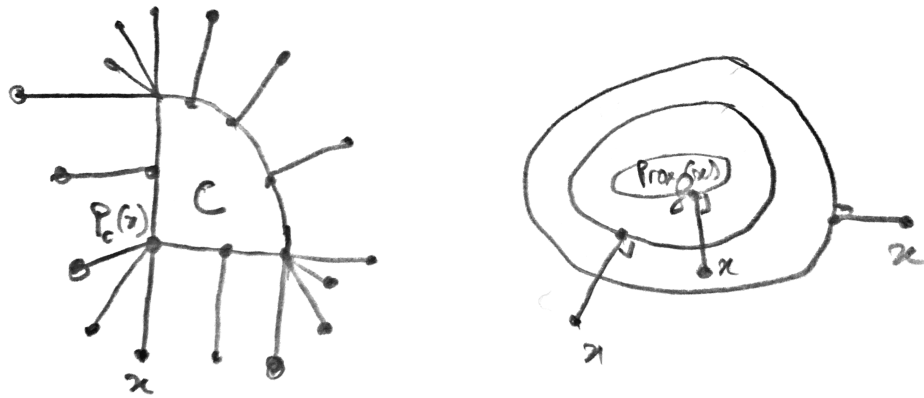
Proximal map and projection map.

Complete figure / Projection and proximal maps

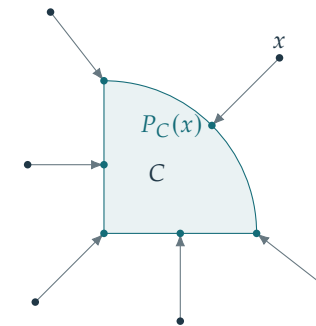
Projection onto a convex set and the sublevel-set interpretation of a proximal point.

Original

TikZ reconstruction



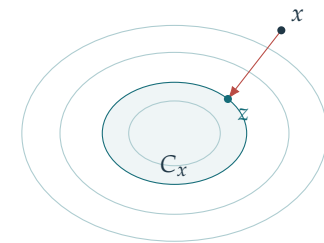
Projection onto a fixed set



$$x - P_C(x) \in N_C(P_C(x))$$

A sublevel set determined by the input

$$f(u, v) = \frac{1}{2}(u^2 + 2v^2)$$



$$z = \text{prox}_f(x), \quad C_x = \{u : f(u) \leq f(z)\}$$

$$z = P_{C_x}(x)$$

Reading the reconstruction. The left panel uses a quarter disk; all displayed input-to-projection segments lie in the appropriate outward normal cones. The right panel explicitly names $f(u,v)=(u^2+2v^2)/2$ and uses its nested sublevel sets with the exact proximal pair $x=(1.4,1.35)$, $z=(0.7,0.45)$. The highlighted ellipse radii are computed from $f(z)$, so z lies exactly on its boundary. The relevant sublevel set depends on the input; a fixed set does not represent the entire proximal map.